

NxM Templates for SuperCDMS SNOLAB Run 4 using Ge Activation Data

K-line event selection · 1x1 and NxM (PCA) templates

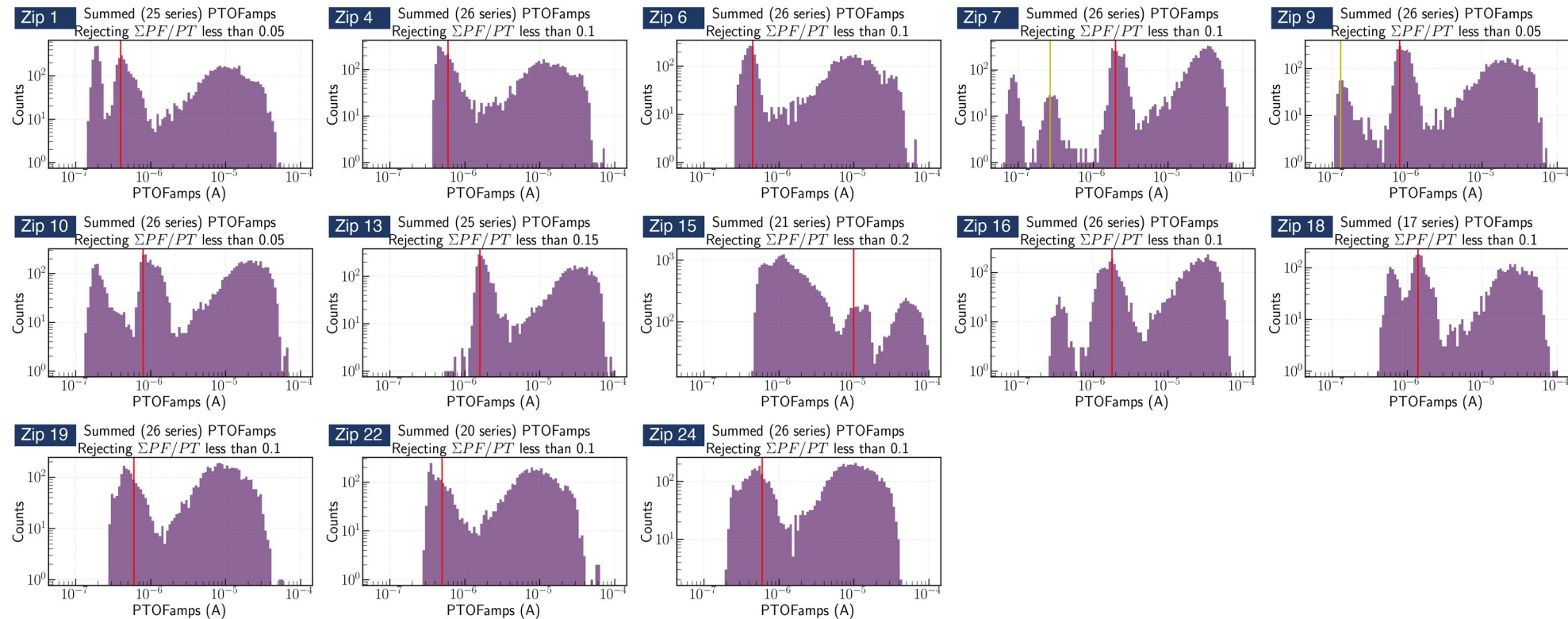
Zhiheng Li — July 2026

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- 4. Cross-checks with real pulses:** cuts remove noise, real pulses stay; slow-fall follow-up
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- 6. Template file and future steps:** cdmsbats ROOT files, ready for merge; three-, four-exponential fits
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Ge activation dataset

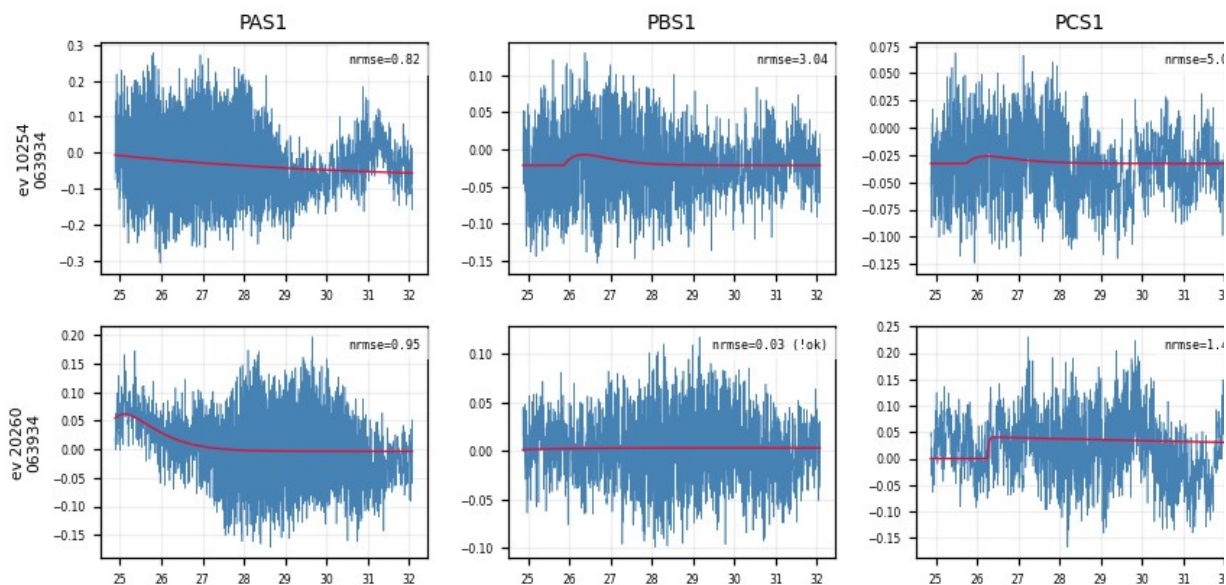
- Per detector, Prof. Saab's study marked the 10.37 keV K-line position in **PTOFamps** (the red line in each panel below)
- A **PTOFamps** window bracketing the red line: position $\times/\div 1.35$, a rough eyeballed choice; deliberately minimal, no other cuts
- For every selected event: cache the fully unprocessed raw MIDAS traces of all channels + event metadata, saved as per-series pickle files (reusable for any later analysis)
- Events in the window per zip: Z1 10.1k Z4 26.1k Z6 17.0k Z7 2.2k Z9 2.6k Z10 7.6k Z13 9.1k Z15 4.2k Z16 1.6k Z18 22.1k Z19 23.3k Z22 25.0k Z24 22.9k (total 174k)
- **Focus: Z7, the best detector; examples use its PBS1 channel; other detectors in the backup slides**



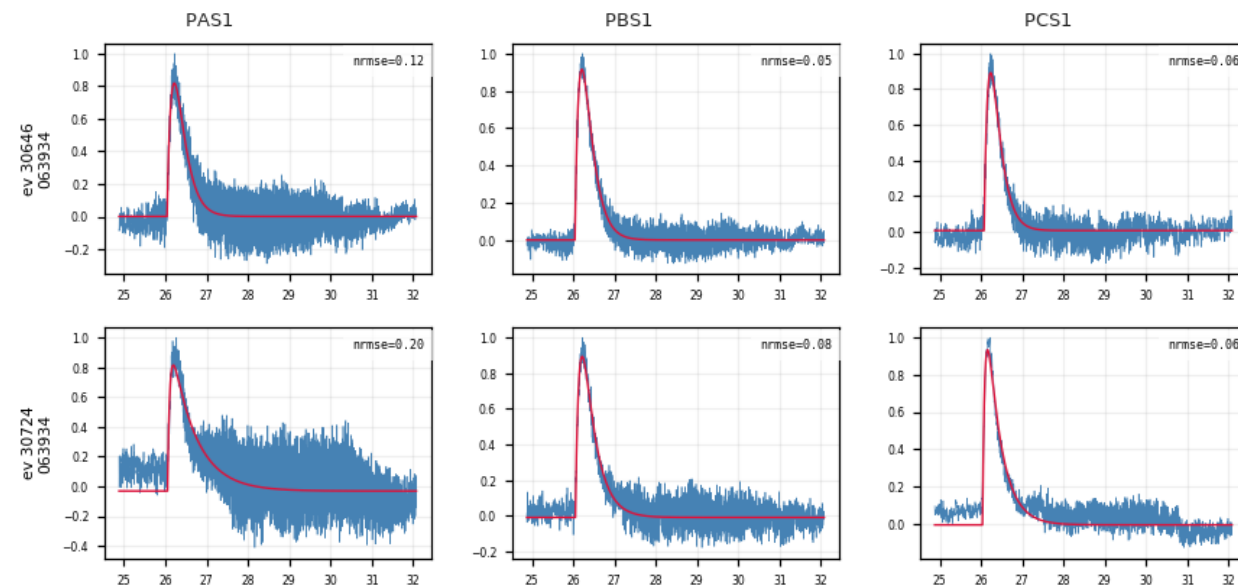
Fit examples

- Real events and noise triggers can be told apart by visual comparison (no cut applied at this stage)

NOISE



REAL PULSE



Per-trace algorithm: low-pass → fit → align

1. **Low-pass**: 100 kHz Butterworth
2. **Normalize**: subtract the baseline (median of the pre-pulse samples), scale the trace to unit peak
3. **Fit**: two-exponential model:

$$y(t) = A[e^{-(t - t_0)/\tau_{\text{fall}}} - e^{-(t - t_0)/\tau_{\text{rise}}}] + b$$

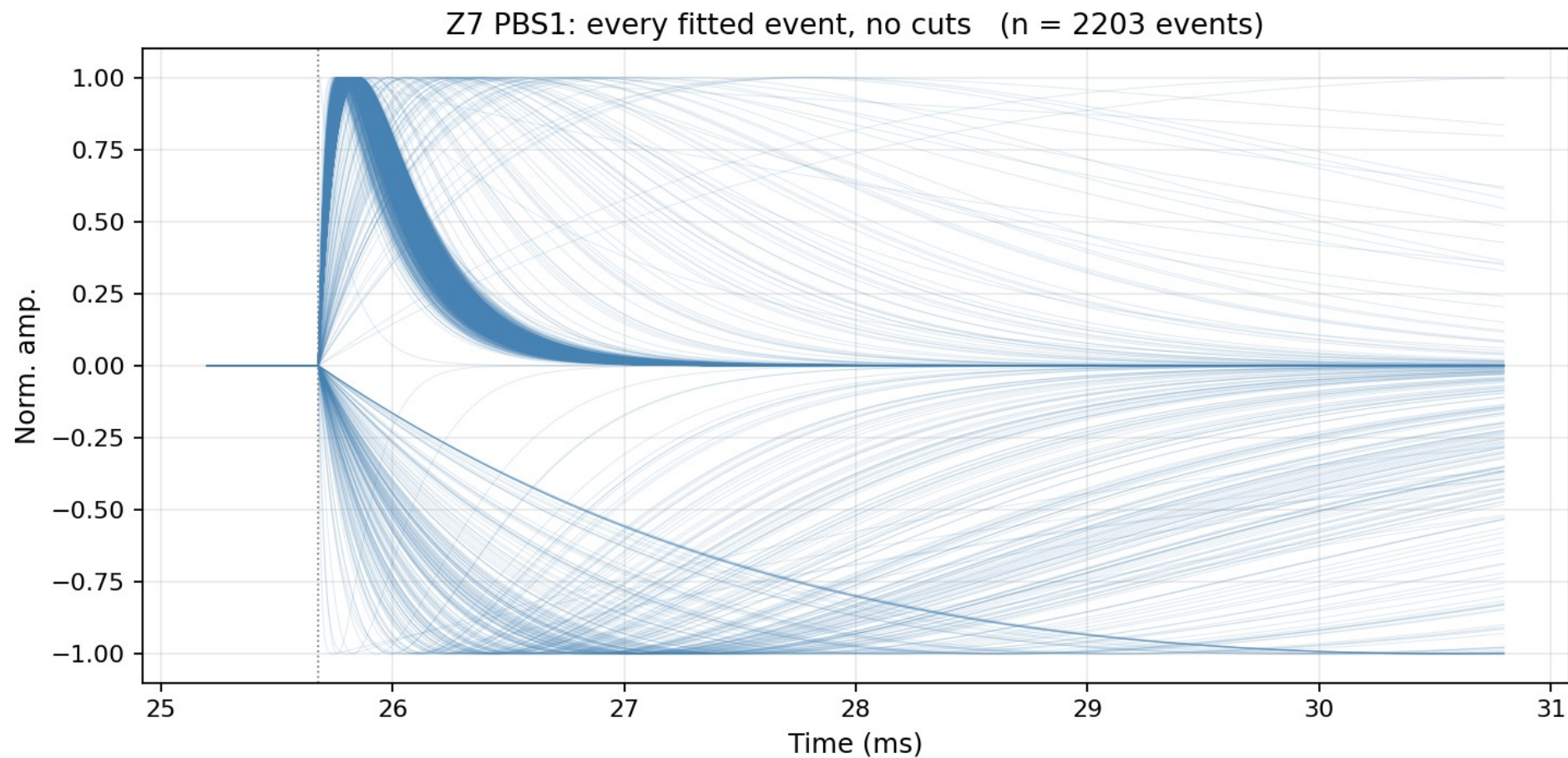
t_0 = pretrigger, free within 16050 ± 3000 samples (τ_{rise} , τ_{fall} : rise / fall time)

Fit parameters: A amplitude · τ_{rise} rise time constant · τ_{fall} fall time constant · t_0 pulse onset (pretrigger) · b baseline

4. **Two quality checks per trace** (calculated and recorded here, not cut yet):

- **fit_ok**: amplitude is positive, and the pulse rises faster than it falls
- **NRMSE**: RMS of the fit residual ÷ pulse peak (smaller = better fit)

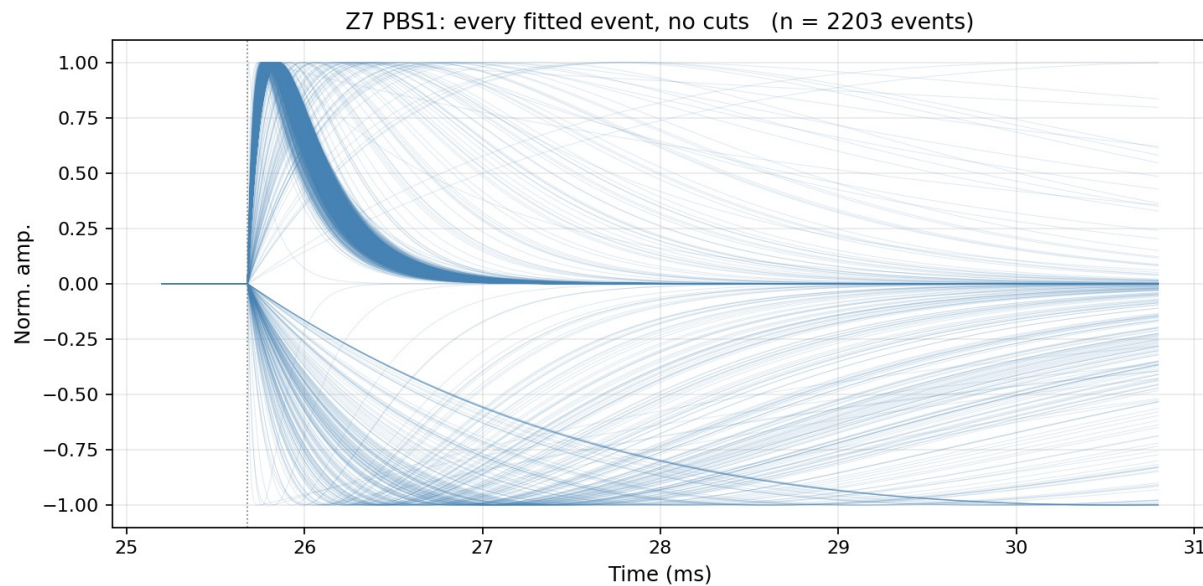
5. Align



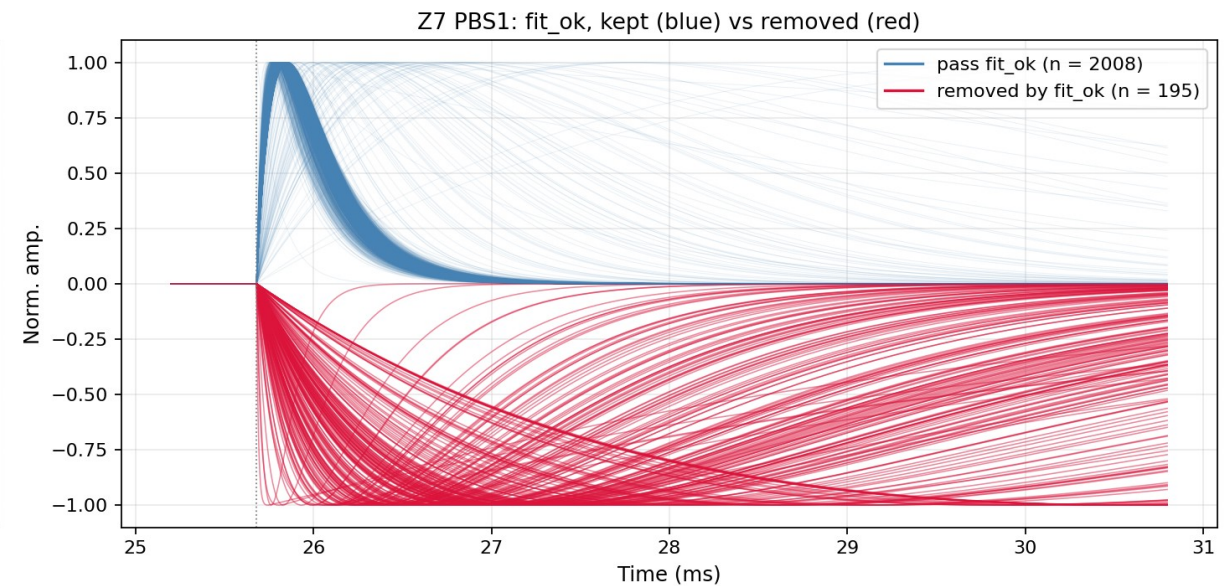
Z7 PBS1: every fitted event drawn at the common pretrigger, no cuts of any kind (n = 2203, inverted fits included)

Quality cut 1: fit_ok

- fit_ok: amplitude > 0 AND rise faster than fall (the fit is a physical pulse)
- Z7 PBS1: 2008 of 2206 events pass; 195 removed (red, inverted or negative shapes), 3 fits did not converge



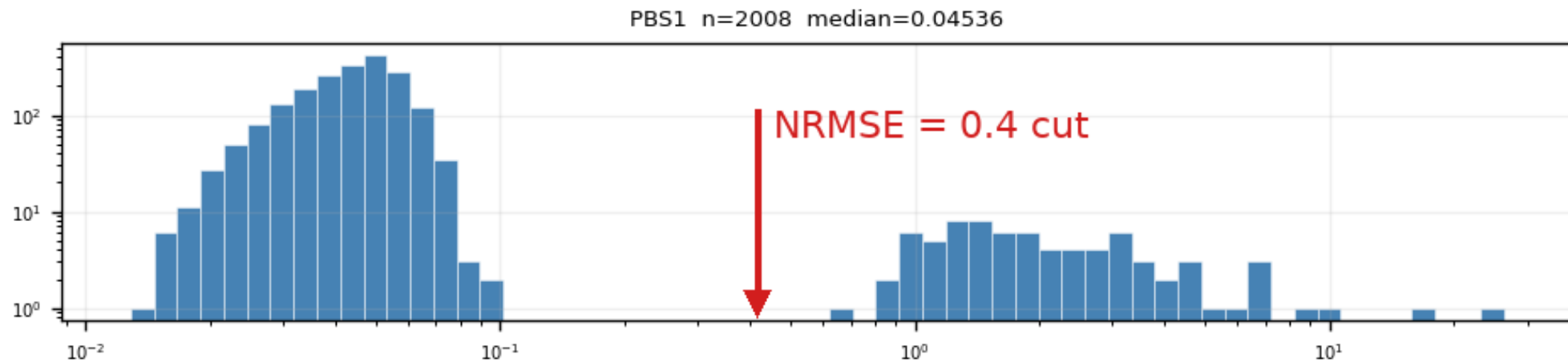
All fitted events, no cuts (n = 2203 drawn, inverted fits included)



fit_ok: kept (blue) vs removed (red, drawn at |peak| normalization)

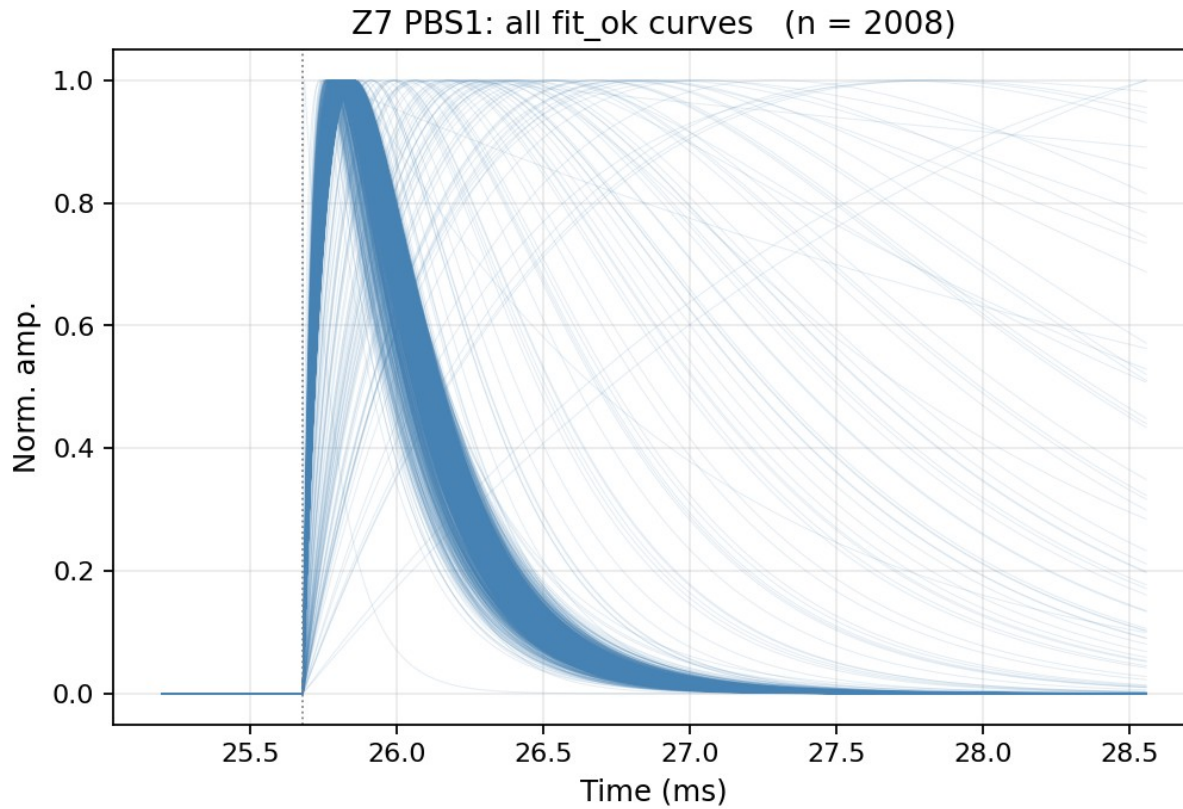
Quality cut 2: $\text{NRMSE} \leq 0.4$ — where the number comes from

- NRMSE of fit_ok events is bimodal: good fits (median ≈ 0.05 – 0.1) vs noise triggers (≈ 1 – 2), valley at ≈ 0.4 – 0.5
- The cut sits in the valley: eyeball read off the distribution
- Worse detectors (Z1, Z4, Z6, Z18, Z19, Z22, Z24): same bimodal picture, noise peak dominates

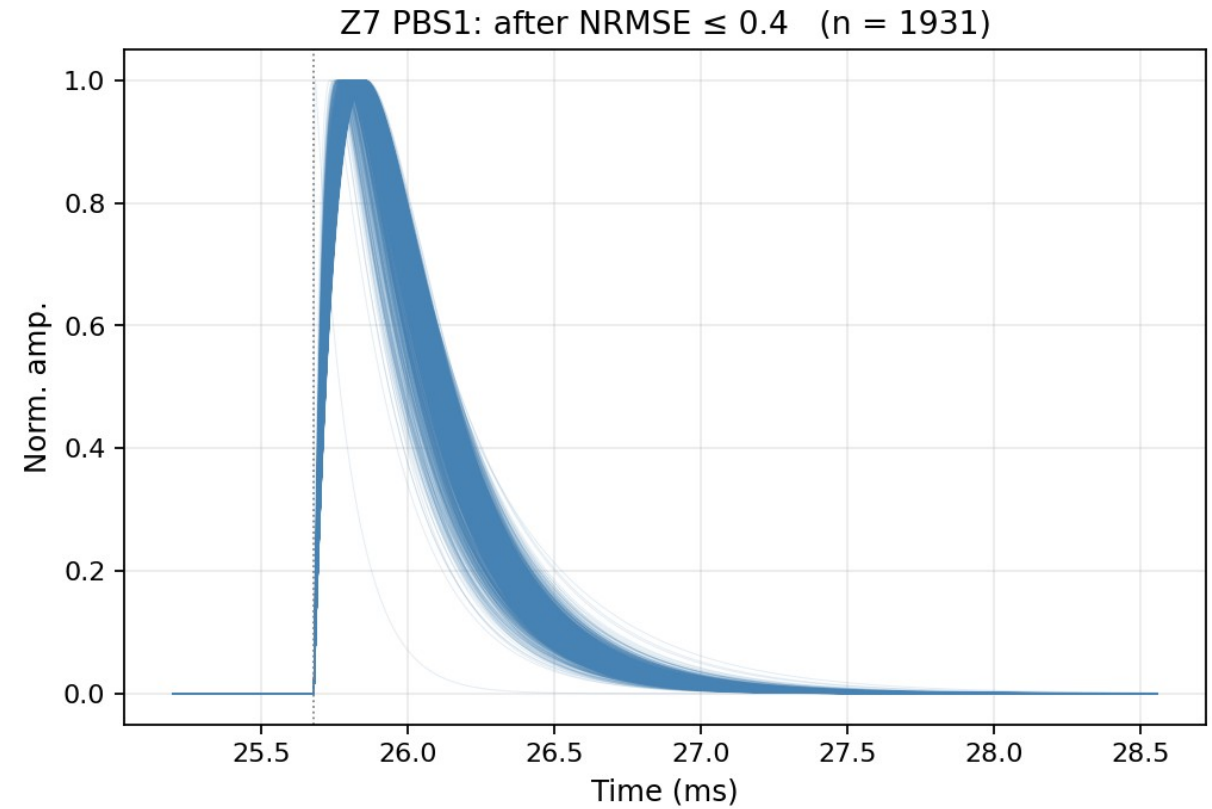


Z7 PBS1: two clean populations, log-log axes

Aligned fitted curves: before and after the NRMSE cut



All fit_ok fitted curves (Z7 PBS1, n = 2008, every curve drawn)

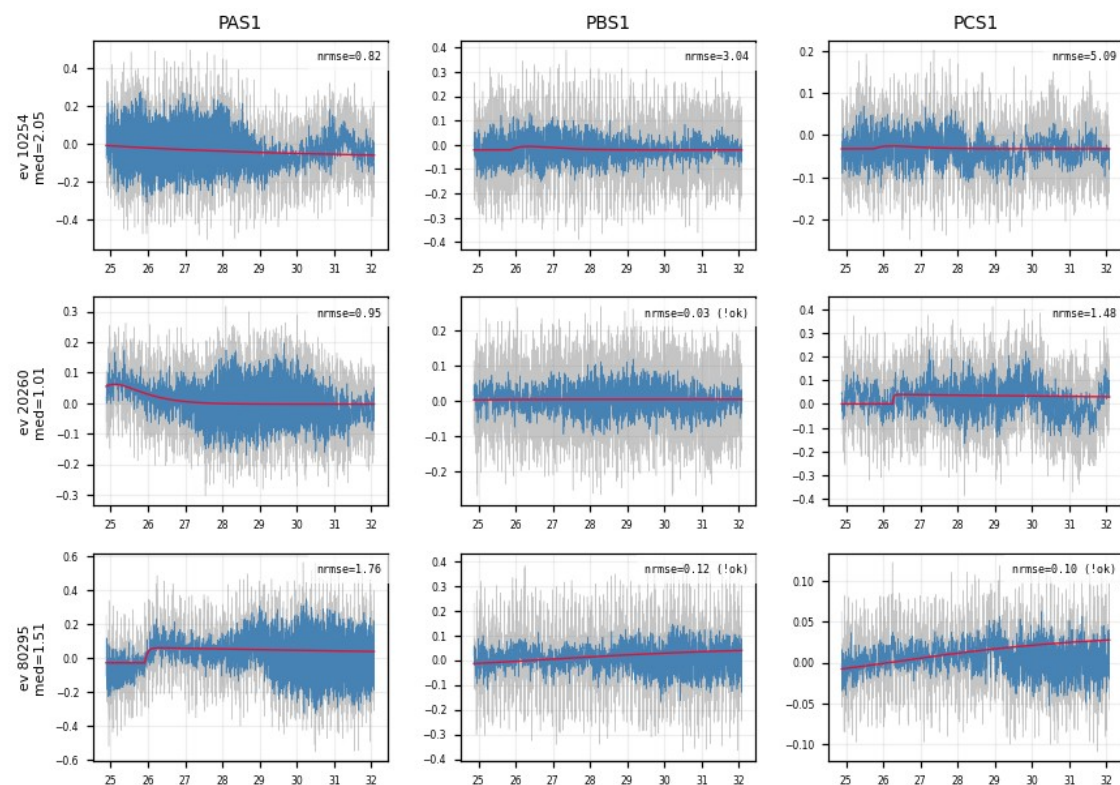


After NRMSE ≤ 0.4 (n = 1931): one tight shape family

The rejected population is noise — verified in the raw traces

- Example of NRMSE-rejected events (median NRMSE > 0.4 across channels): the raw traces show no pulse
- the cut removes noise triggers, not physics

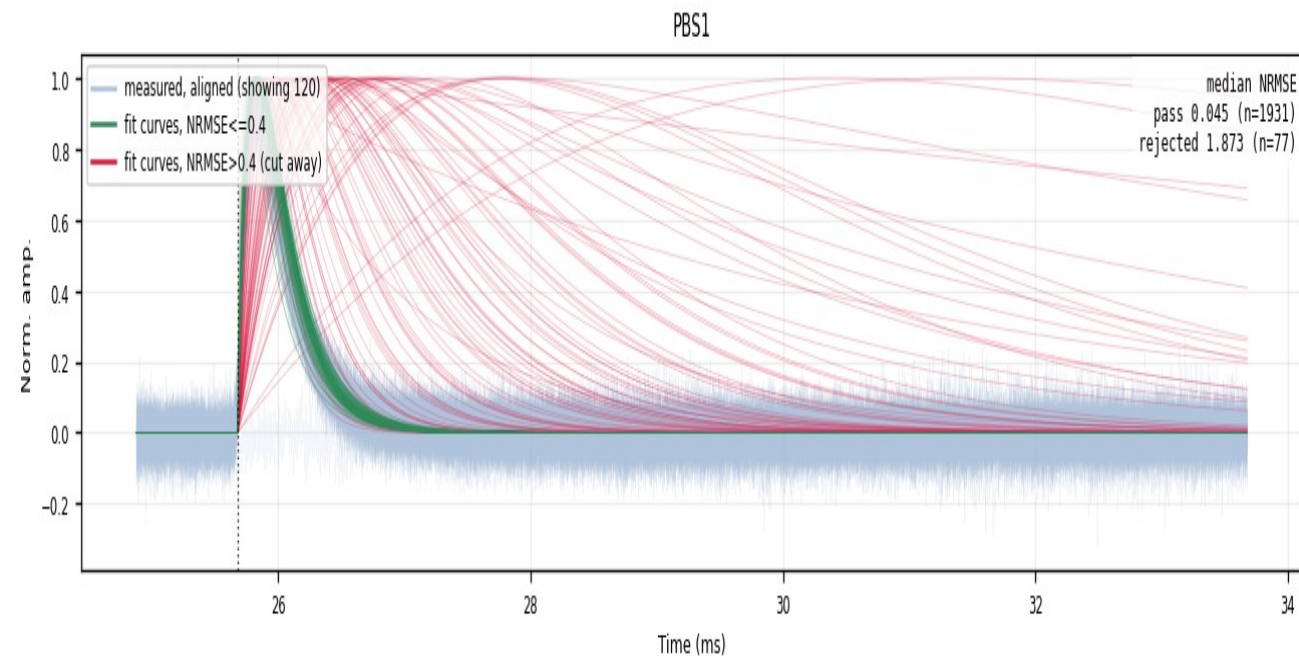
Rejected events: raw vs fit



Z7: NRMSE-rejected events — raw traces are noise

Aligned curves: kept vs rejected

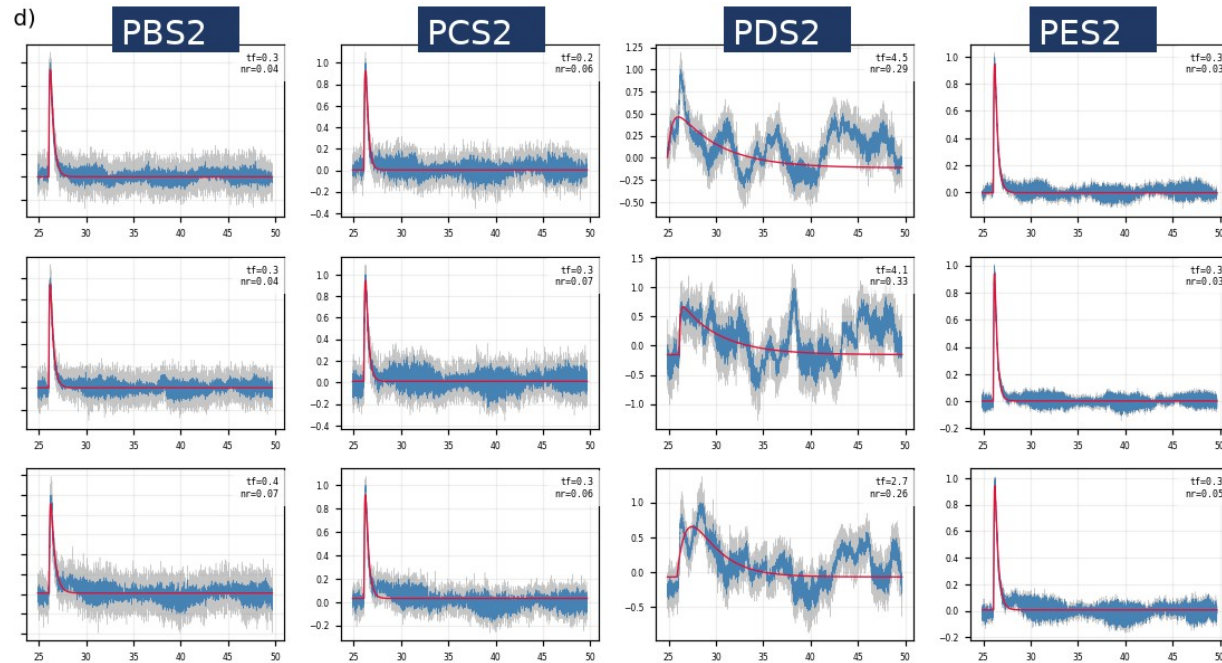
median NRMSE: **kept 0.045 (n = 1931)** · **rejected 1.873 (n = 77)**



Z7 PBS1 — **passes the cut (kept)** — **cut away = rejected**
faint blue = measured traces 1931 kept / 77 cut (~3.8%)

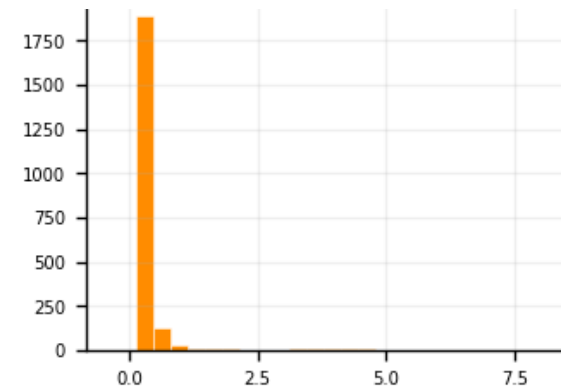
Slow-fall tail in one particular channel

- PDS2 alone: can not make useful templates; hard to fix in analysis/software, the noise itself needs to be fixed
- Temporary solution: use the PDS1 template in place of PDS2 (applied in the ROOT files)

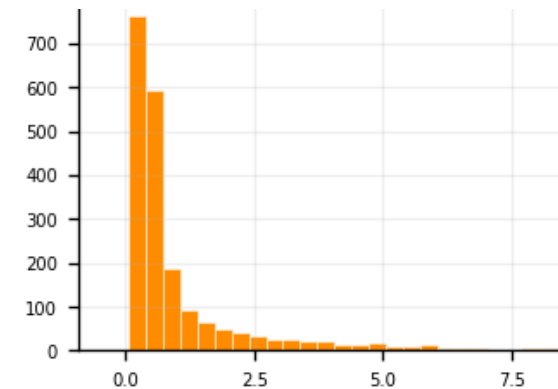


Z7, 3 sampled slow-fall events: normal in PBS2/PCS2/PES2, swings only in PDS2

Fitted τ_{fall} distribution, same 0–7 ms axis:



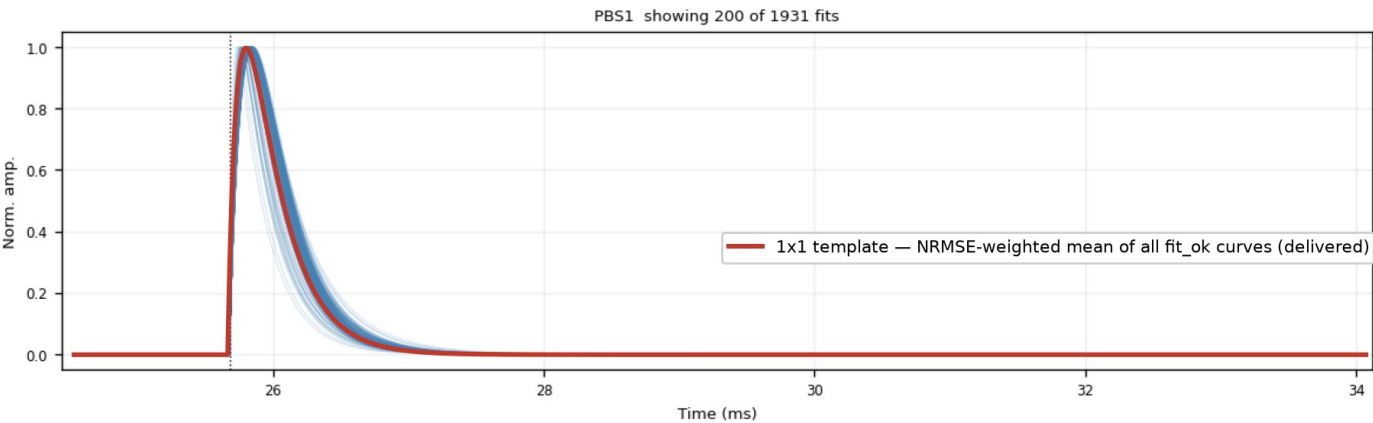
Z7 PAS1 — normal channel: narrow
(median 0.25 ms)



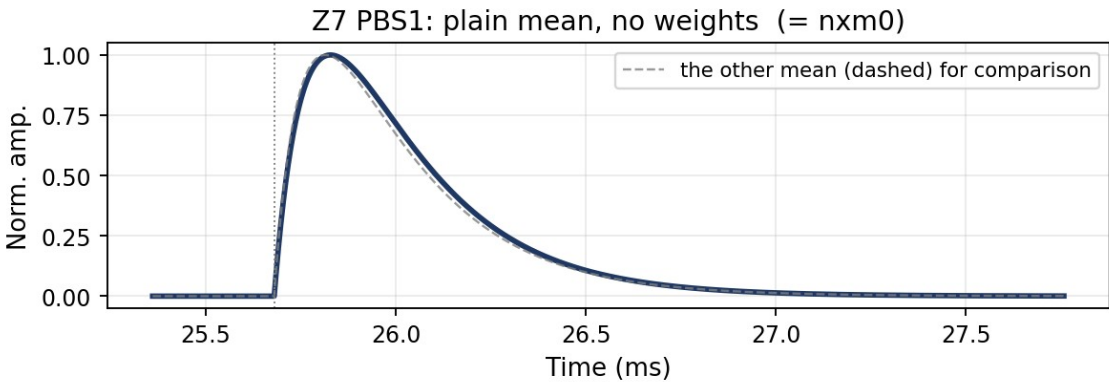
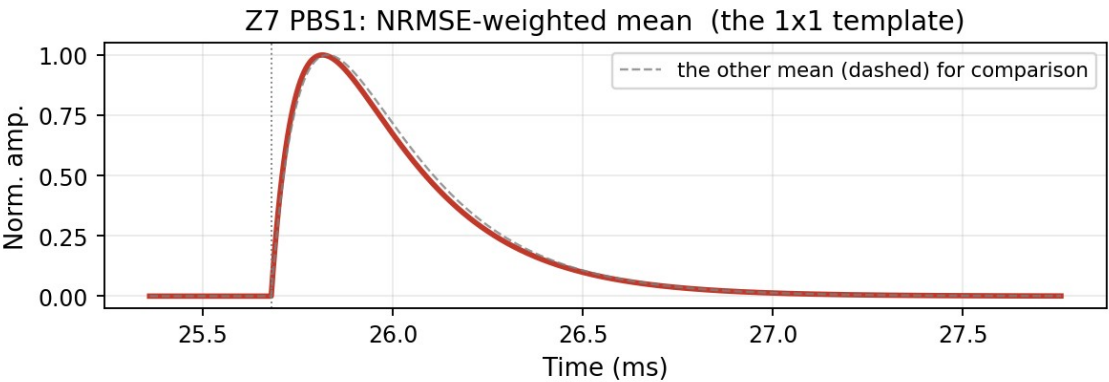
Z7 PDS2 — bad channel: broad tail
(median 0.51 ms)

Results: 1x1 Templates

- NRMSE-weighted mean of the fit_ok 2-exp curves
- Smooth & noise-free by construction

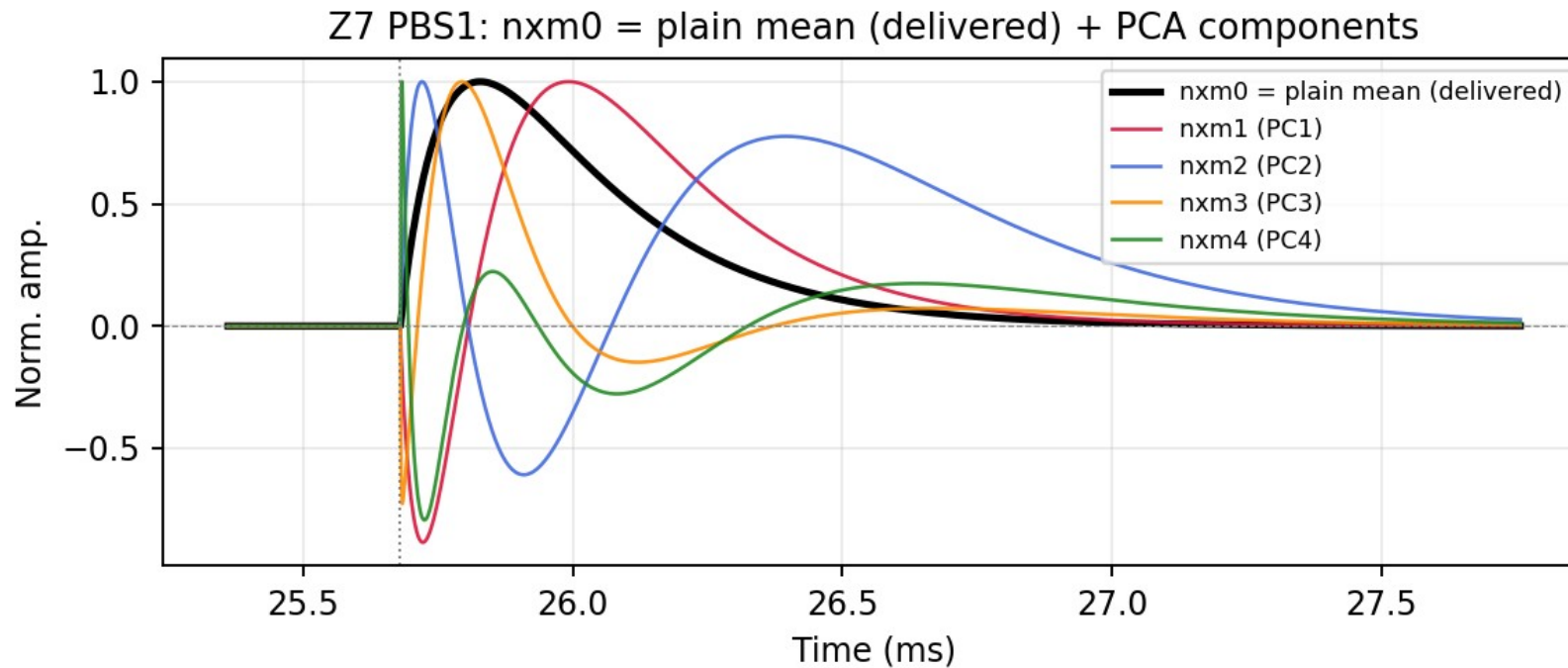


Z7 PBS1 — blue: the fitted curves (200 of 1931 drawn); red: the 1x1 template



Results: NxM Templates

- channel-specific PCA over the fitted curves ($\text{fit_ok} + \text{NRMSE} \leq 0.4$)
- nxm0 = mean shape; nxm1 – 4 = principal components; $\text{real pulse} = \sum_i \text{amp}_i \cdot \text{nxm}_i$
- $\text{PC1} + \text{PC2} \approx 96\text{--}98\%$ of the shape variance



Z7 PBS1: nxm0 (plain mean over the $\text{fit_ok} + \text{NRMSE}$ -cut curves) + nxm1 – 4 , zoomed to the pulse

Template file

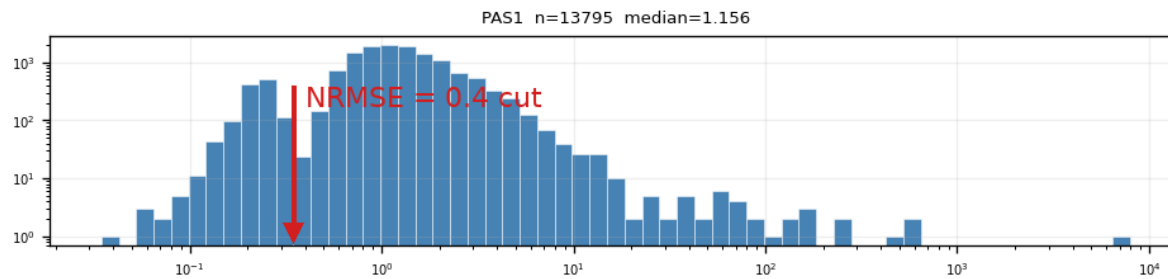
- Two template sets for all 13 detectors: 1x1 (2-exp weighted) + NxM PCA
- In cdmsbats_config feature branch, ready for merge

Future Steps

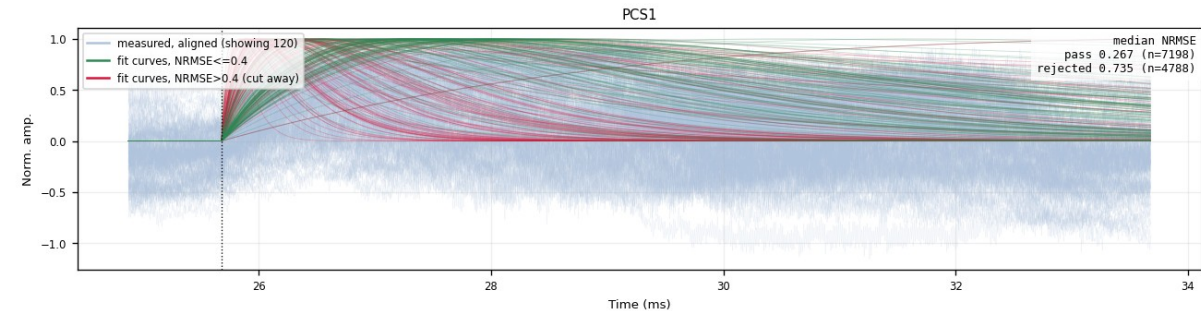
- Three-, four-exponential fits
- Exercise NxM processing
- Other detectors: worse noise, a few more cuts added, in the backup slides

Backup · weak detectors (example: Z22)

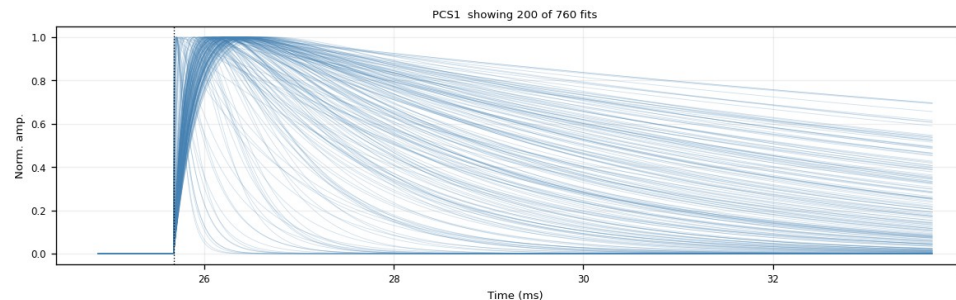
- Z1 / Z4 / Z6 / Z18 / Z19 / Z22 / Z24: K-line inside the noise population, the window admits a noise-dominated mixture
- Same bimodal NRMSE picture, noise peak dominant; the 0.4 cut digs the real pulses out (Z22 PCS1: 7198 kept / 4788 cut, ~40%)
- Kept bundle broader; steep red curves = fits latching onto noise spikes, no real pulses lost
- $\tau_{\text{rise}} \leq 0.3$ ms (after NRMSE): removes 55–74% on weak zips (Z22: 71%) vs 1.6% on Z7; removes residual slow drift



Z22 PAS1 NRMSE histogram: noise dominates



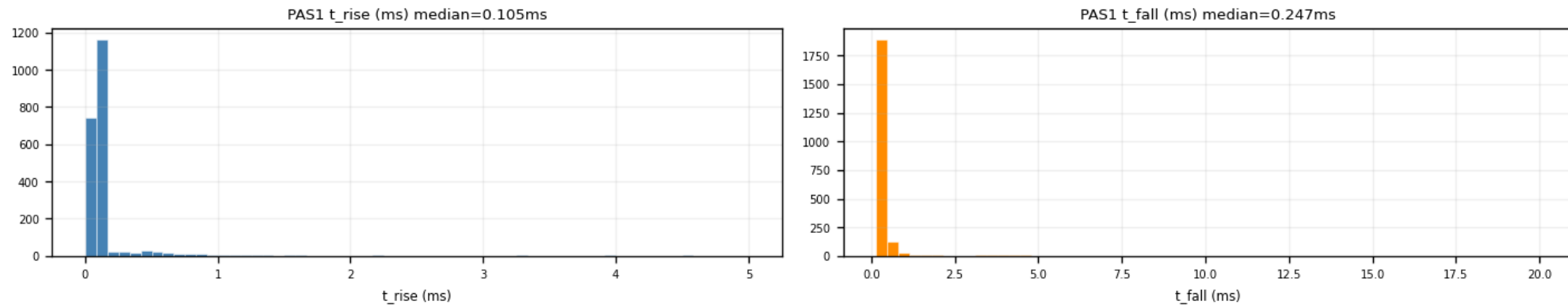
Z22 PCS1: kept (green) vs cut (red)



Z22 PCS1 after $\text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3$ ms: final shape family (760 fits)

Backup · $\tau_{\text{rise}} \leq 0.3$ ms ceiling (PCA input)

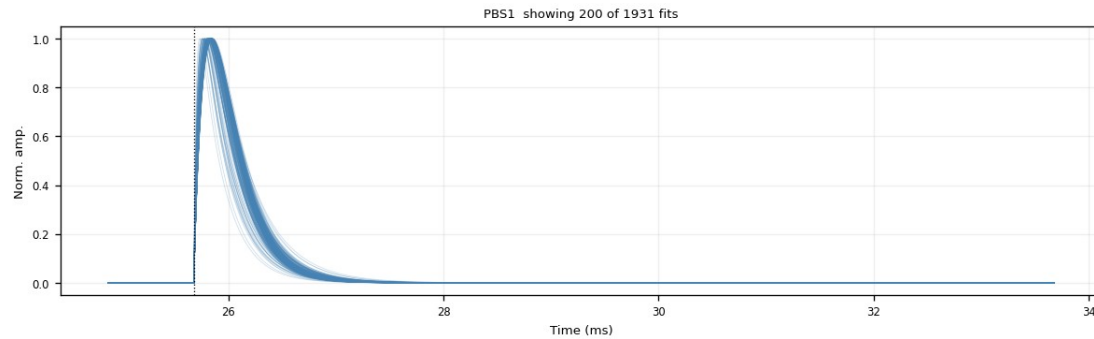
- Slow baseline drift survives NRMSE (a slow 2-exp hugs it) → mimics a slow rise
- Fast pulses at $\tau_{\text{rise}} \approx 0.1$ ms (p90 ≈ 0.15 ms) → ceiling at 0.3 ms blocks the drift tail
- Removed after the NRMSE cut: Z7 1.6% (quiet 2–6%), weak 55–74% (Z22 71%), all zips 54%
- Trade-off: trims the very slowest genuine pulses → decision pending



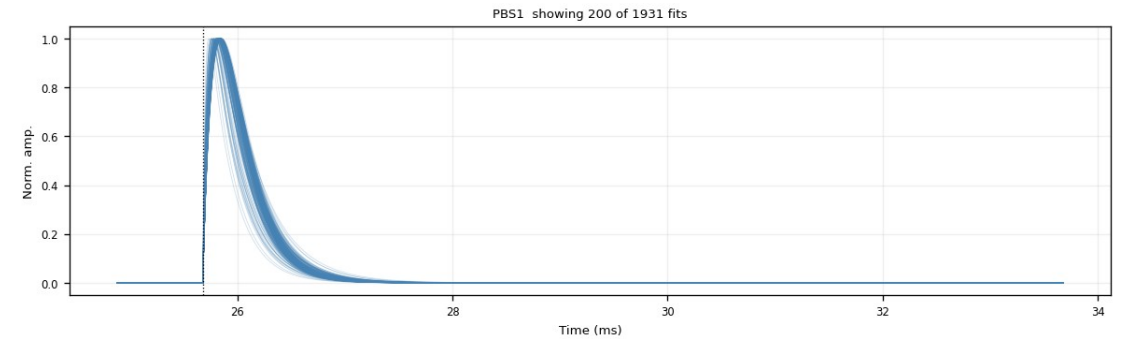
Z7 PAS1: fitted τ_{rise} (median 0.105 ms) and τ_{fall} distributions

Backup · τ_{rise} cut: good vs bad channel (Z7)

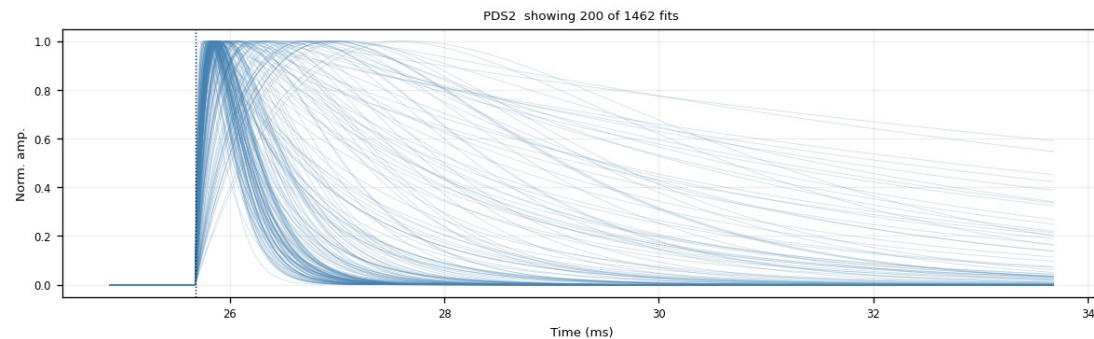
- Good channel PBS1: tight clean bundle; the cut removes ~0% (nearly free on quiet channels)
- Bad channel PDS2: broad messy fan, the channel itself is bad; the cut trims slow risers, 1462 $\rightarrow$ 1154 (21%)



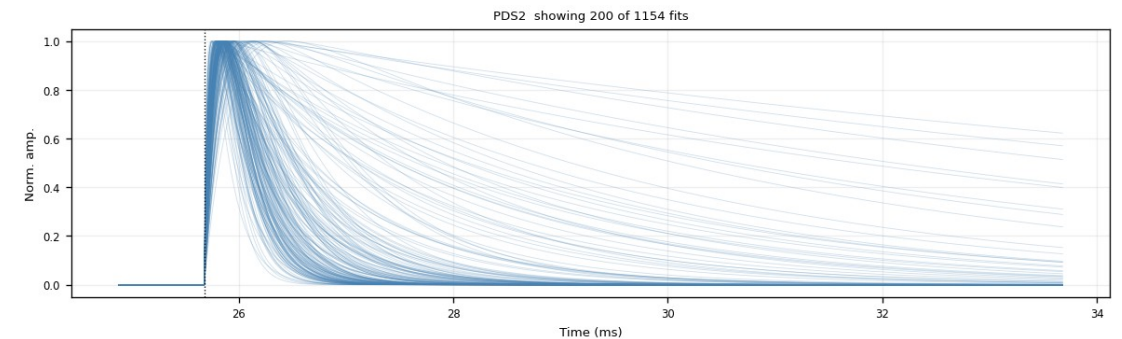
PBS1 (good): $\text{NRMSE} \leq 0.4$



PBS1 (good): $+\tau_{\text{rise}} \leq 0.3 \text{ ms}$ (unchanged)



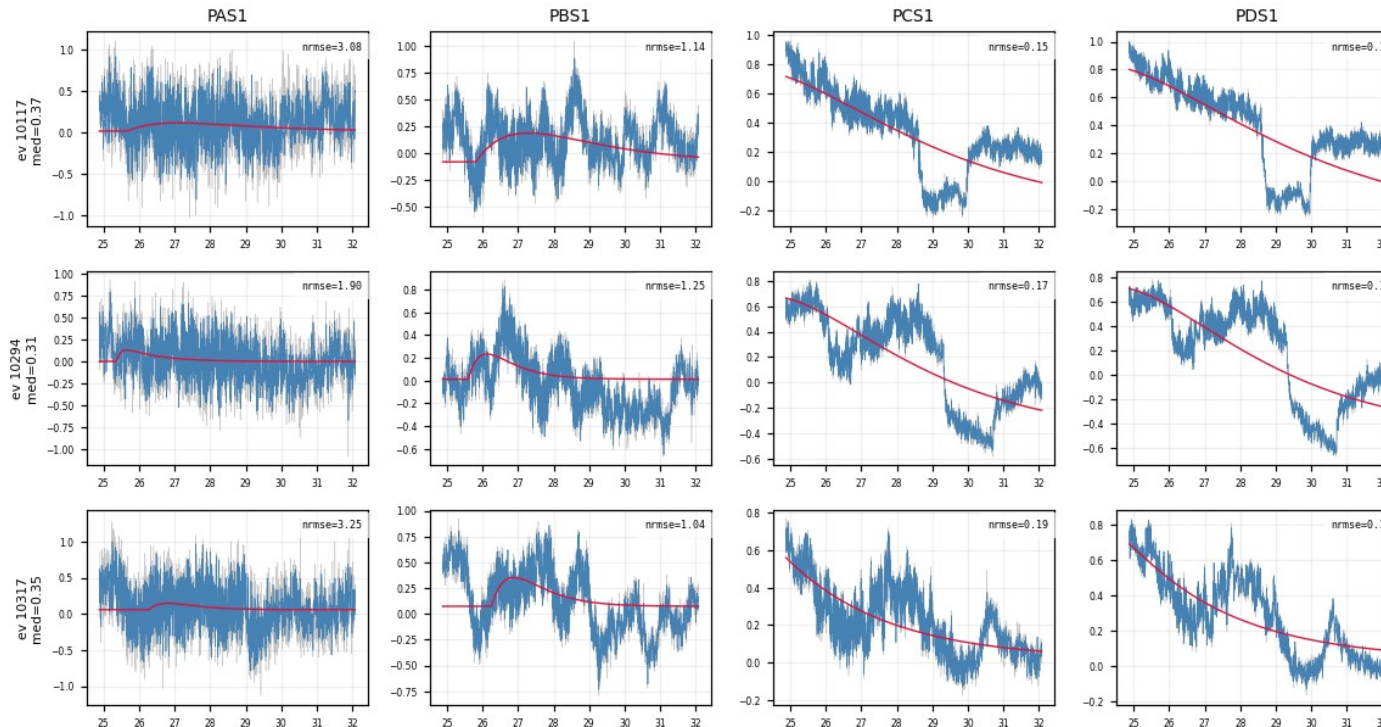
PDS2 (bad): $\text{NRMSE} \leq 0.4$



PDS2 (bad): $+\tau_{\text{rise}} \leq 0.3 \text{ ms}$ (slow risers trimmed)

Backup · what the τ_{rise} cut removes (Z22)

- Events that PASS $\text{NRMSE} \leq 0.4$ but are REMOVED by $\tau_{\text{rise}} \leq 0.3$ ms: the noise 0.4 let through, 0.3 catches
- PCS1 / PDS1 pass at $\text{NRMSE} \approx 0.15$ (a slow 2-exp hugs the drift); the raw trace has no clean fast pulse
- Trade-off: also trims genuine slow-rise pulses; decision pending



Z22, 3 events $\times$ PAS1/PBS1/PCS1/PDS1: pass $\text{NRMSE} \leq 0.4$, removed by $\tau_{\text{rise}} \leq 0.3$ ms; PCS1/PDS1 = pure slow drift

Backup · Z7 full results gallery

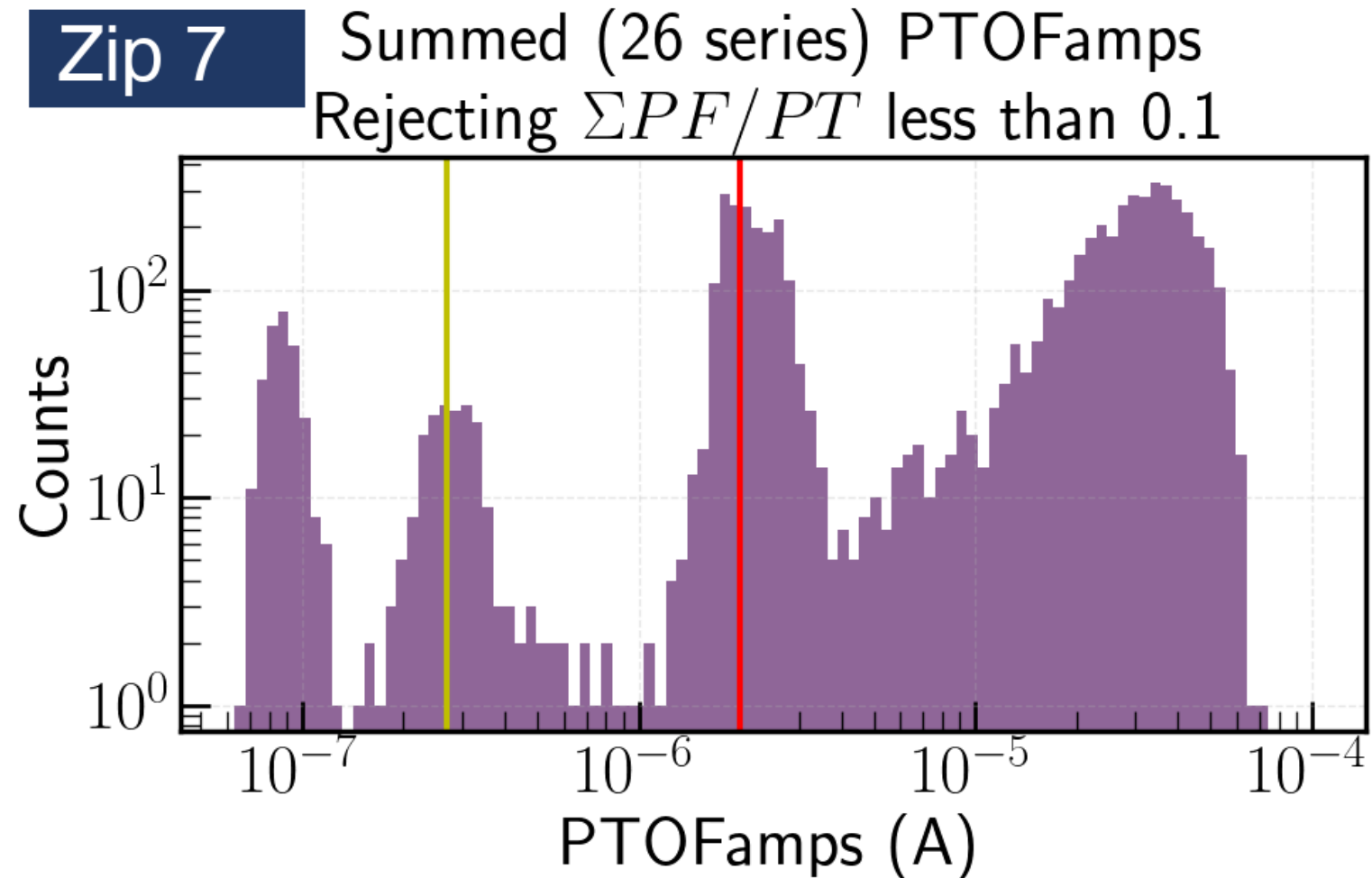
every diagnostic and result figure of the Z7 pipeline, in processing order

- Dataset: the K-line selection window
- Example grids: LP + fit and raw vs fit, event × channel
- Fit diagnostics: NRMSE, fitted pretrigger, τ_{rise} / τ_{fall} distributions
- Aligned fans: no cut $\rightarrow$ $\text{NRMSE} \leq 0.4 \rightarrow + \tau_{\text{rise}} \leq 0.3$ ms; kept vs rejected; measured traces
- Special populations: rejected events, well-fit slow-rise events, PDS2 slow-fall study
- Templates: PCA nxm0–4 per channel, summed PT / PS1 / PS2
- Tall figures are cut at white space into full-width panels, two per page; grids into 3-event × half-channel blocks

Backup · Z7 · K-line selection window

figure: kline_zip7.png

Summed PTOFamps spectrum of Z7; red line = Prof. Saab's 10.37 keV K-line position, shaded band = the $\times/\div 1.35$ selection window; everything inside is cached raw.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (1/10)

figure: zip7_fit_examples.png, event rows 1–3 of 15, left half of the channels

First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.

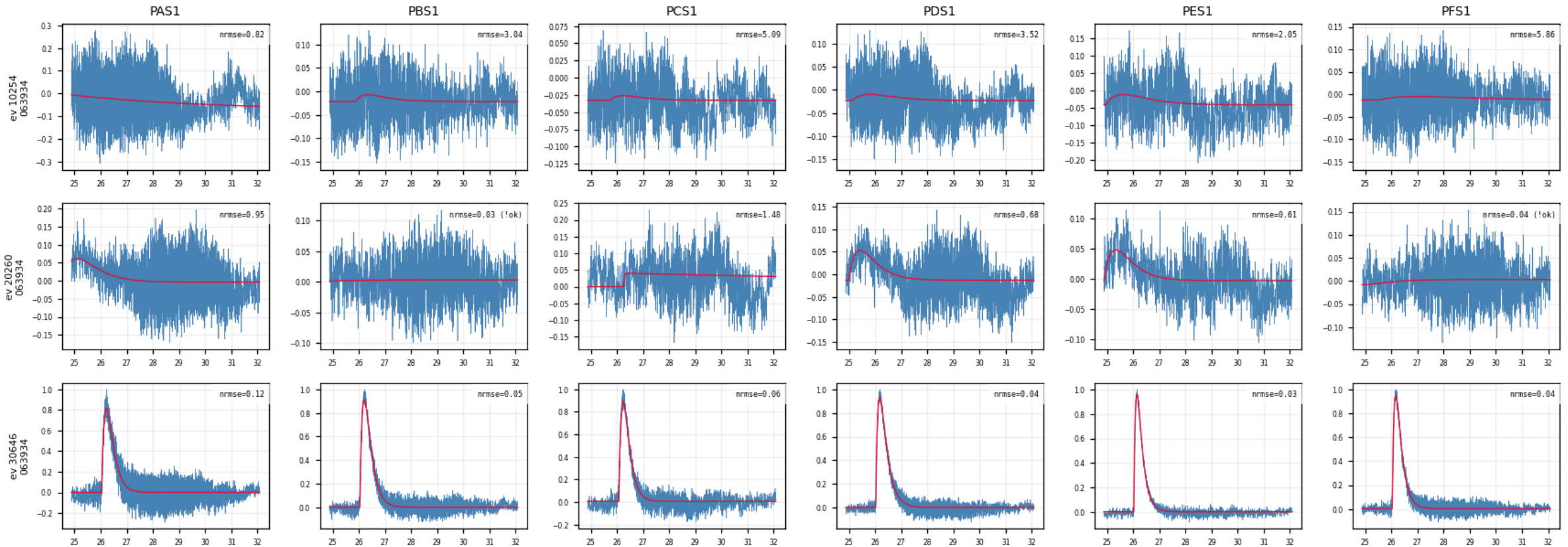


figure: zip7_fit_examples.png, event rows 1–3 of 15, right half of the channels

Figure 1 displays a 3x6 grid of plots showing time series data and fitted models for six variables: PAS2, PBS2, PCS2, PDS2, PES2, and PFS2. The first five columns show time series plots with fitted models and their respective nrmse values. The sixth column shows missing data plots for PFS2.

The first five columns (PAS2, PBS2, PCS2, PDS2, PES2) show time series plots with fitted models and their respective nrmse values:

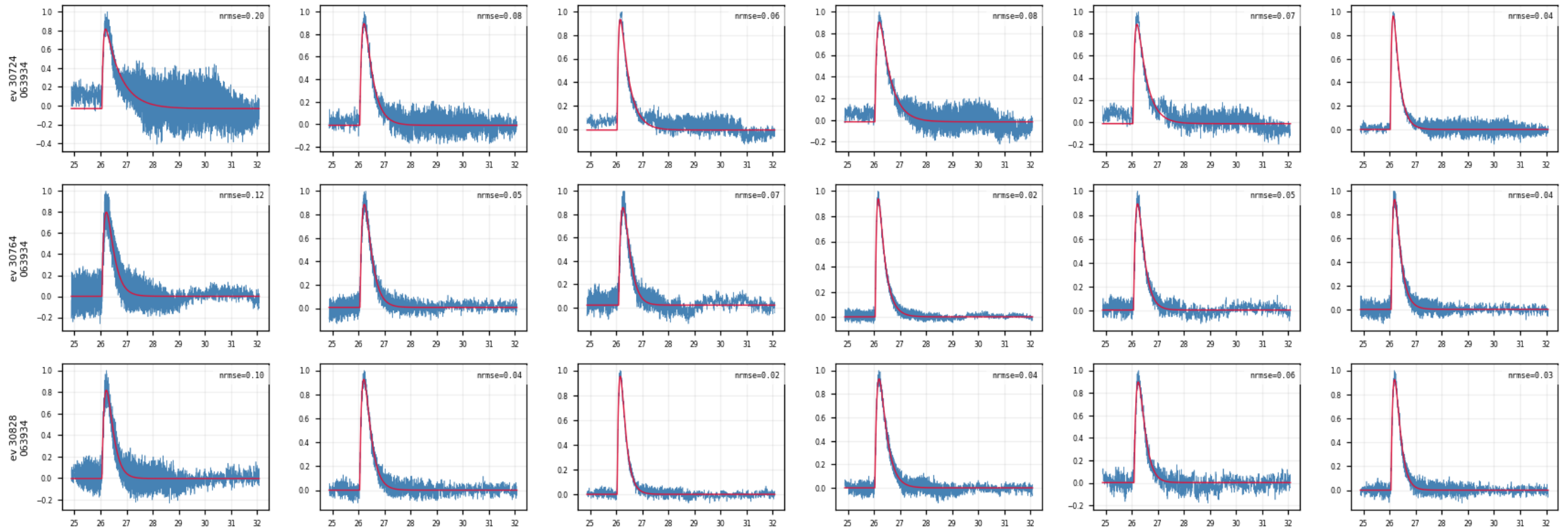
- PAS2: nrmse=0.55
- PBS2: nrmse=3.38
- PCS2: nrmse=0.29
- PDS2: nrmse=0.81
- PES2: nrmse=0.53

The sixth column (PFS2) shows missing data plots, with the text "missing" displayed in the center of each plot.

Backup · Z7 · fit_examples: LP trace vs 2-exp fit (3/10)

figure: zip7_fit_examples.png, event rows 4–6 of 15, left half of the channels

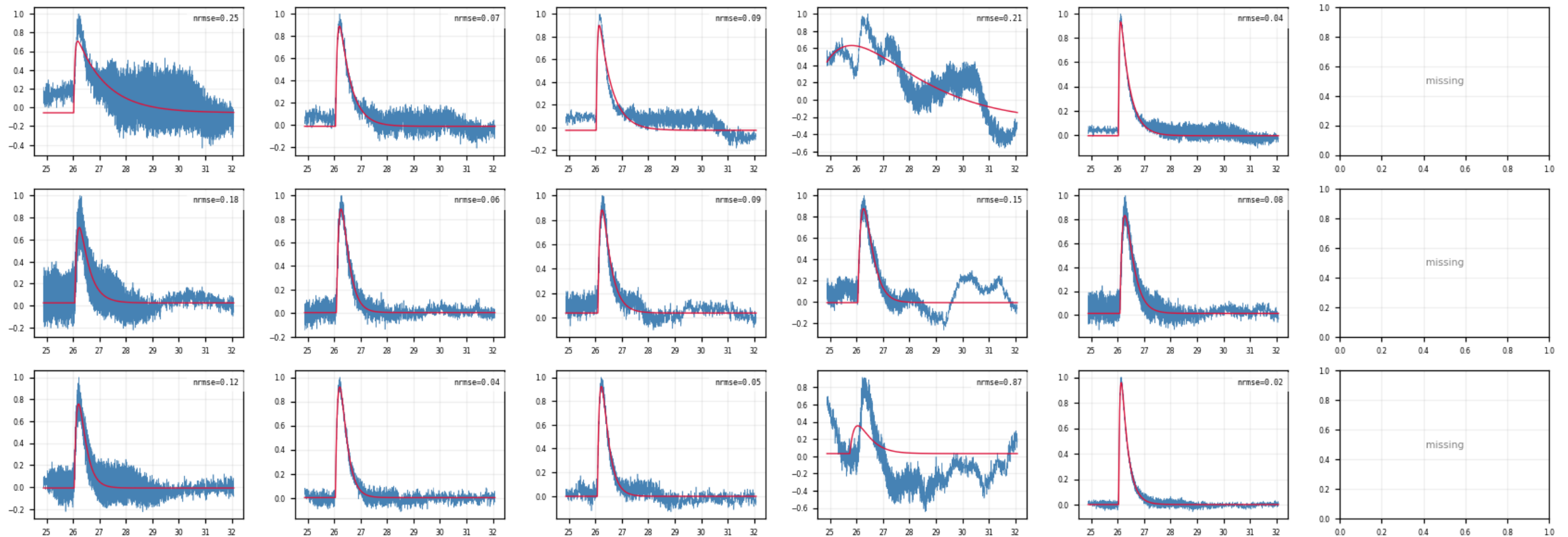
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (4/10)

figure: zip7_fit_examples.png, event rows 4–6 of 15, right half of the channels

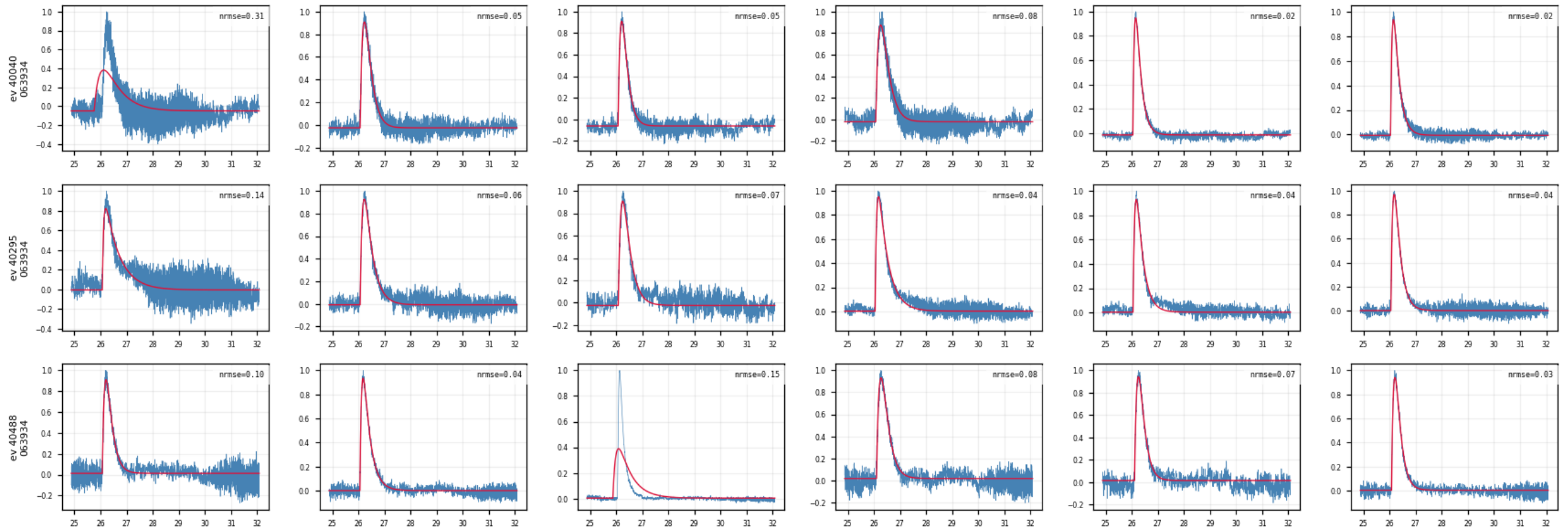
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (5/10)

figure: zip7_fit_examples.png, event rows 7–9 of 15, left half of the channels

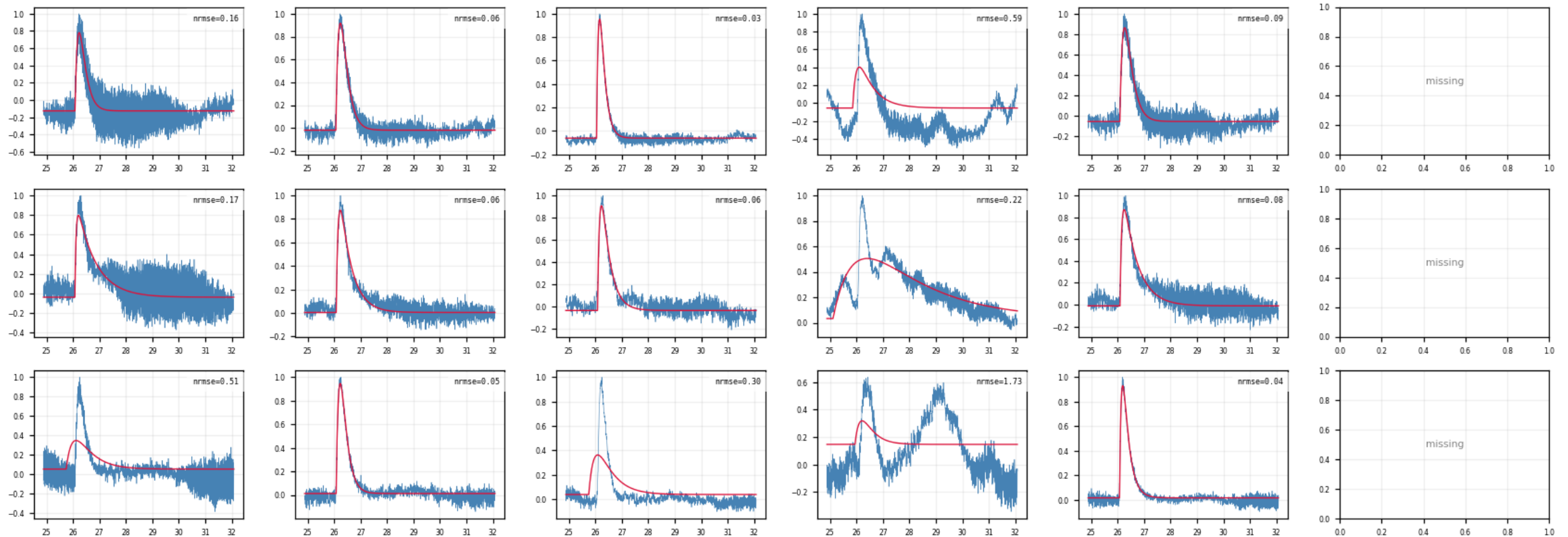
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (6/10)

figure: zip7_fit_examples.png, event rows 7–9 of 15, right half of the channels

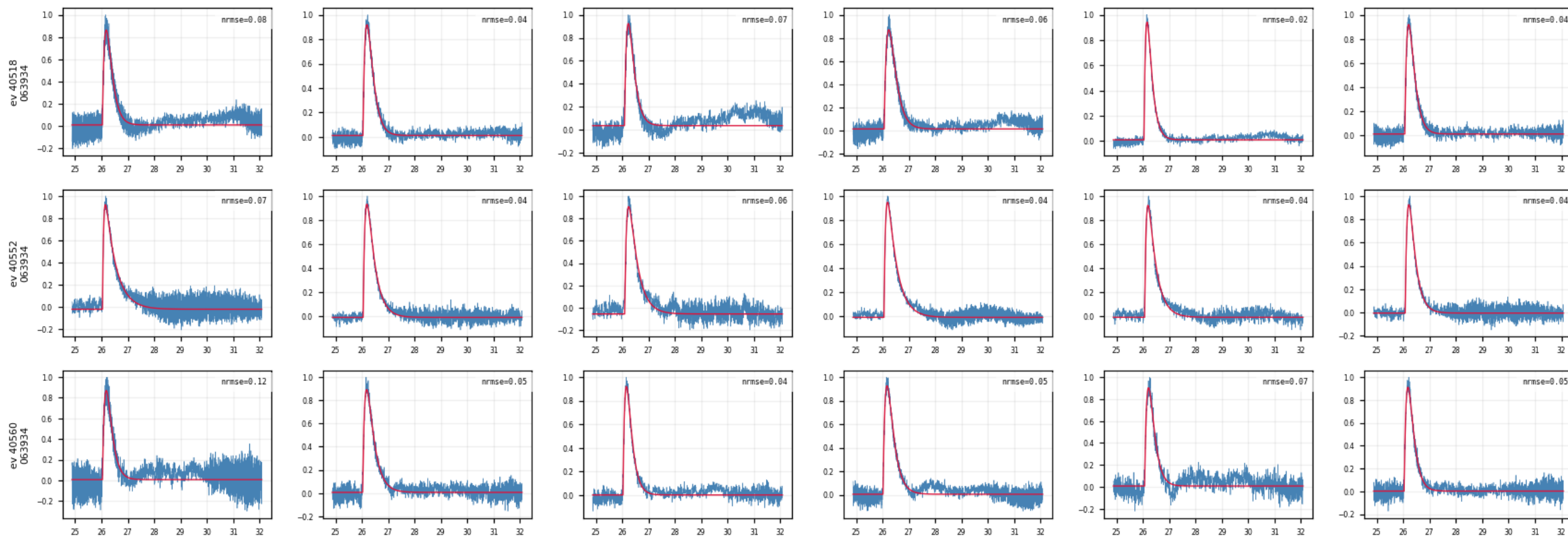
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (7/10)

figure: zip7_fit_examples.png, event rows 10–12 of 15, left half of the channels

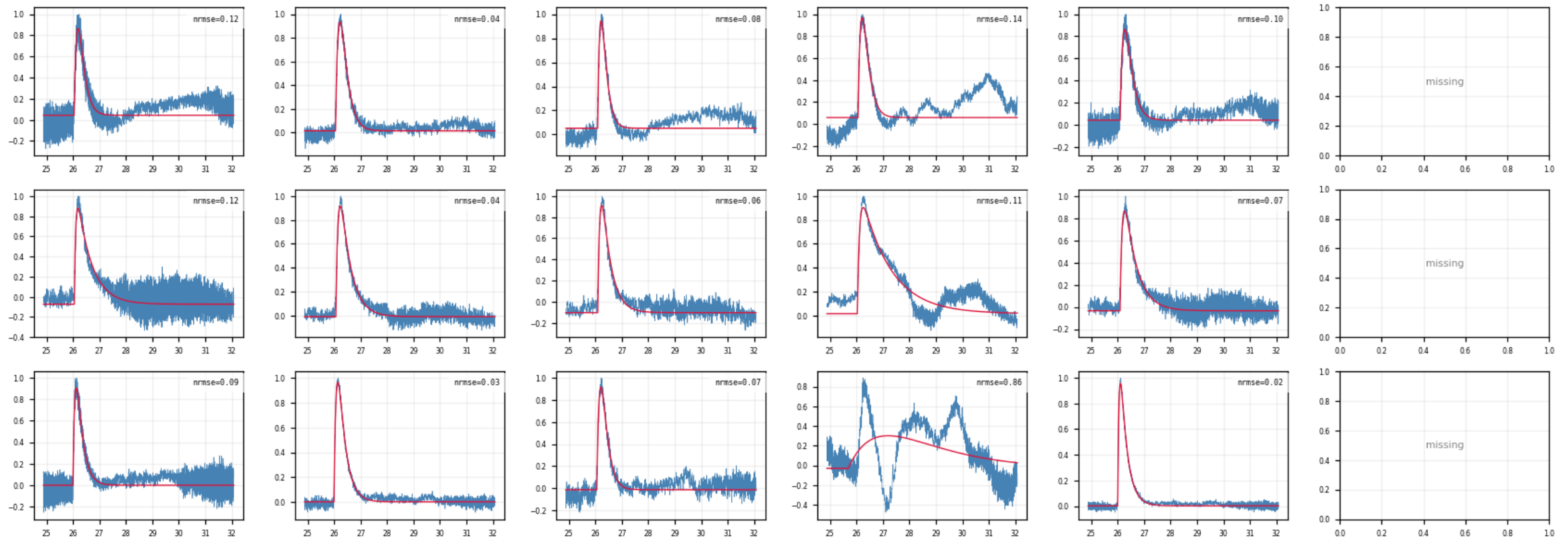
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (8/10)

figure: zip7_fit_examples.png, event rows 10–12 of 15, right half of the channels

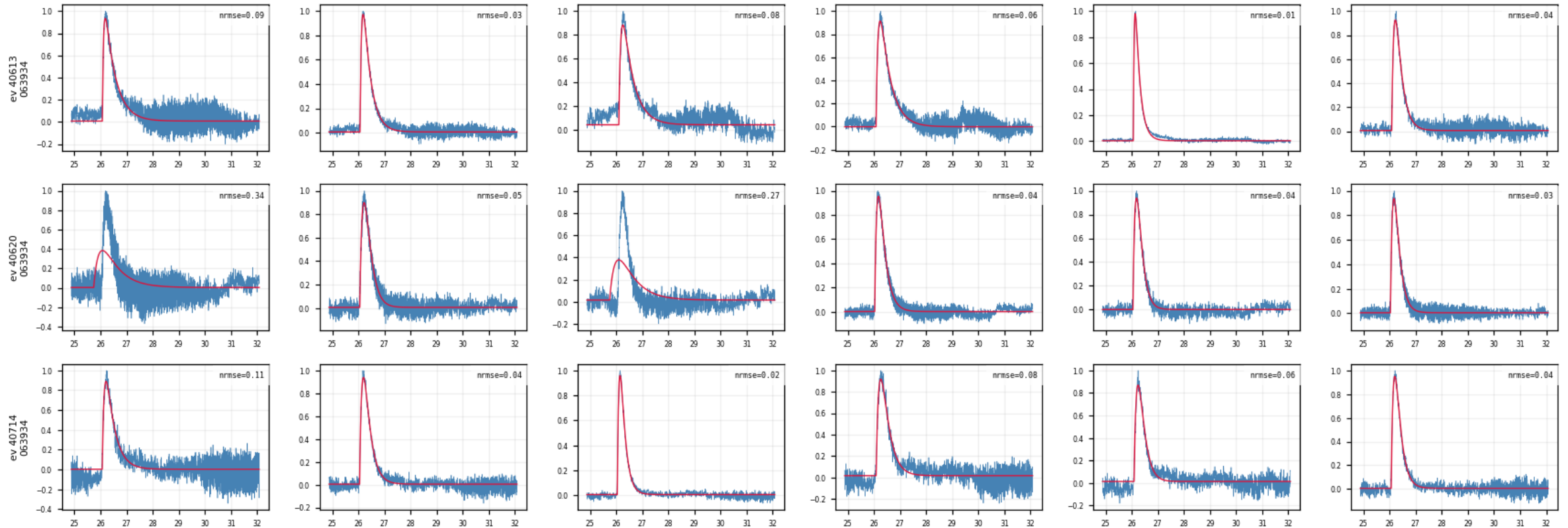
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (9/10)

figure: zip7_fit_examples.png, event rows 13–15 of 15, left half of the channels

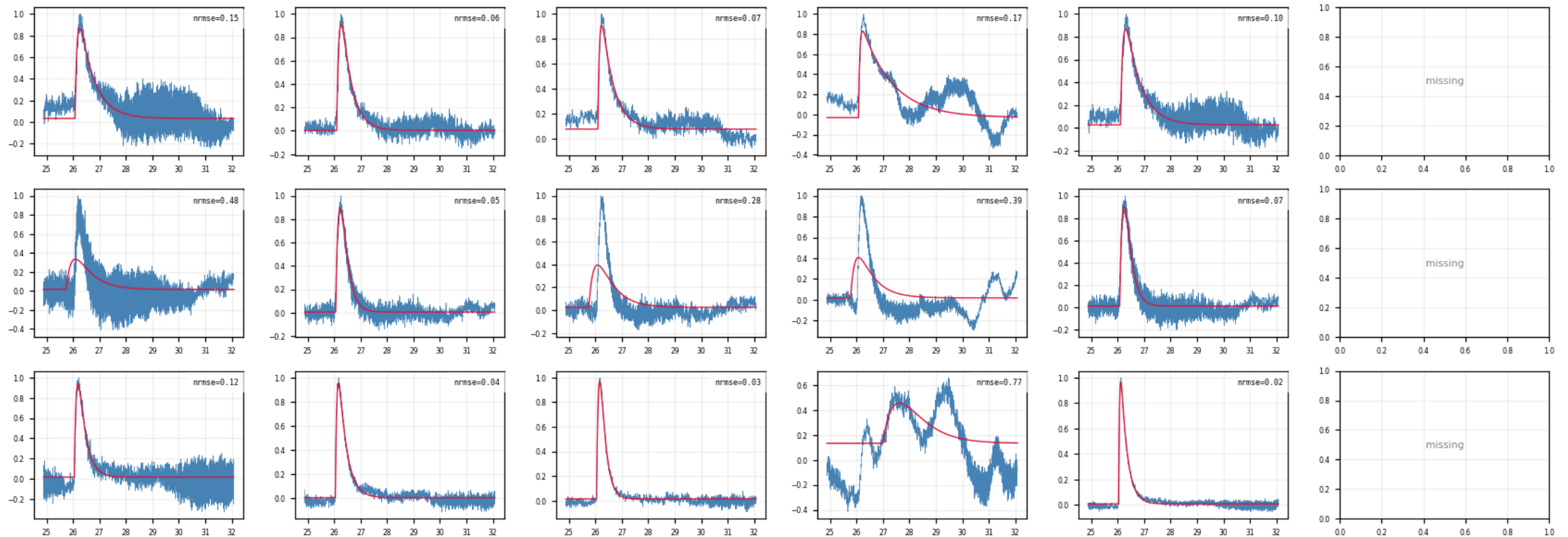
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · fit_examples: LP trace vs 2-exp fit (10/10)

figure: zip7_fit_examples.png, event rows 13–15 of 15, right half of the channels

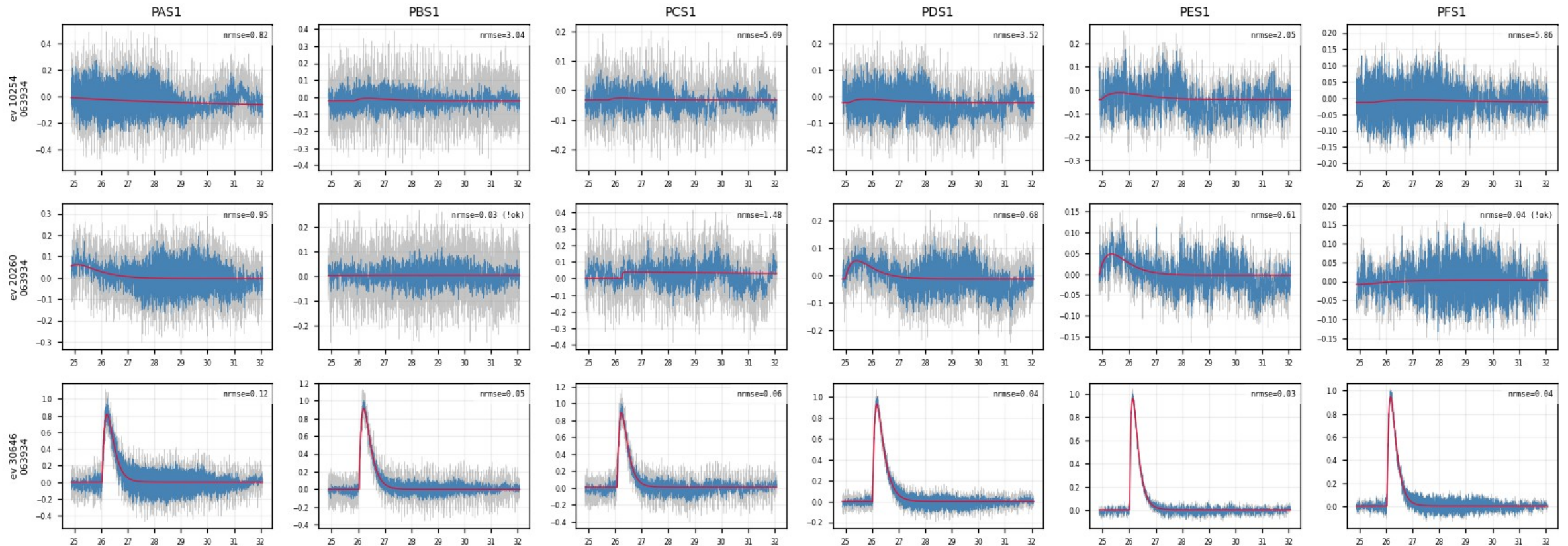
First 15 cached events × all channels; blue = 100 kHz LP trace, red = 2-exp fit. A real event fits well in every channel at once; a noise trigger fails across the whole row.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (1/10)

figure: zip7_raw_vs_fit_examples.png, event rows 1–3 of 15, left half of the channels

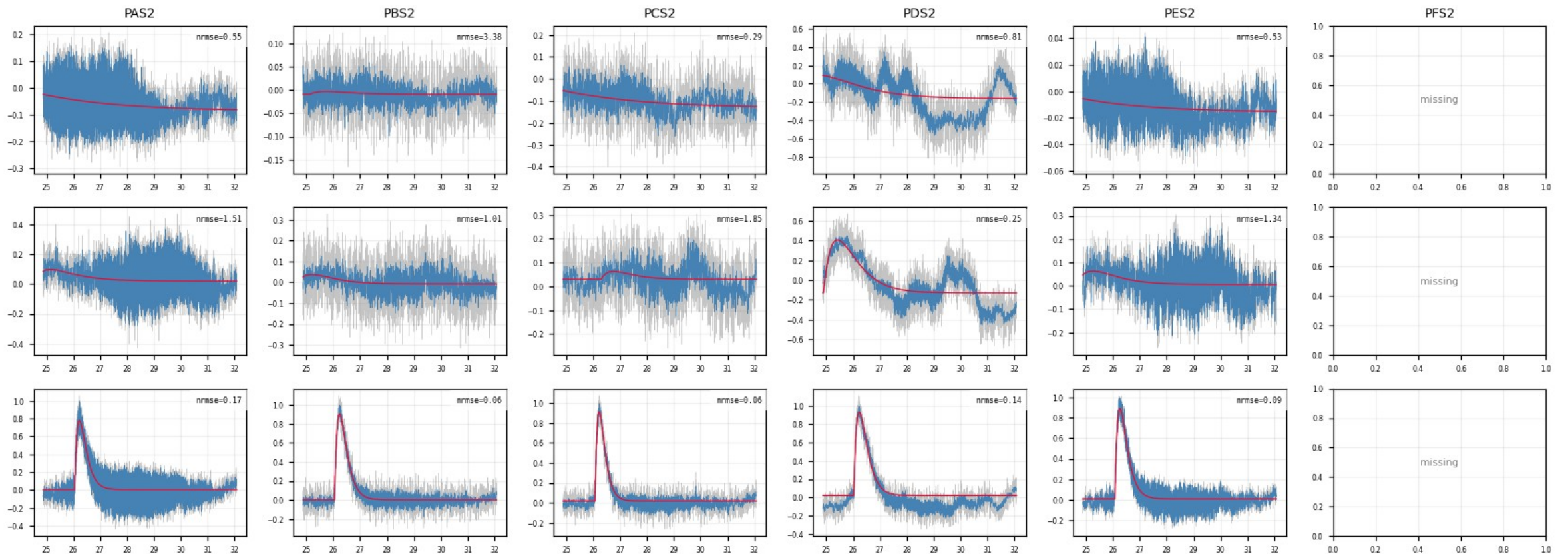
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (2/10)

figure: zip7_raw_vs_fit_examples.png, event rows 1–3 of 15, right half of the channels

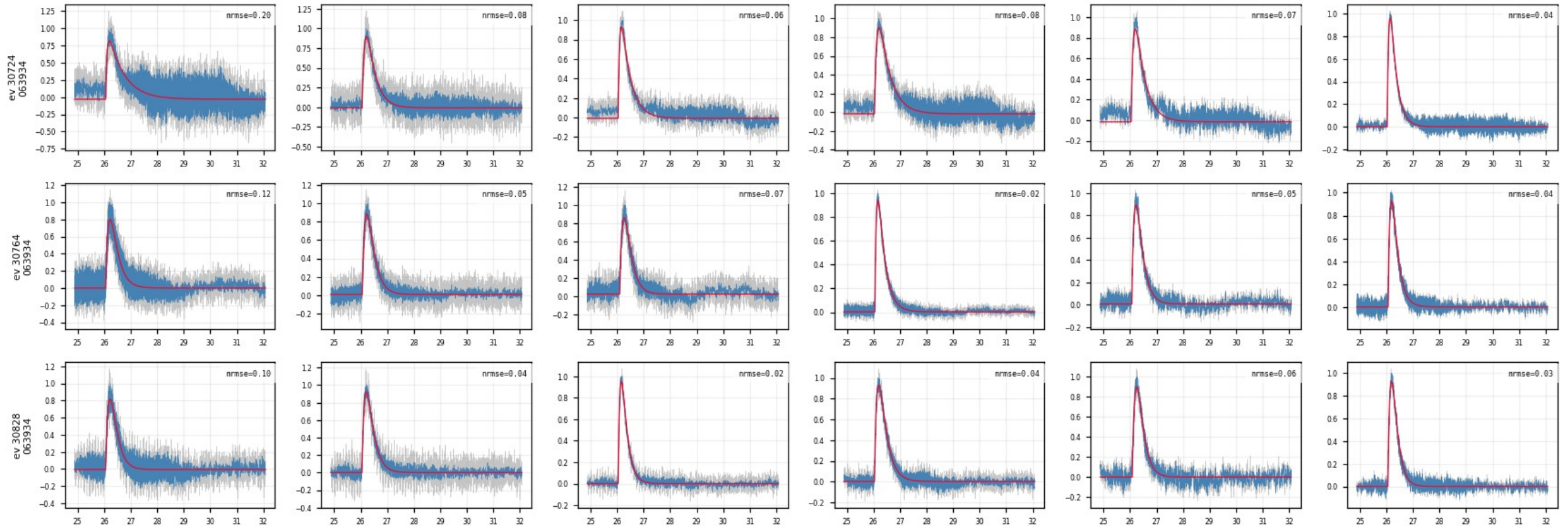
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (3/10)

figure: zip7_raw_vs_fit_examples.png, event rows 4–6 of 15, left half of the channels

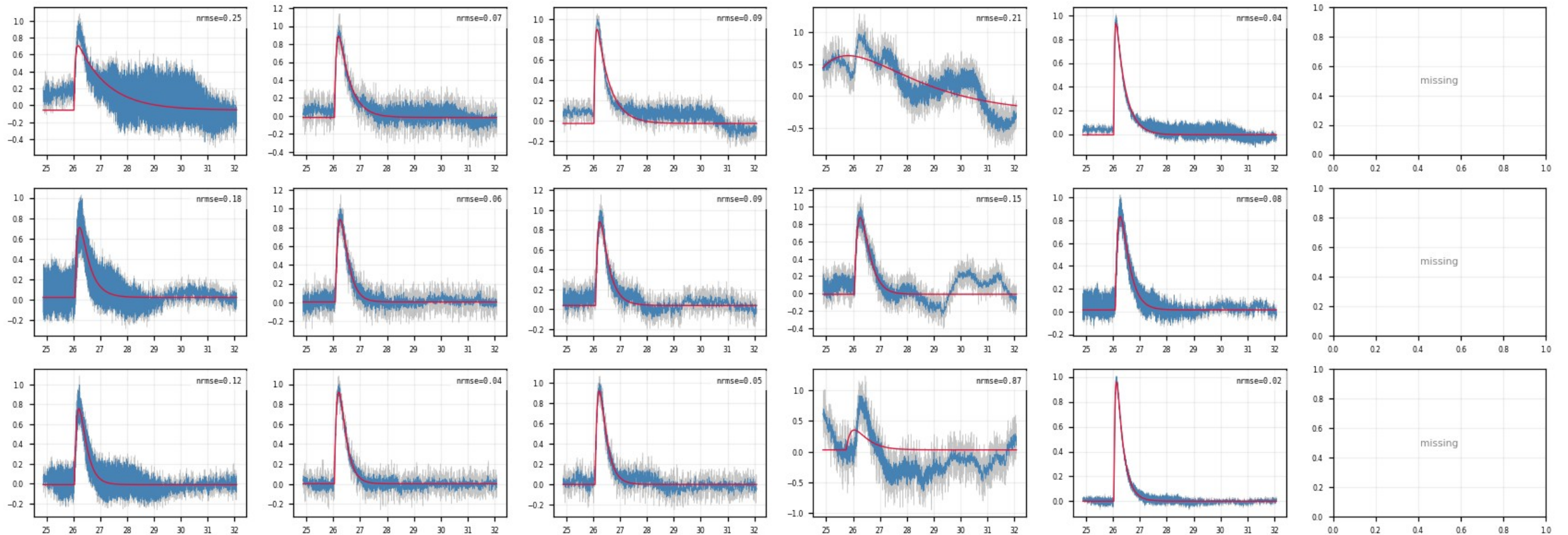
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (4/10)

figure: zip7_raw_vs_fit_examples.png, event rows 4–6 of 15, right half of the channels

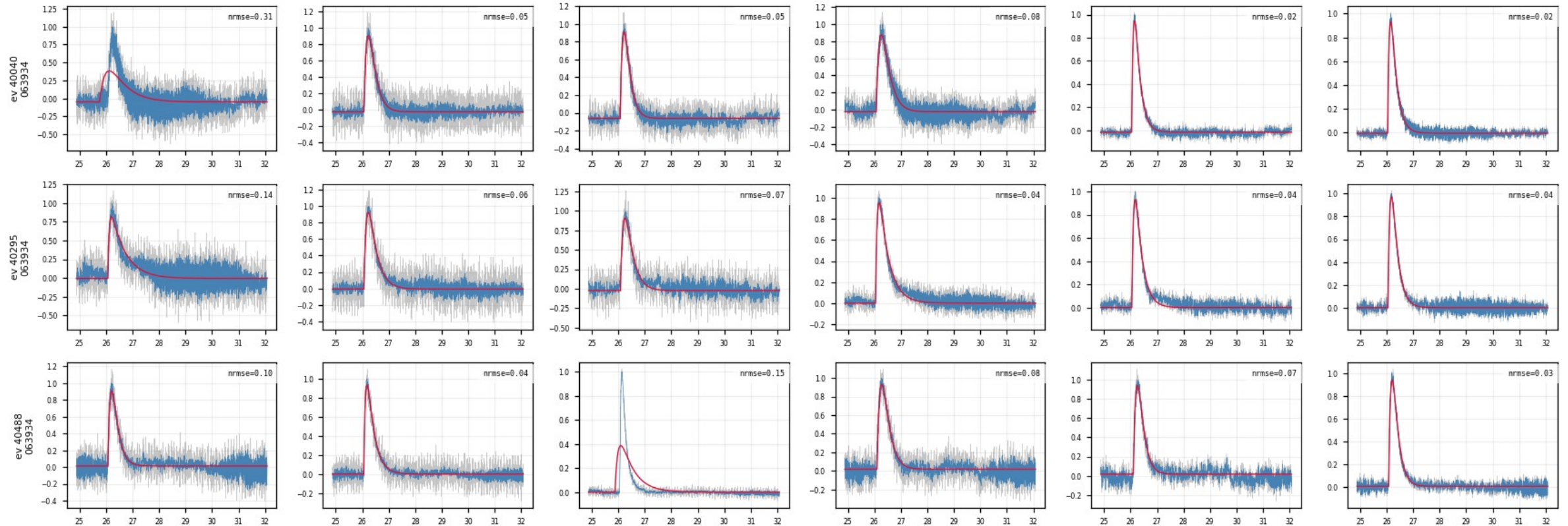
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (5/10)

figure: zip7_raw_vs_fit_examples.png, event rows 7–9 of 15, left half of the channels

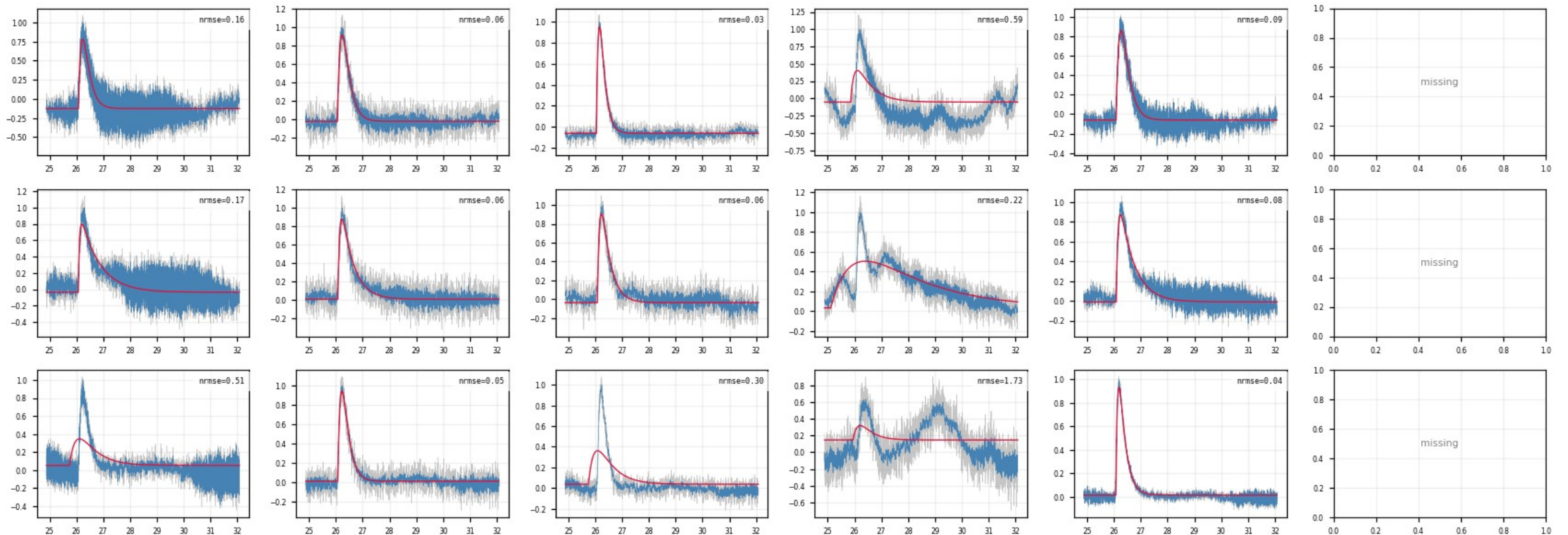
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (6/10)

figure: zip7_raw_vs_fit_examples.png, event rows 7–9 of 15, right half of the channels

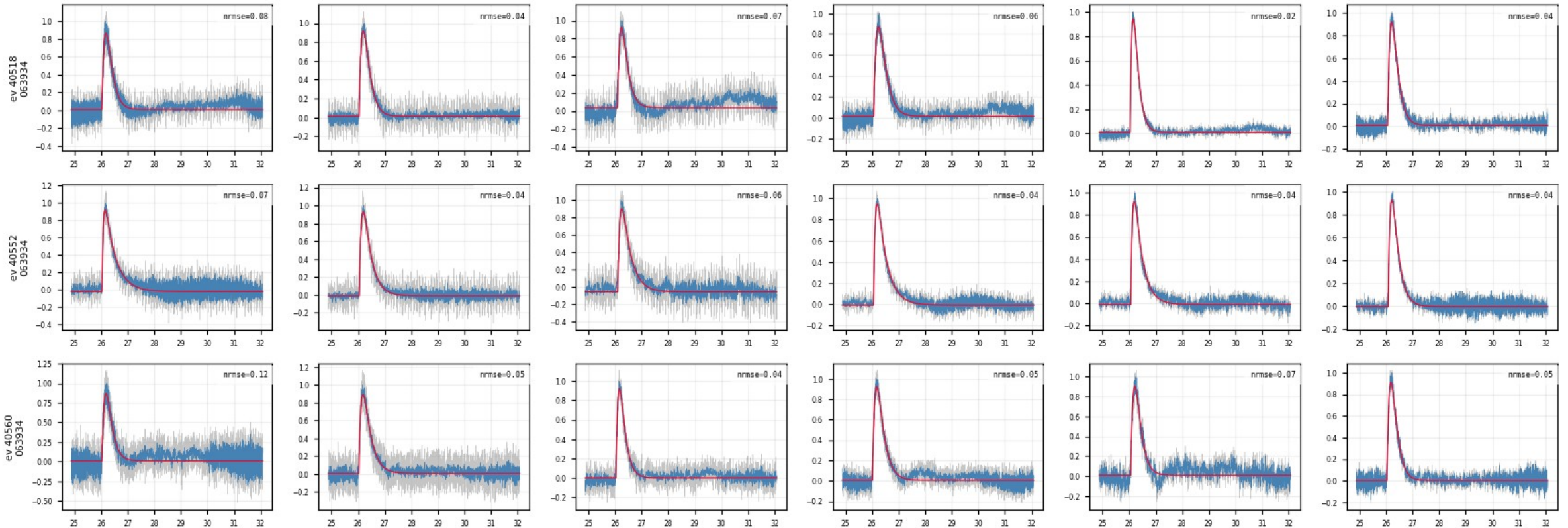
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (7/10)

figure: zip7_raw_vs_fit_examples.png, event rows 10–12 of 15, left half of the channels

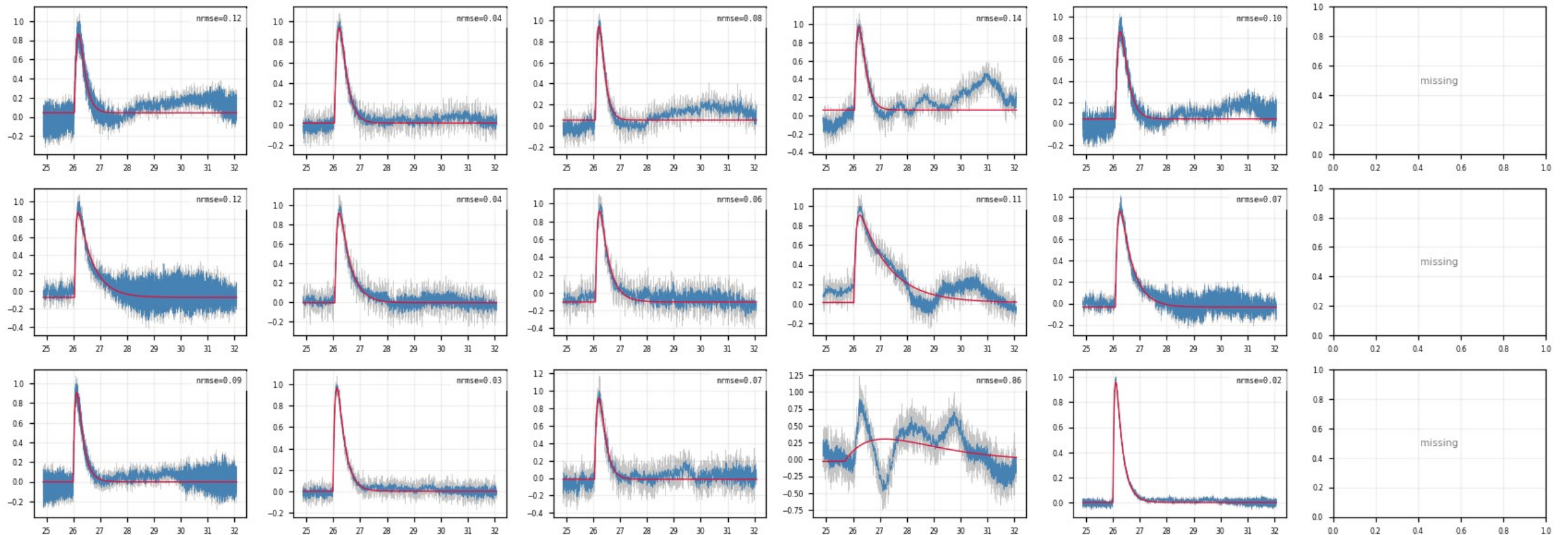
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (8/10)

figure: zip7_raw_vs_fit_examples.png, event rows 10–12 of 15, right half of the channels

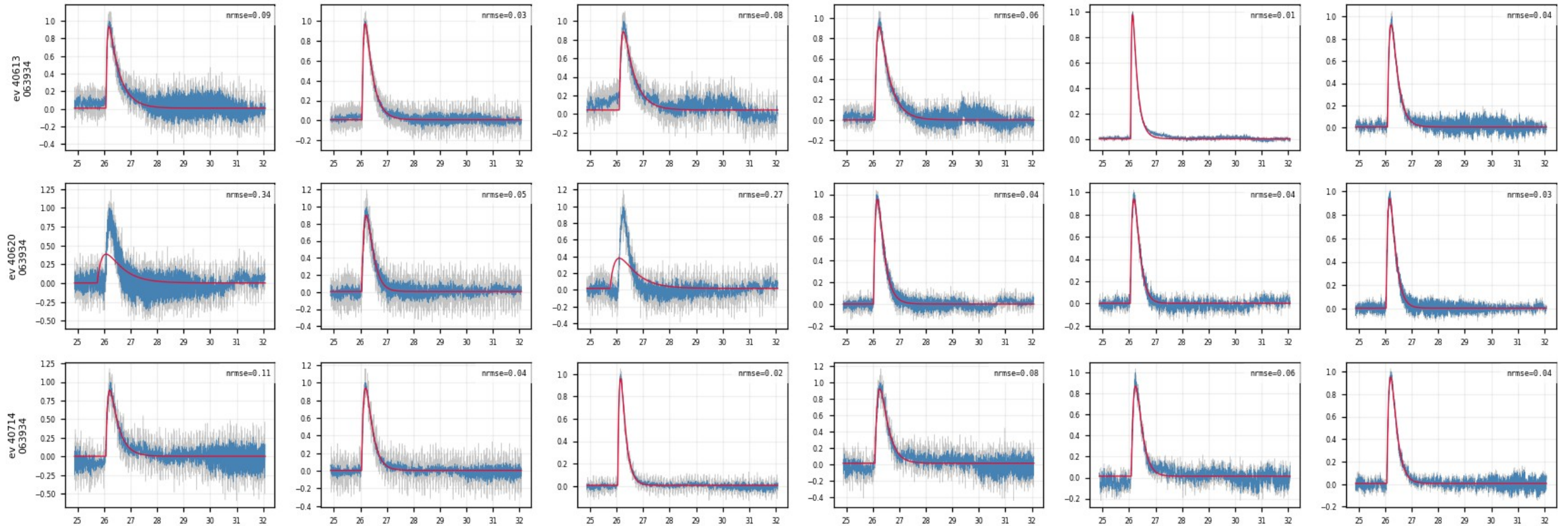
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (9/10)

figure: zip7_raw_vs_fit_examples.png, event rows 13–15 of 15, left half of the channels

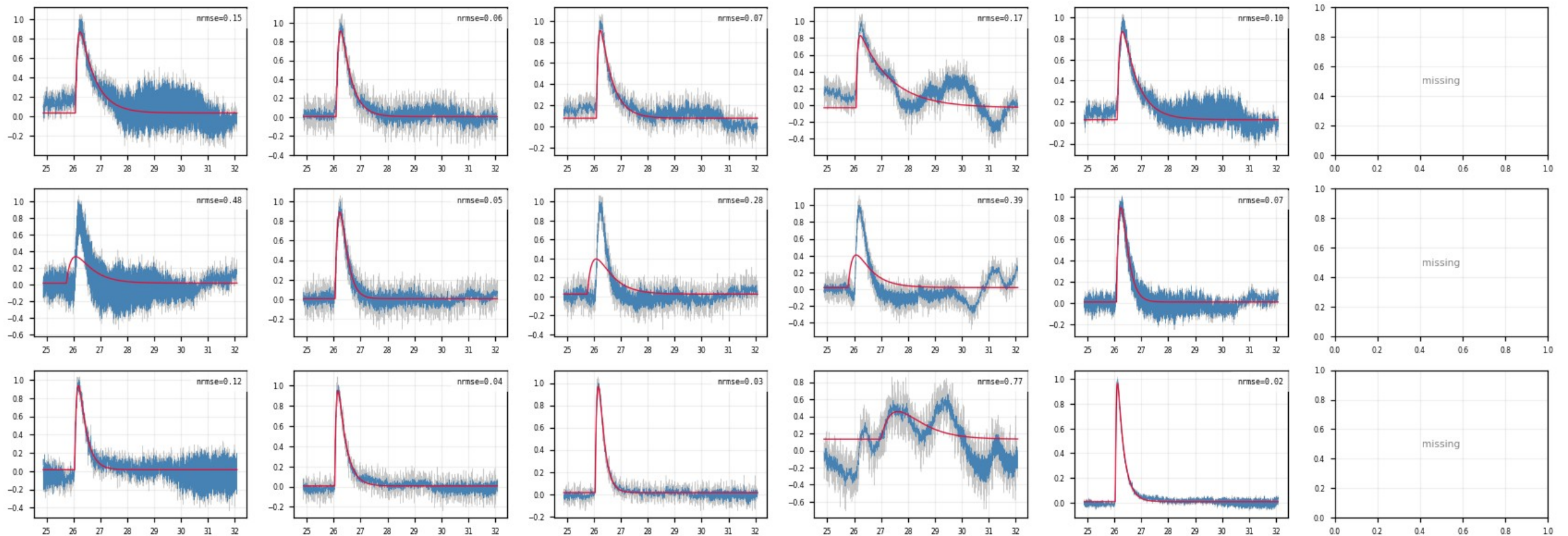
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · raw_vs_fit_examples: raw trace vs fit (10/10)

figure: zip7_raw_vs_fit_examples.png, event rows 13–15 of 15, right half of the channels

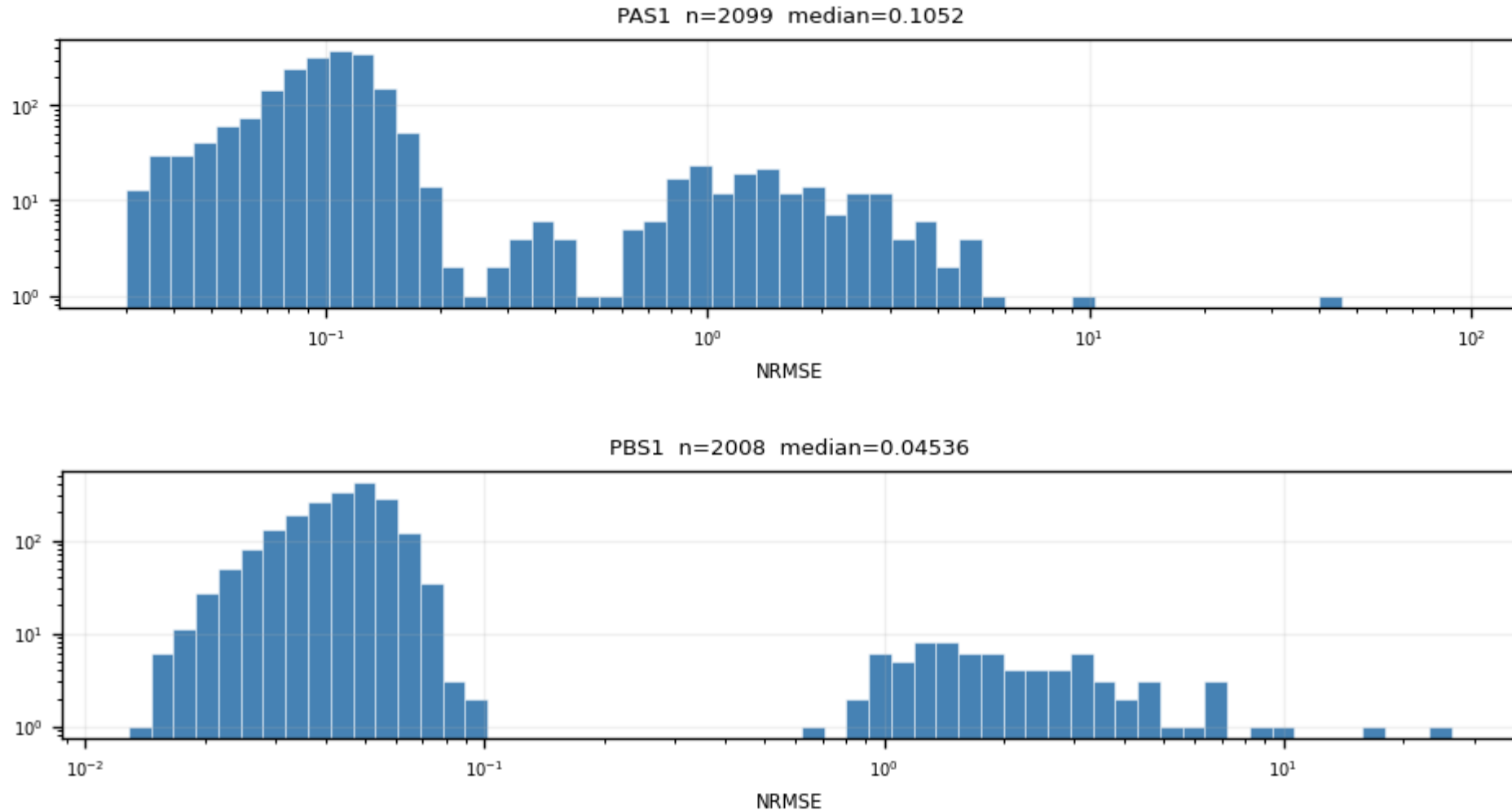
Same 15 events × all channels; gray = raw unfiltered trace, blue = LP, red = fit, per-panel NRMSE top-right. Noise vs pulse size is directly visible in the raw data.



Backup · Z7 · nrmse: NRMSE distribution per channel (1/6)

figure: zip7_nrmse.png, channel panels 1–2 of 11, top to bottom

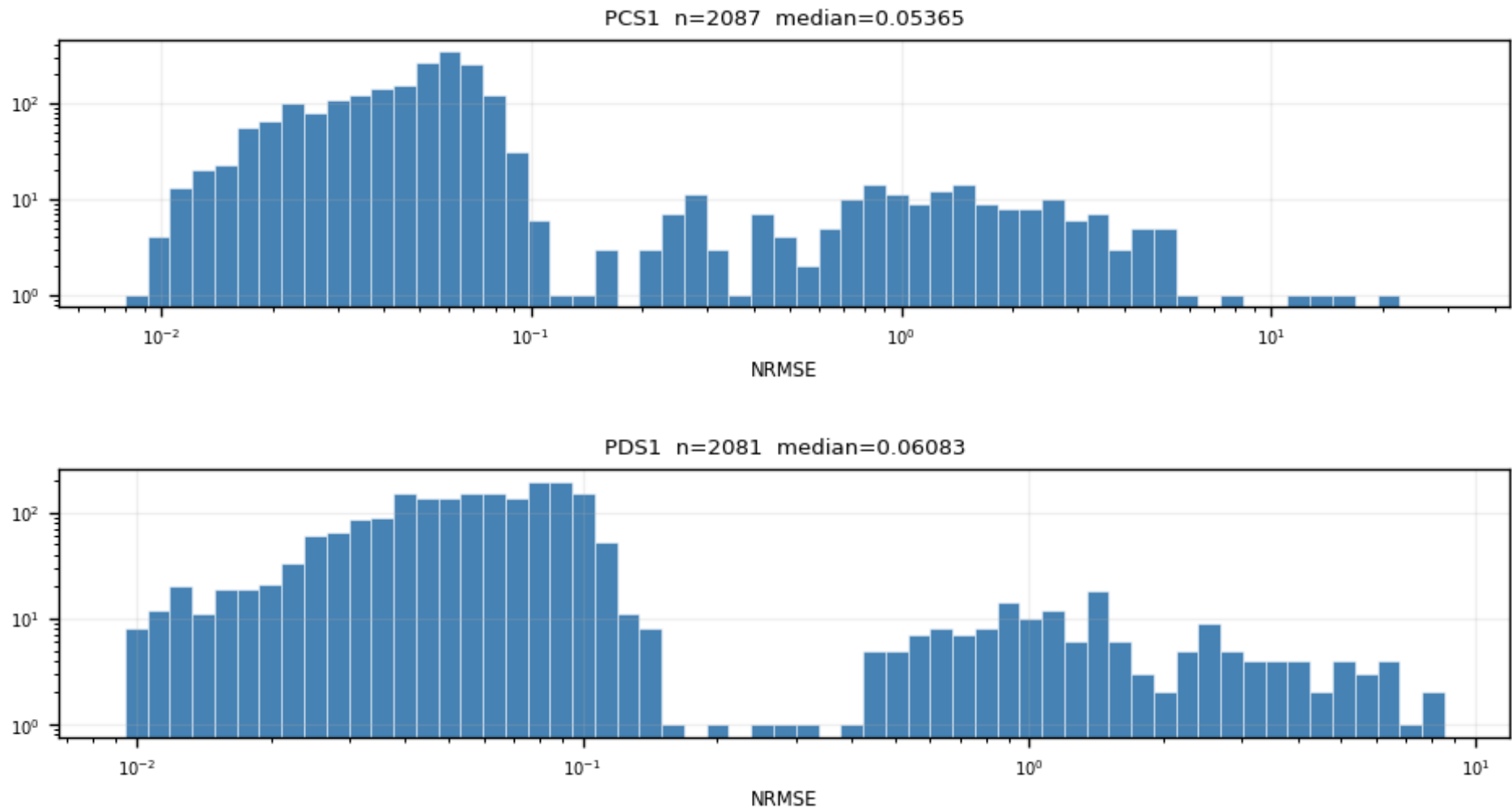
NRMSE = $\text{RMS}(\text{fit residual}) / \text{fitted peak}$, log-log. Bimodal: good fits ($\sim 0.05\text{--}0.1$) vs noise triggers ($\sim 1\text{--}2$); the valley at ≈ 0.4 sets quality cut 2.



Backup · Z7 · nrmse: NRMSE distribution per channel (2/6)

figure: zip7_nrmse.png, channel panels 3–4 of 11, top to bottom

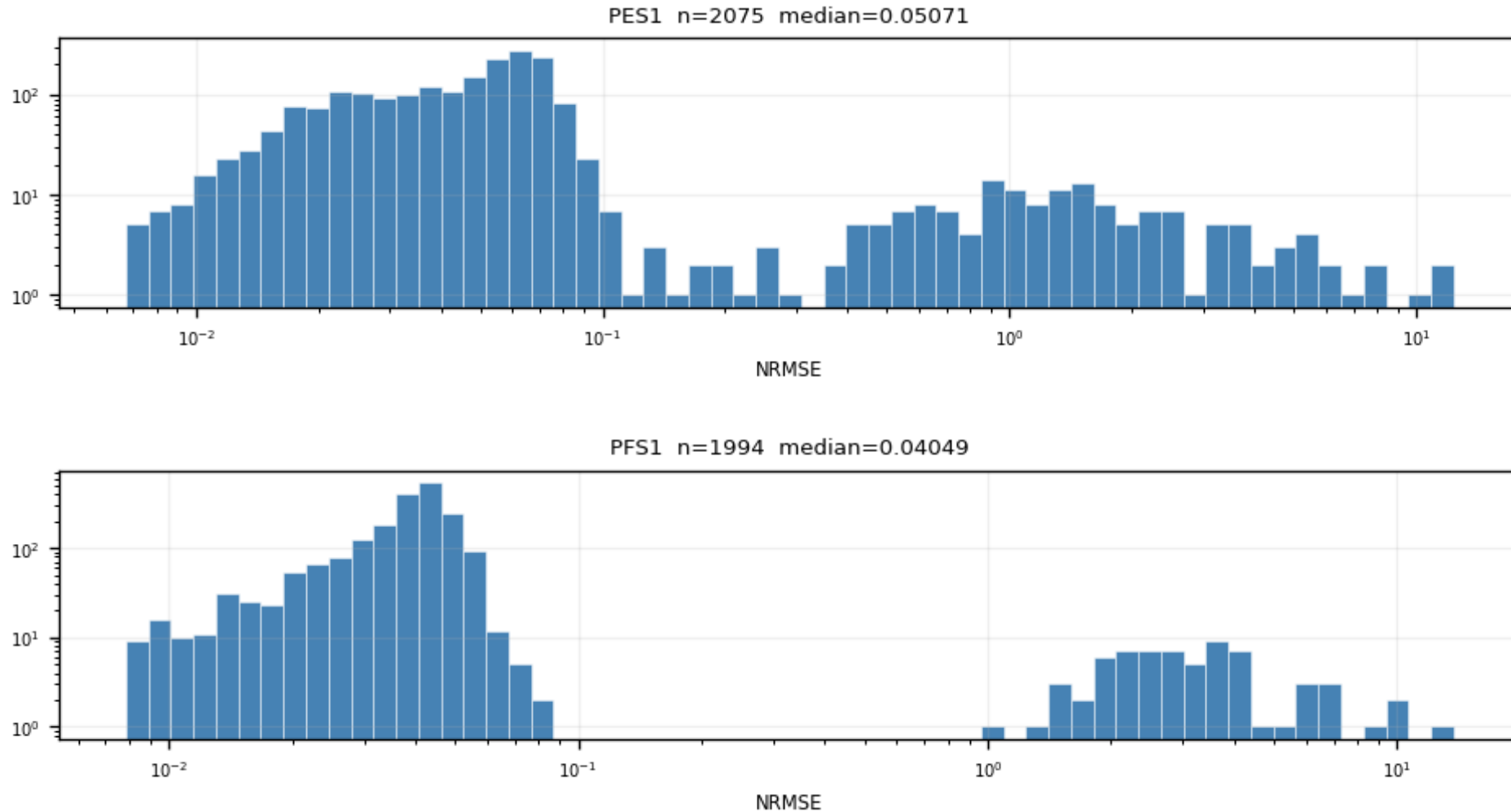
NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits (~0.05–0.1) vs noise triggers (~1–2); the valley at ≈0.4 sets quality cut 2.



Backup · Z7 · nrmse: NRMSE distribution per channel (3/6)

figure: zip7_nrmse.png, channel panels 5–6 of 11, top to bottom

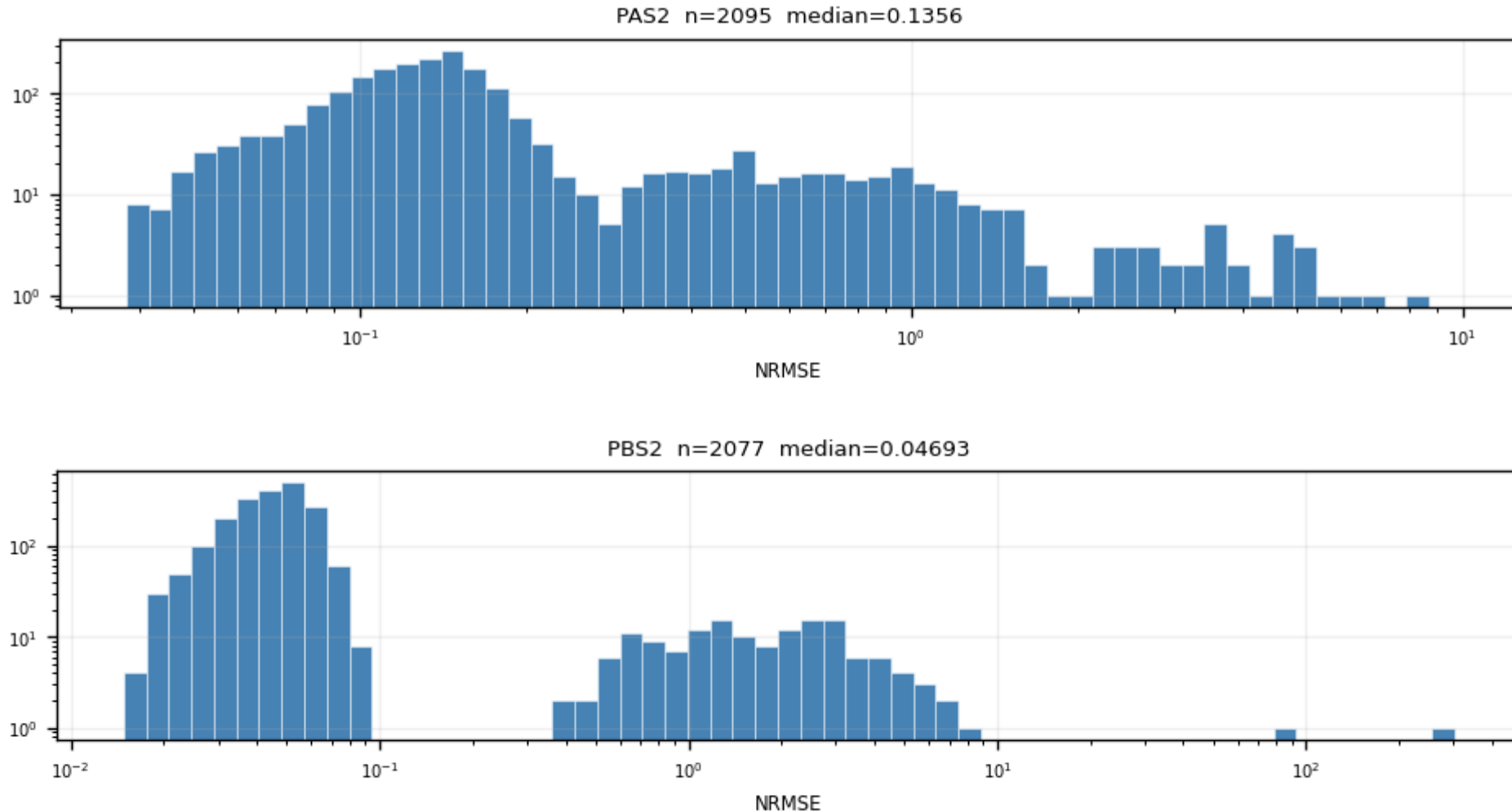
NRMSE = $\text{RMS}(\text{fit residual}) / \text{fitted peak}$, log-log. Bimodal: good fits ($\sim 0.05\text{--}0.1$) vs noise triggers ($\sim 1\text{--}2$); the valley at ≈ 0.4 sets quality cut 2.



Backup · Z7 · nrmse: NRMSE distribution per channel (4/6)

figure: zip7_nrmse.png, channel panels 7–8 of 11, top to bottom

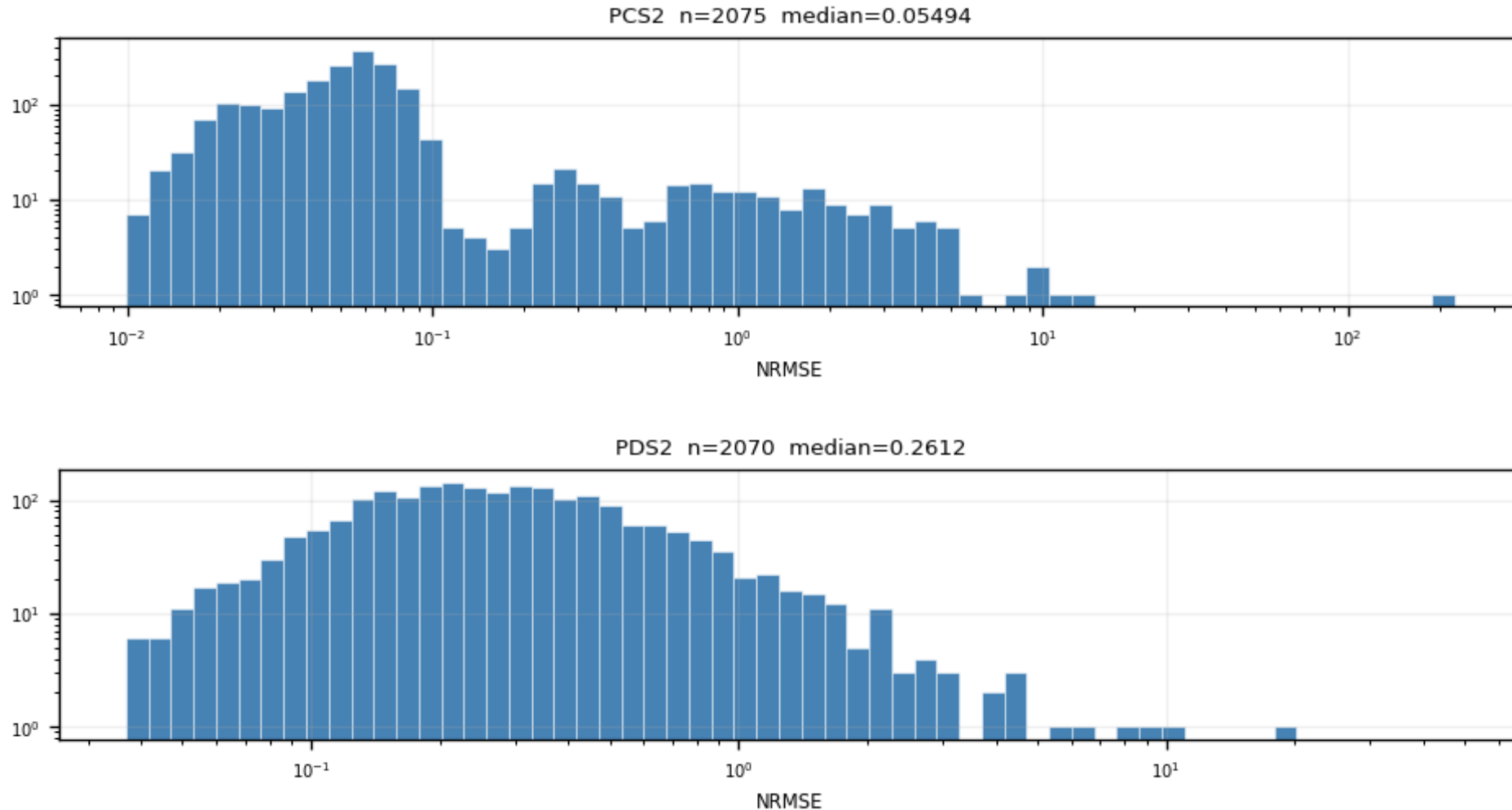
NRMSE = $\text{RMS}(\text{fit residual}) / \text{fitted peak}$, log-log. Bimodal: good fits ($\sim 0.05\text{--}0.1$) vs noise triggers ($\sim 1\text{--}2$); the valley at ≈ 0.4 sets quality cut 2.



Backup · Z7 · nrmse: NRMSE distribution per channel (5/6)

figure: zip7_nrmse.png, channel panels 9–10 of 11, top to bottom

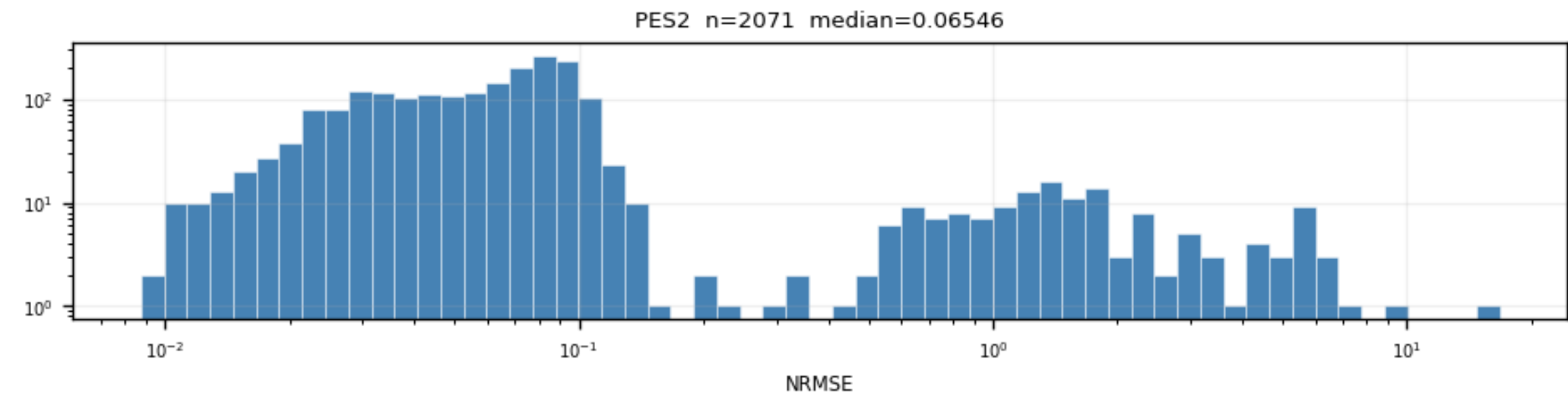
NRMSE = $\text{RMS}(\text{fit residual}) / \text{fitted peak}$, log-log. Bimodal: good fits ($\sim 0.05\text{--}0.1$) vs noise triggers ($\sim 1\text{--}2$); the valley at ≈ 0.4 sets quality cut 2.



Backup · Z7 · nrmse: NRMSE distribution per channel (6/6)

figure: zip7_nrmse.png, channel panels 11–11 of 11, top to bottom

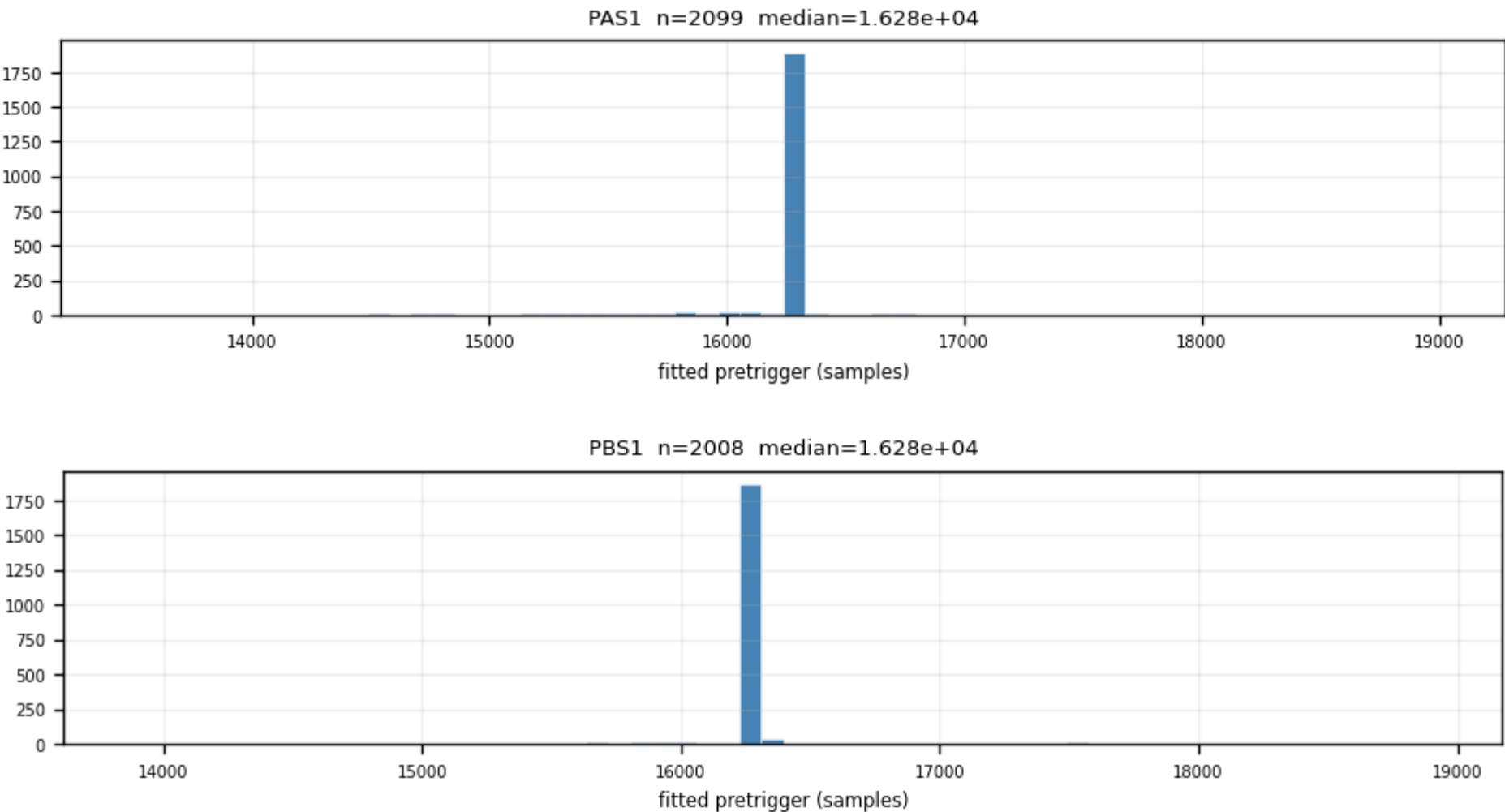
NRMSE = RMS(fit residual)/fitted peak, log-log. Bimodal: good fits ($\sim 0.05\text{--}0.1$) vs noise triggers ($\sim 1\text{--}2$); the valley at ≈ 0.4 sets quality cut 2.



Backup · Z7 · pretrigger: fitted onset per channel (1/6)

figure: zip7_pretrigger.png, channel panels 1–2 of 11, top to bottom

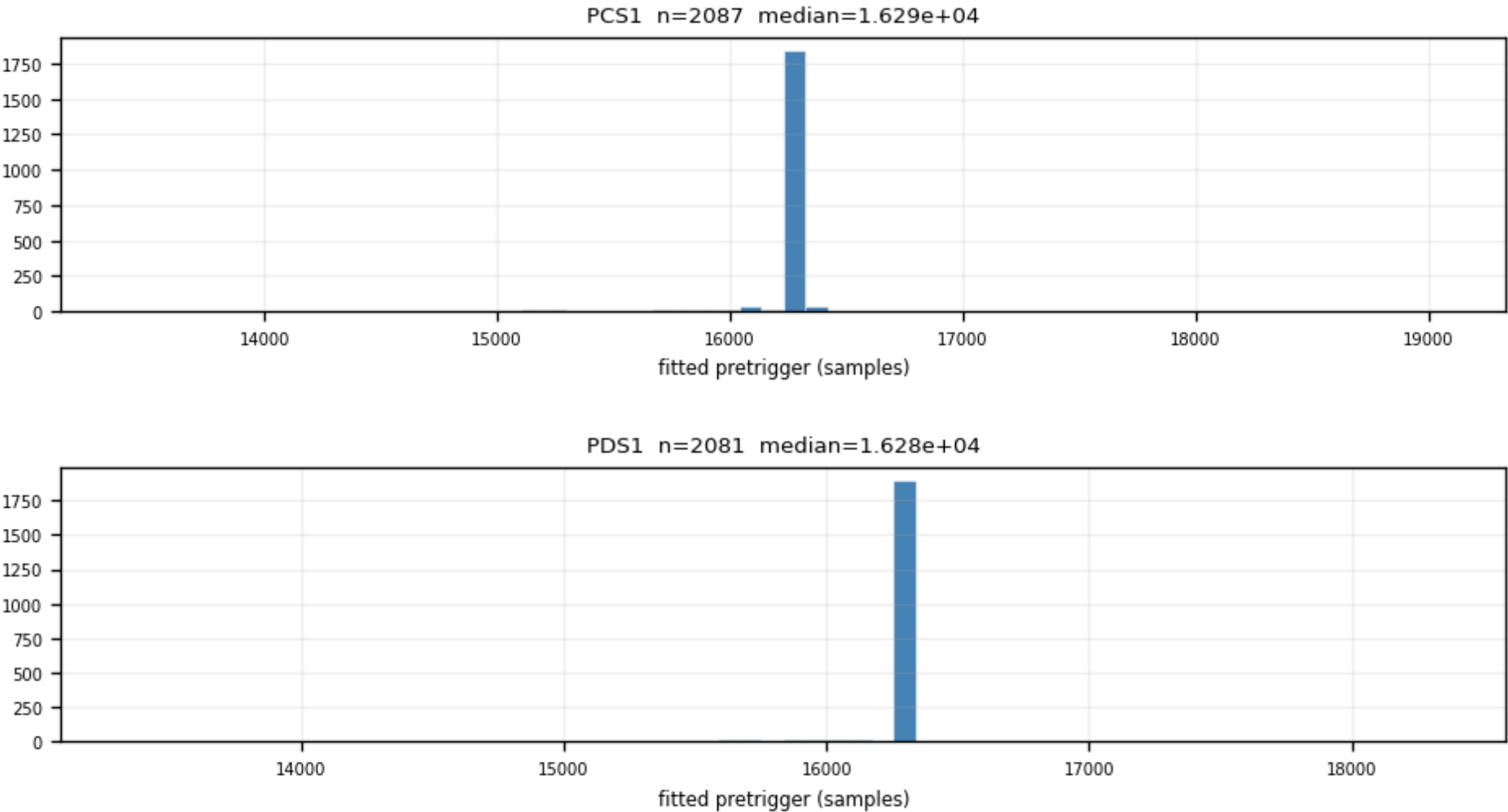
The onset is a FREE fit parameter; fitted values cluster ≈ 230 samples after the nominal 16050; pinning it inside the fit would bias every other parameter.



Backup · Z7 · pretrigger: fitted onset per channel (2/6)

figure: zip7_pretrigger.png, channel panels 3–4 of 11, top to bottom

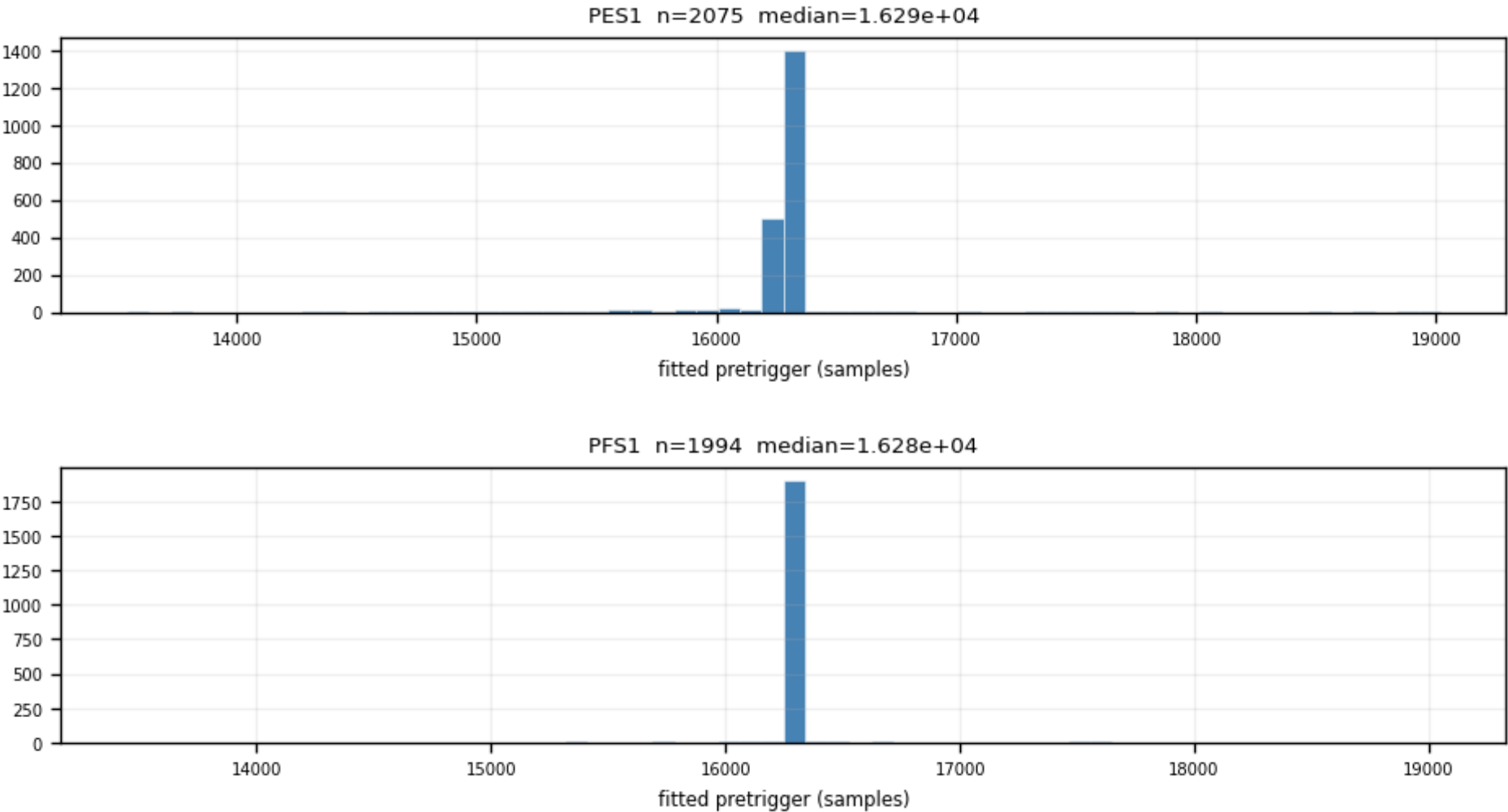
The onset is a FREE fit parameter; fitted values cluster ≈ 230 samples after the nominal 16050; pinning it inside the fit would bias every other parameter.



Backup · Z7 · pretrigger: fitted onset per channel (3/6)

figure: zip7_pretrigger.png, channel panels 5–6 of 11, top to bottom

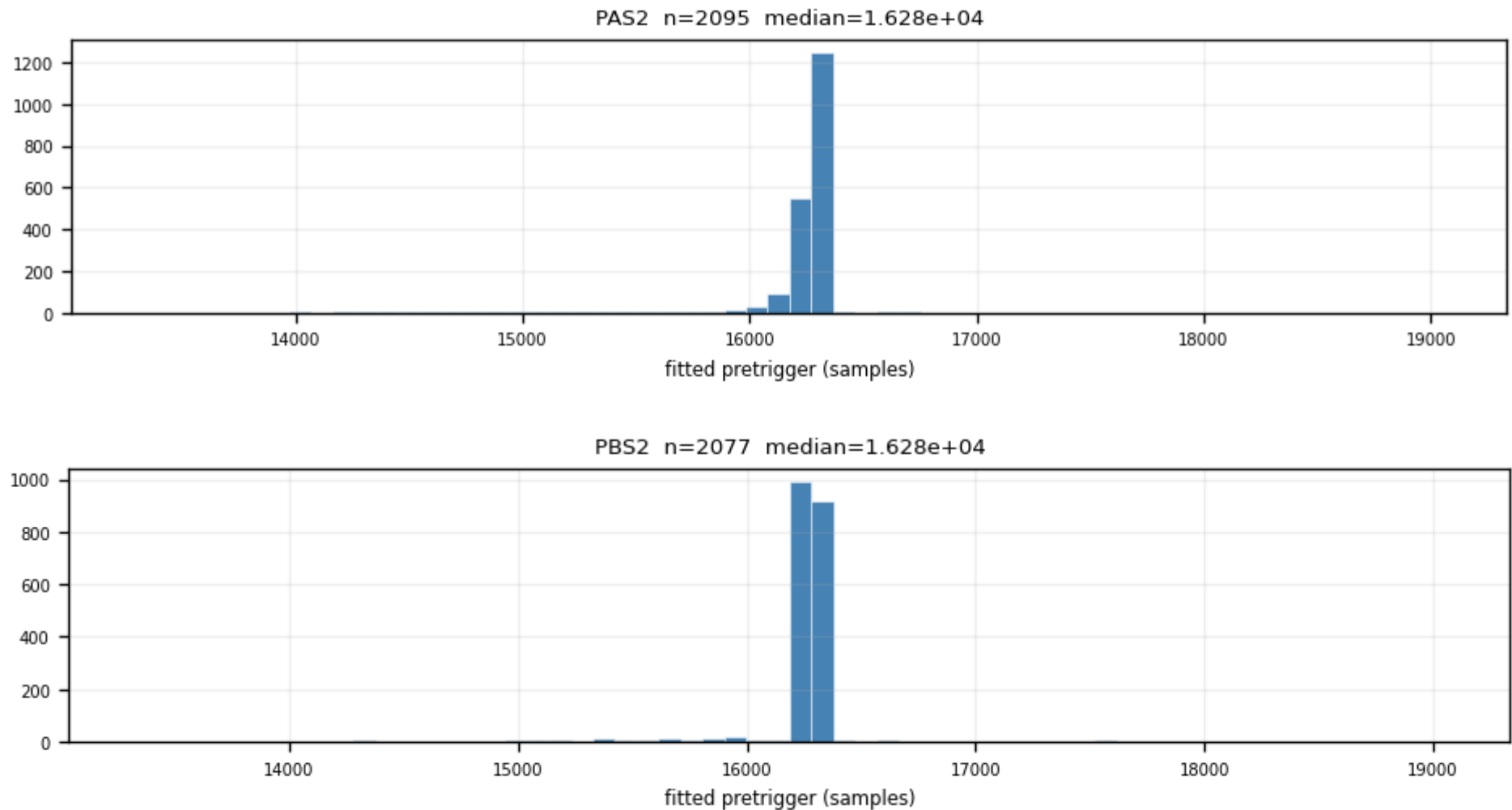
The onset is a FREE fit parameter; fitted values cluster ≈ 230 samples after the nominal 16050; pinning it inside the fit would bias every other parameter.



Backup · Z7 · pretrigger: fitted onset per channel (4/6)

figure: zip7_pretrigger.png, channel panels 7–8 of 11, top to bottom

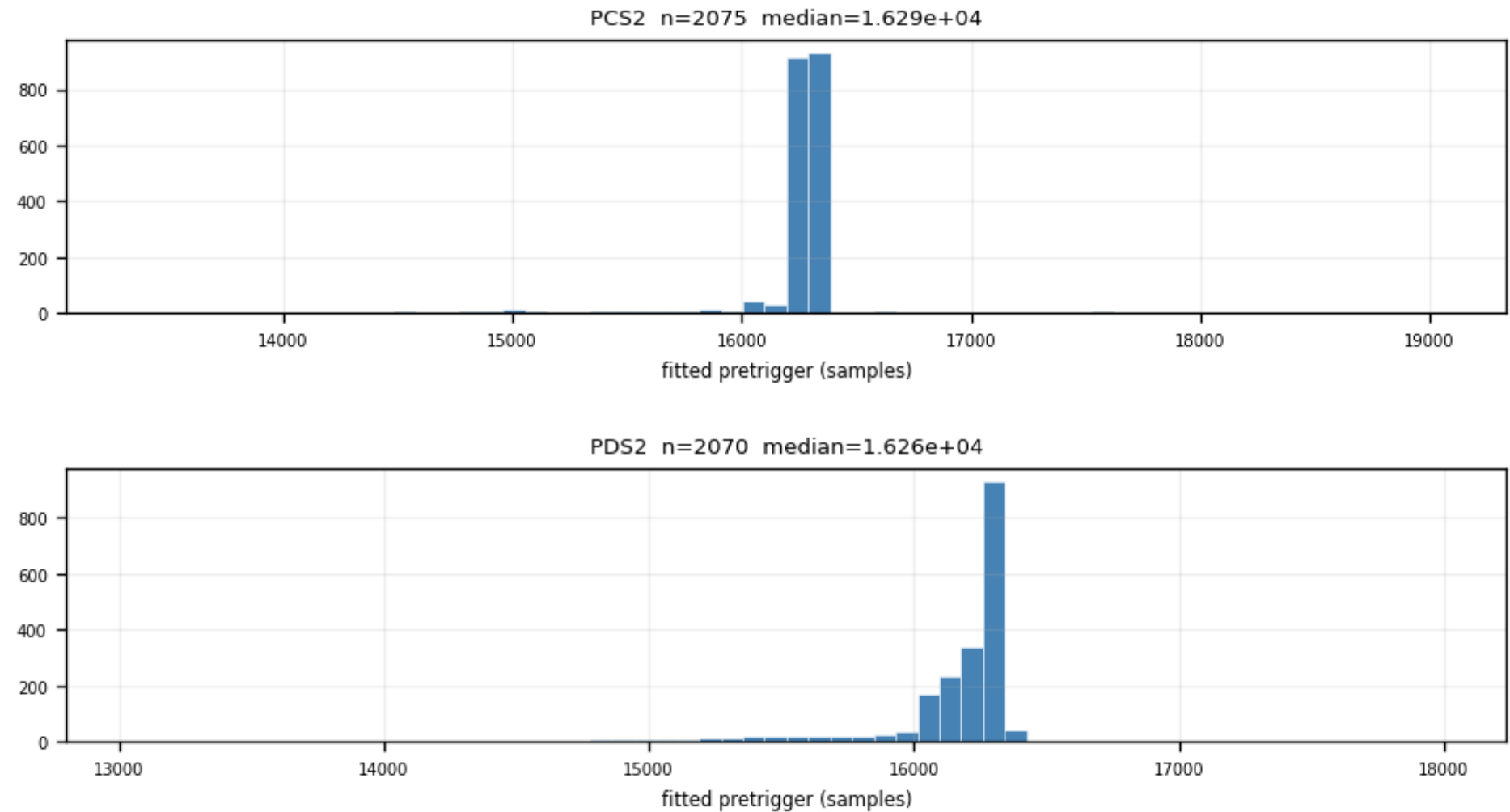
The onset is a FREE fit parameter; fitted values cluster ≈ 230 samples after the nominal 16050; pinning it inside the fit would bias every other parameter.



Backup · Z7 · pretrigger: fitted onset per channel (5/6)

figure: zip7_pretrigger.png, channel panels 9–10 of 11, top to bottom

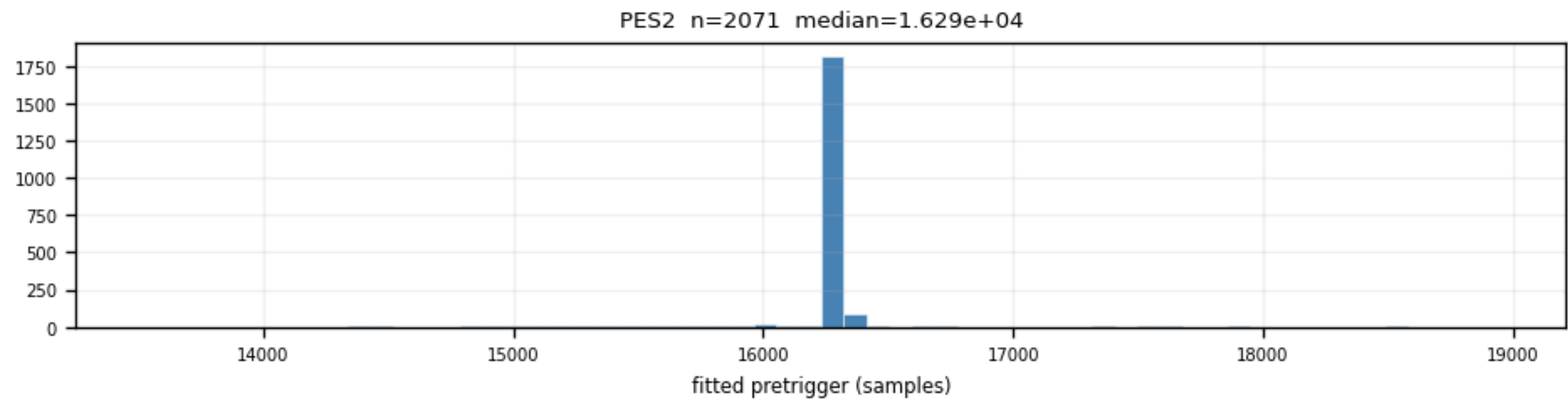
The onset is a FREE fit parameter; fitted values cluster ≈ 230 samples after the nominal 16050; pinning it inside the fit would bias every other parameter.



Backup · Z7 · pretrigger: fitted onset per channel (6/6)

figure: zip7_pretrigger.png, channel panels 11–11 of 11, top to bottom

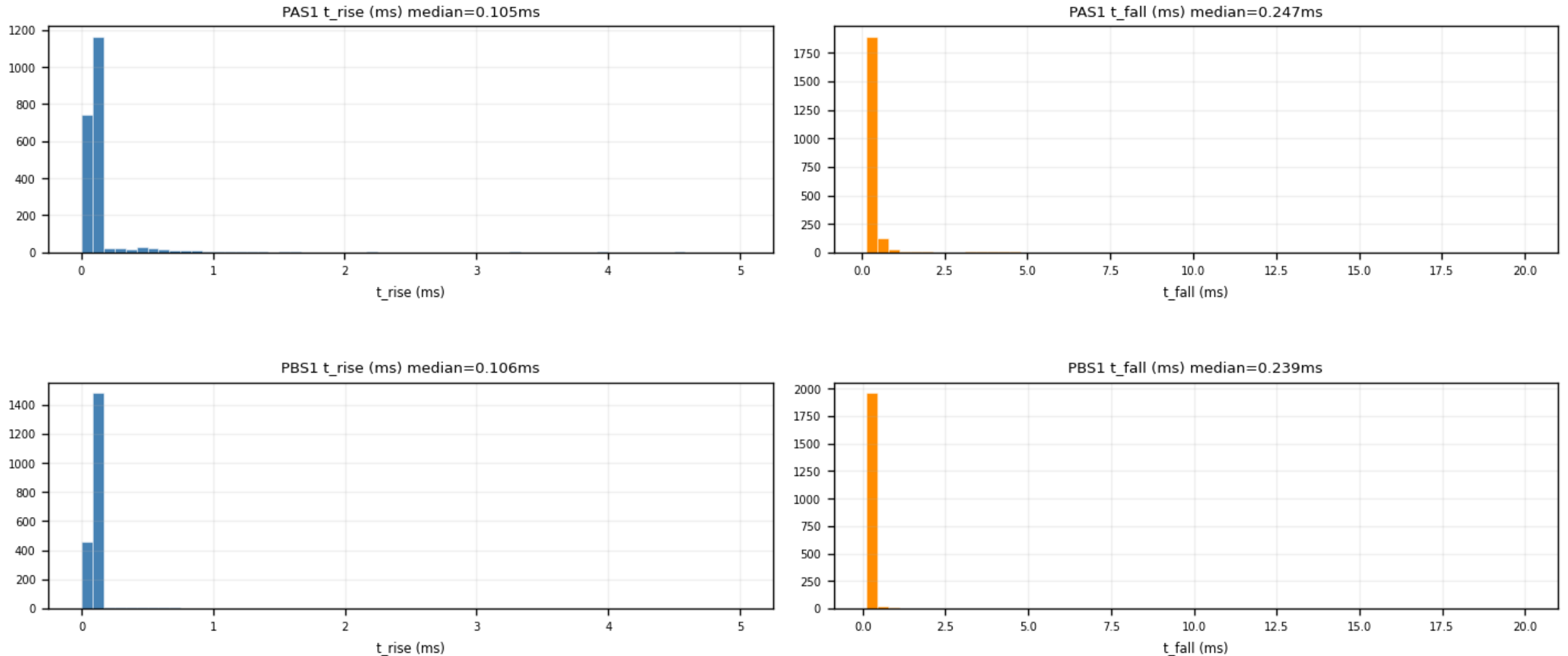
The onset is a FREE fit parameter; fitted values cluster ≈ 230 samples after the nominal 16050; pinning it inside the fit would bias every other parameter.



Backup · Z7 · time_constants: τ_{rise} / τ_{fall} per channel (1/6)

figure: zip7_time_constants.png, channel panels 1–2 of 11, top to bottom

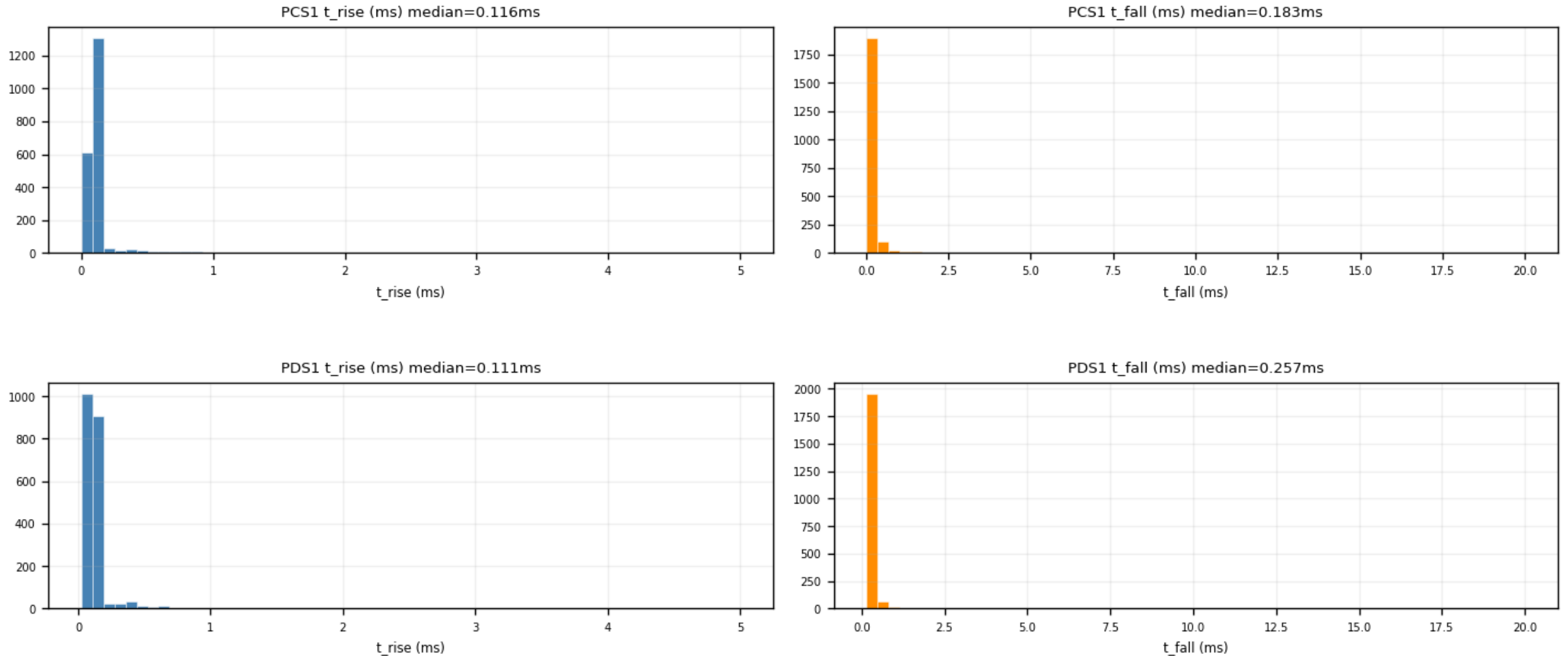
Fast-pulse population at $\tau_{\text{rise}} \approx 0.1$ ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Backup · Z7 · time_constants: τ_{rise} / τ_{fall} per channel (2/6)

figure: zip7_time_constants.png, channel panels 3–4 of 11, top to bottom

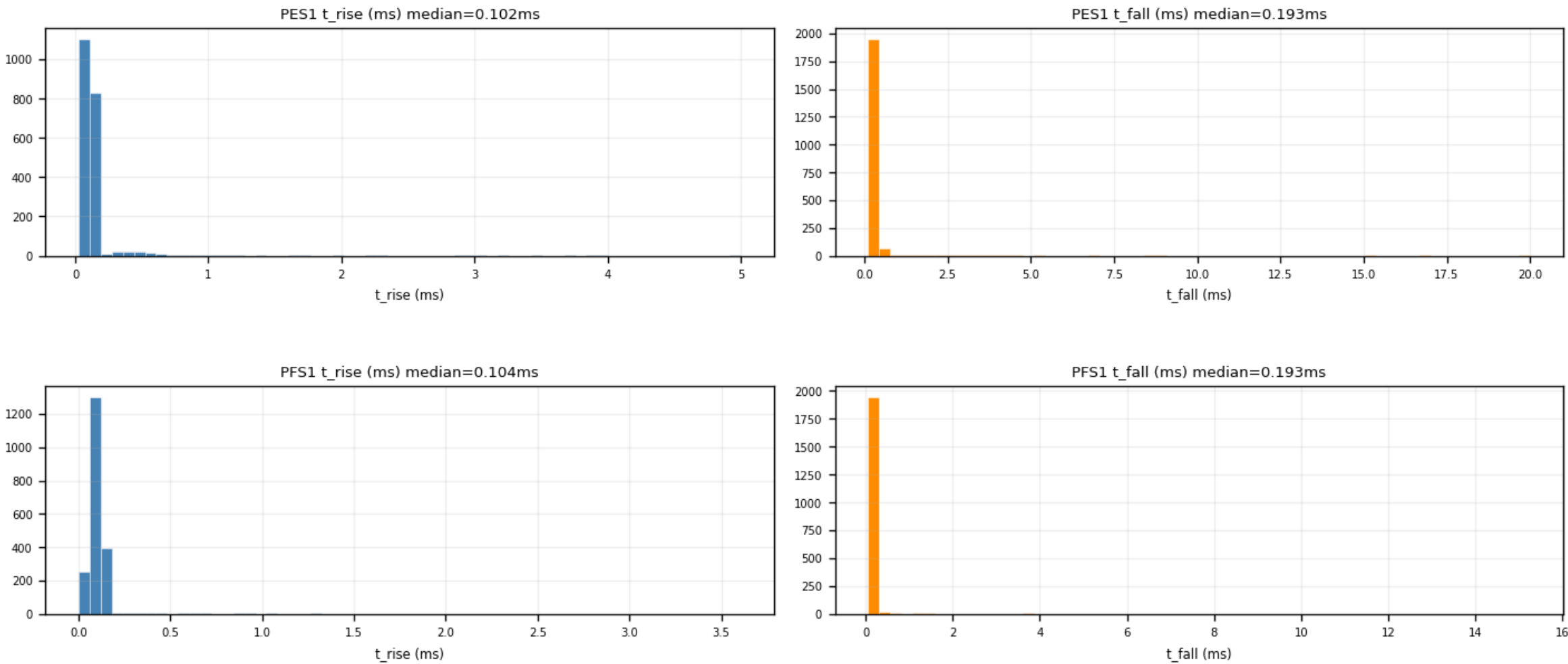
Fast-pulse population at $\tau_{\text{rise}} \approx 0.1$ ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Backup · Z7 · time_constants: τ_{rise} / τ_{fall} per channel (3/6)

figure: zip7_time_constants.png, channel panels 5–6 of 11, top to bottom

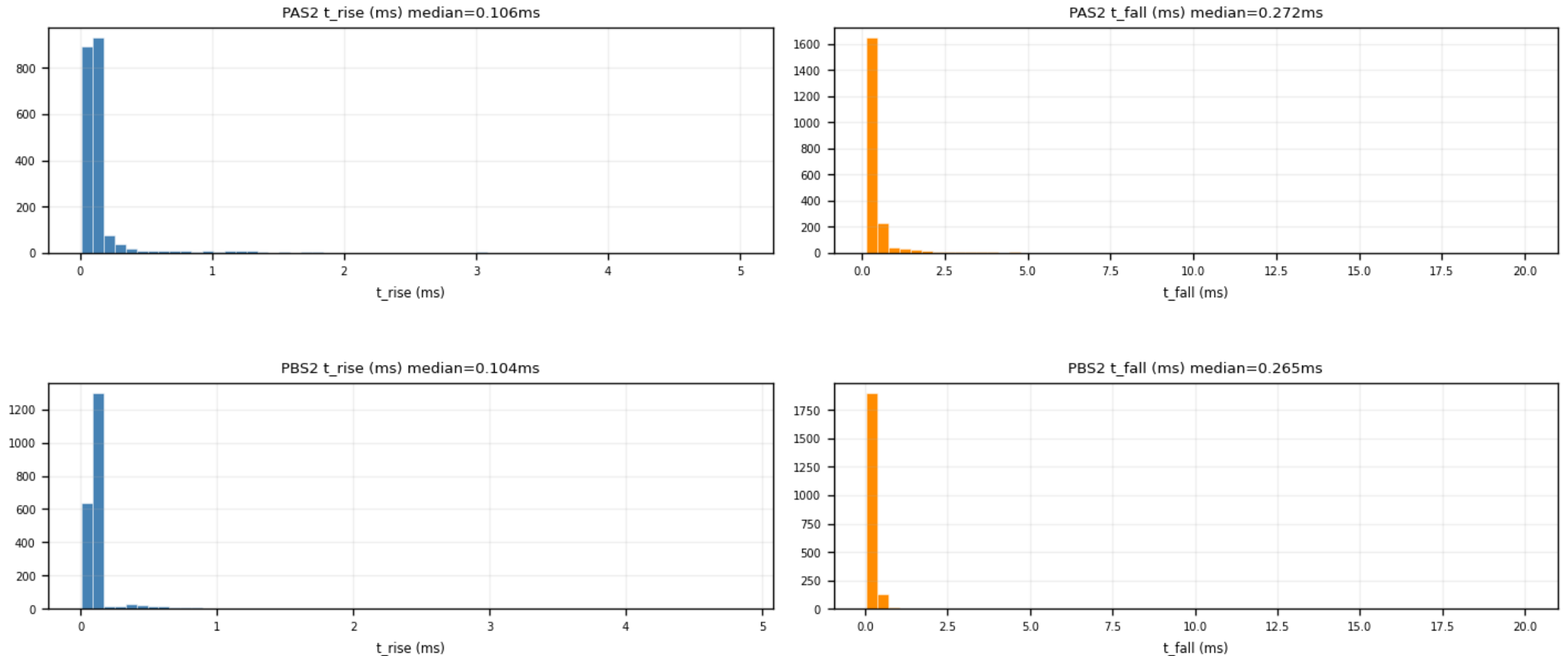
Fast-pulse population at $\tau_{\text{rise}} \approx 0.1$ ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Backup · Z7 · time_constants: τ_{rise} / τ_{fall} per channel (4/6)

figure: zip7_time_constants.png, channel panels 7–8 of 11, top to bottom

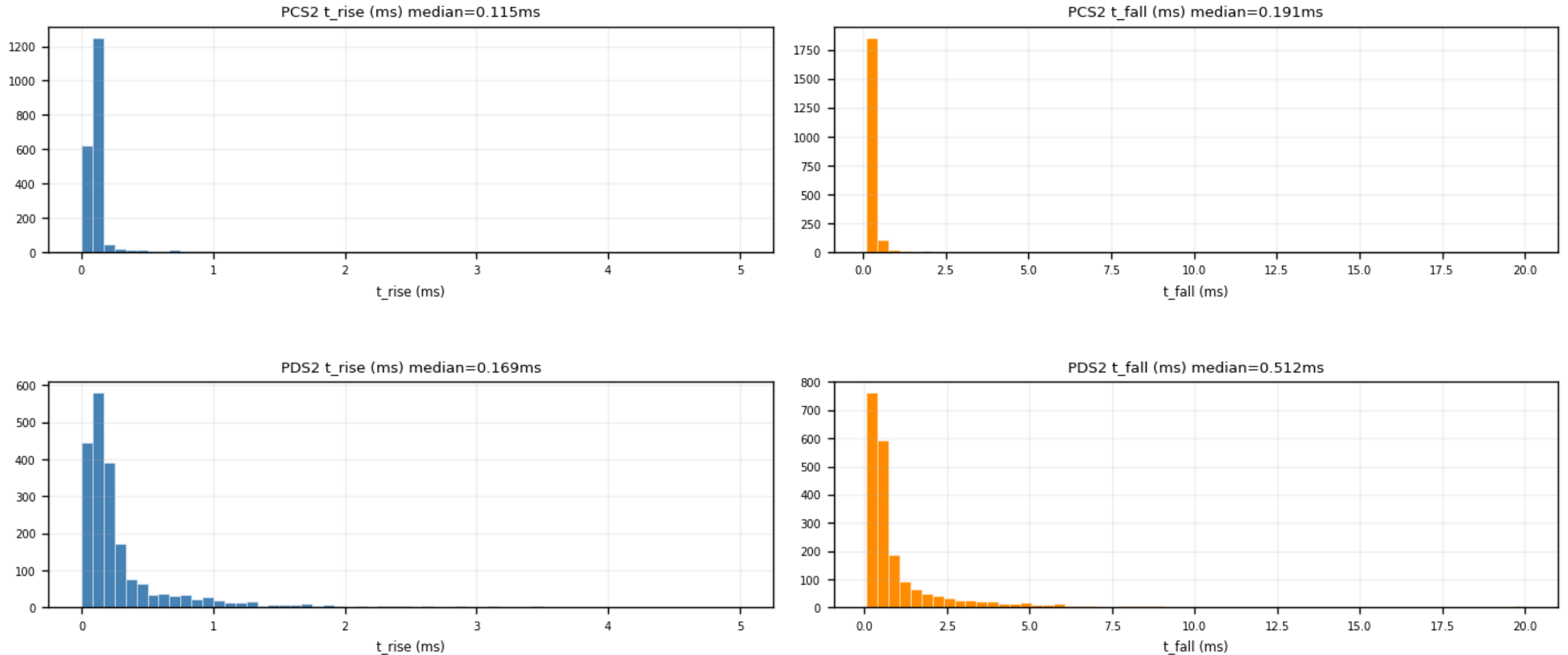
Fast-pulse population at $\tau_{\text{rise}} \approx 0.1$ ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Backup · Z7 · time_constants: τ_{rise} / τ_{fall} per channel (5/6)

figure: zip7_time_constants.png, channel panels 9–10 of 11, top to bottom

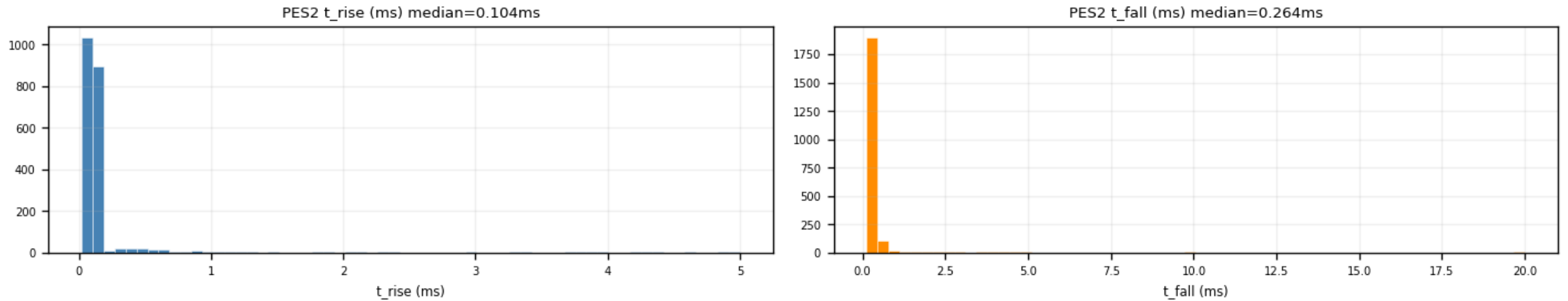
Fast-pulse population at $\tau_{\text{rise}} \approx 0.1$ ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Backup · Z7 · time_constants: τ_{rise} / τ_{fall} per channel (6/6)

figure: zip7_time_constants.png, channel panels 11–11 of 11, top to bottom

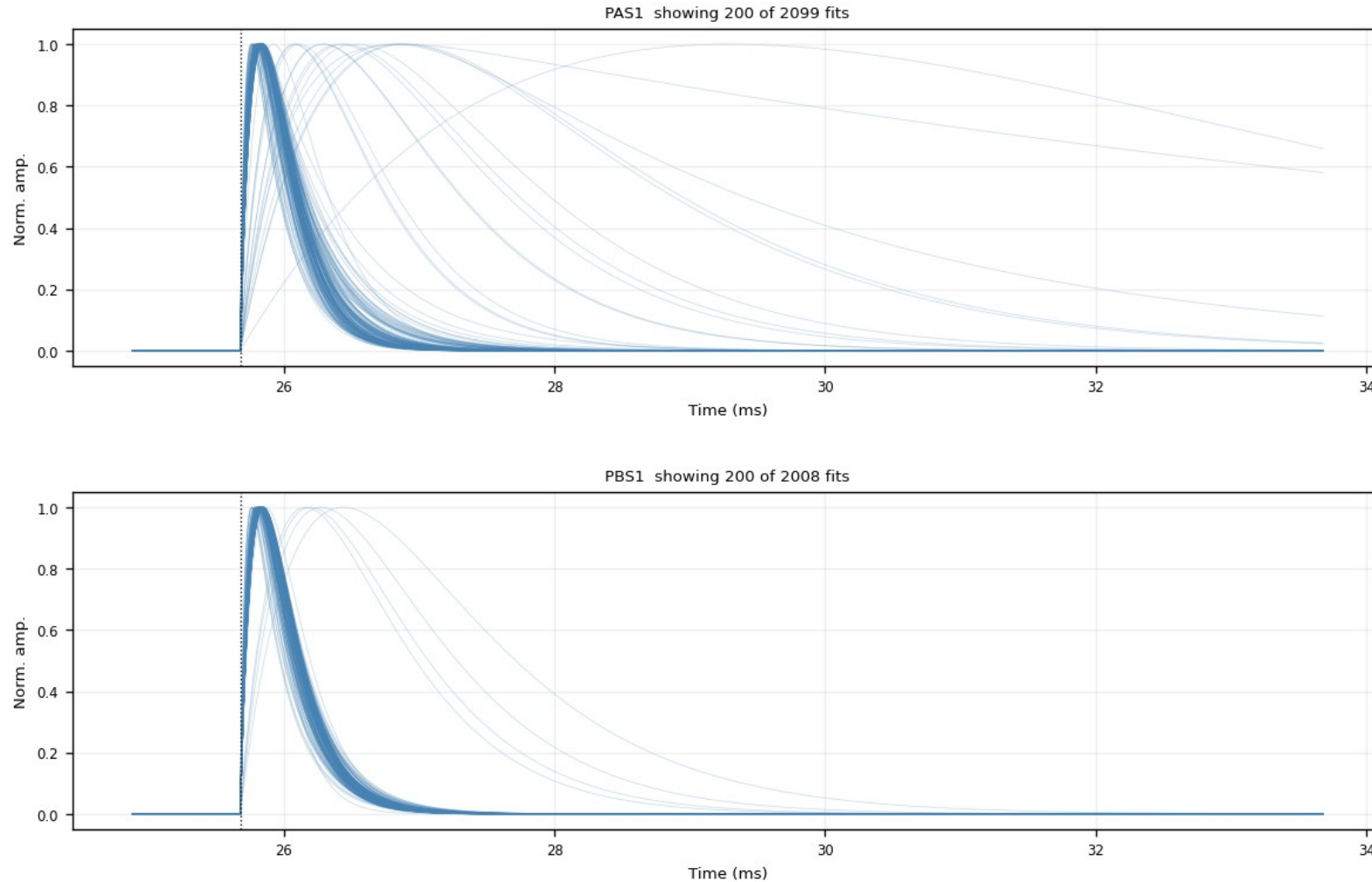
Fast-pulse population at $\tau_{\text{rise}} \approx 0.1$ ms; isolated spikes at the parameter bounds are noise fits pinned at the fit limits, removed by the NRMSE cut.



Backup · Z7 · fitted_curves_overlay: fan, no cut (1/6)

figure: zip7_fitted_curves_overlay.png, channel panels 1–2 of 11, top to bottom

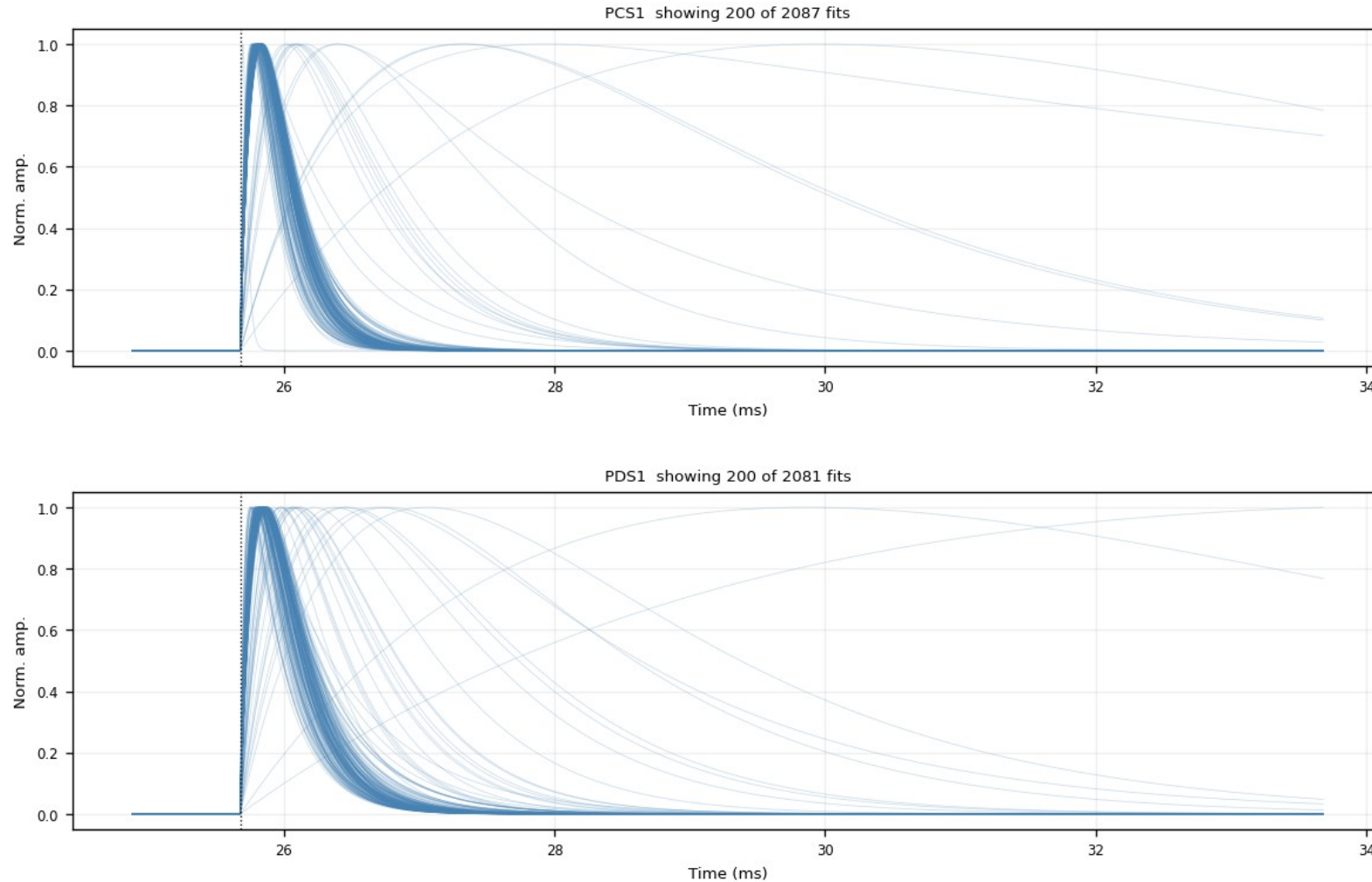
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Backup · Z7 · fitted_curves_overlay: fan, no cut (2/6)

figure: zip7_fitted_curves_overlay.png, channel panels 3–4 of 11, top to bottom

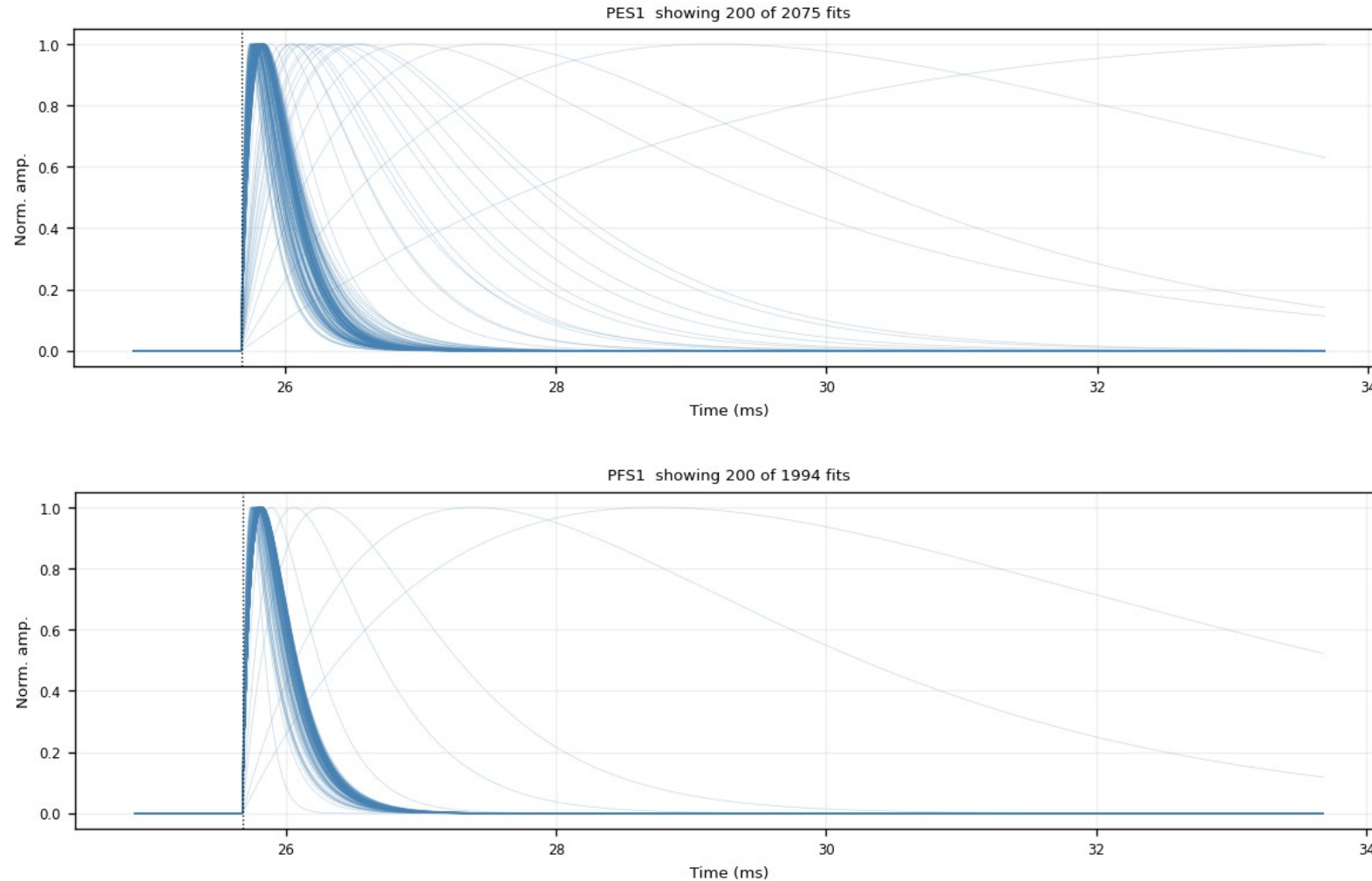
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Backup · Z7 · fitted_curves_overlay: fan, no cut (3/6)

figure: zip7_fitted_curves_overlay.png, channel panels 5–6 of 11, top to bottom

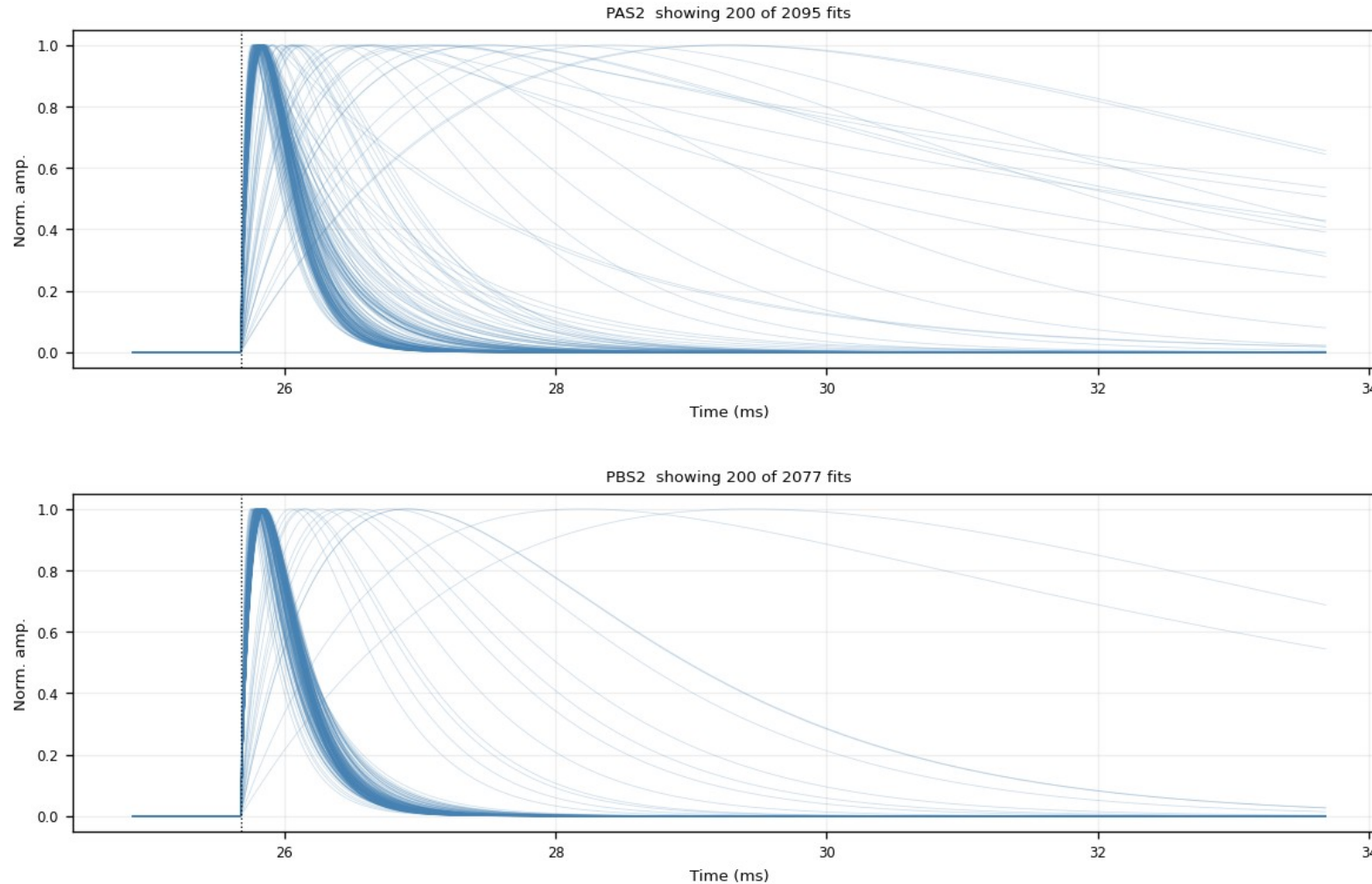
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Backup · Z7 · fitted_curves_overlay: fan, no cut (4/6)

figure: zip7_fitted_curves_overlay.png, channel panels 7–8 of 11, top to bottom

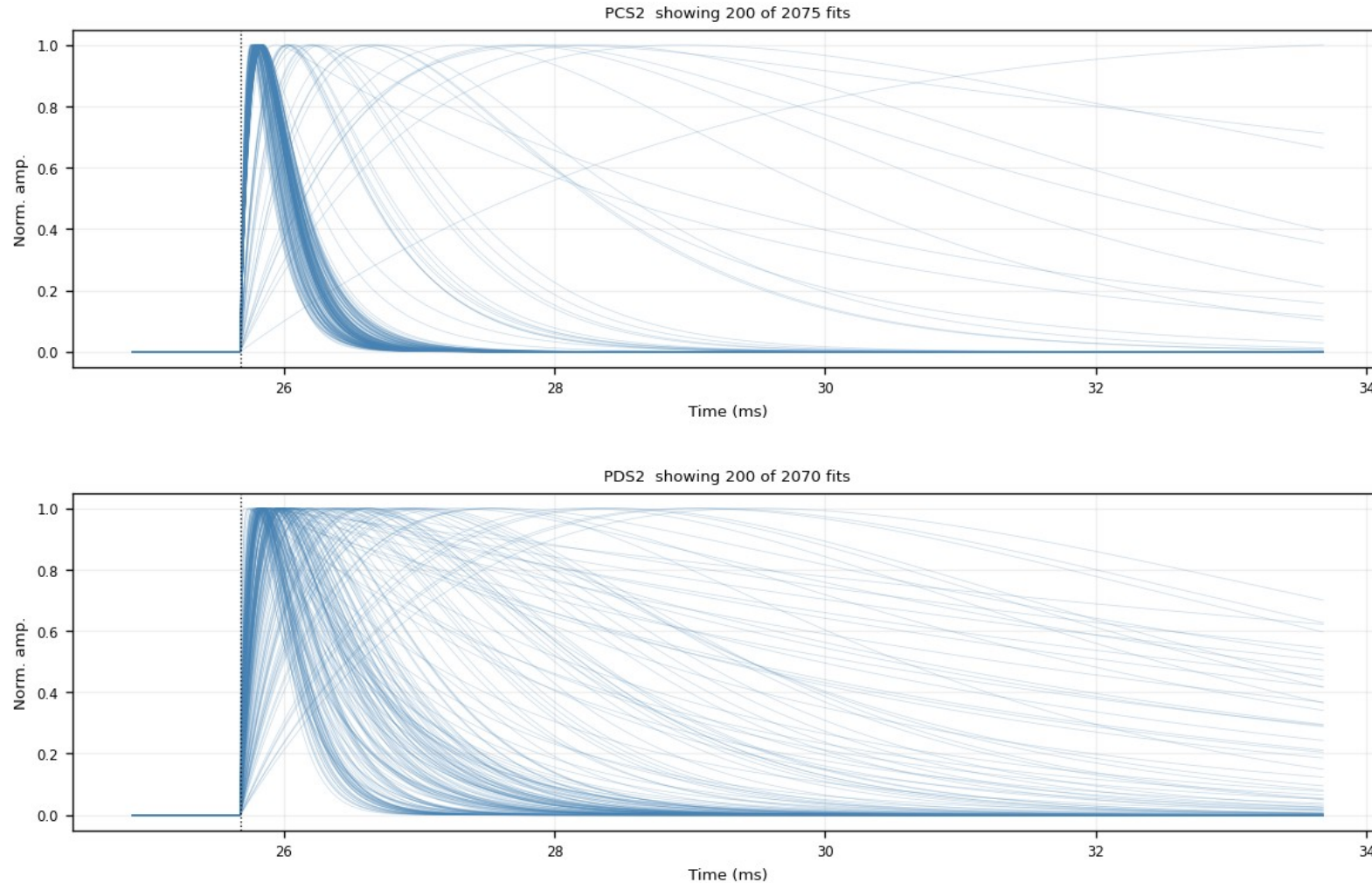
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Backup · Z7 · fitted_curves_overlay: fan, no cut (5/6)

figure: zip7_fitted_curves_overlay.png, channel panels 9–10 of 11, top to bottom

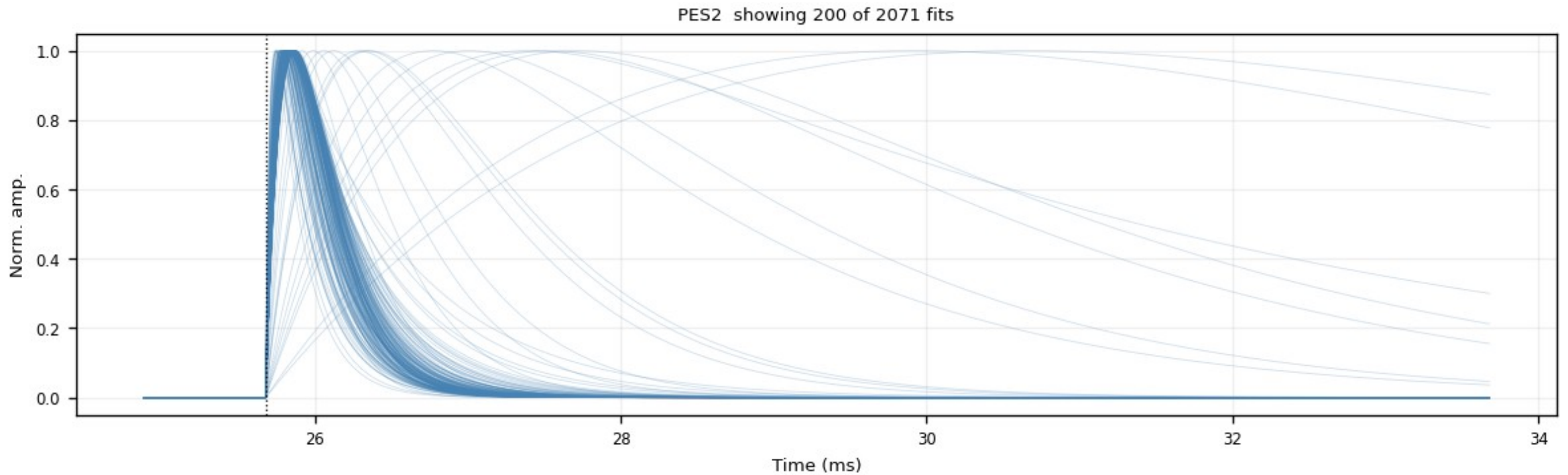
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Backup · Z7 · fitted_curves_overlay: fan, no cut (6/6)

figure: zip7_fitted_curves_overlay.png, channel panels 11–11 of 11, top to bottom

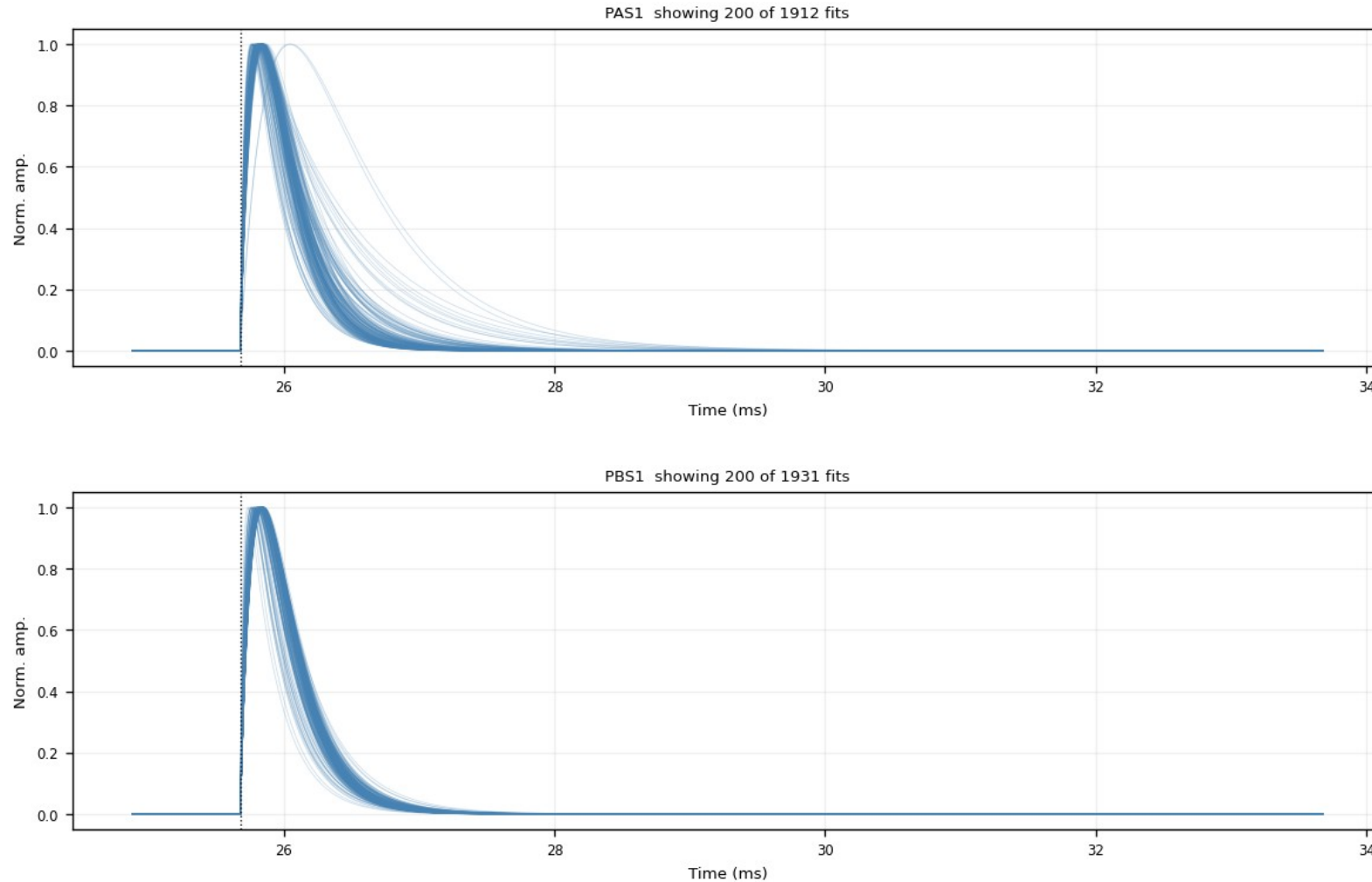
Every fit_ok fitted curve at the common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.



Backup · Z7 · fitted_curves_overlay: after $\text{NRMSE} \leq 0.4$ (1/6)

figure: zip7_fitted_curves_overlay_nrmse0.4.png, channel panels 1–2 of 11, top to bottom

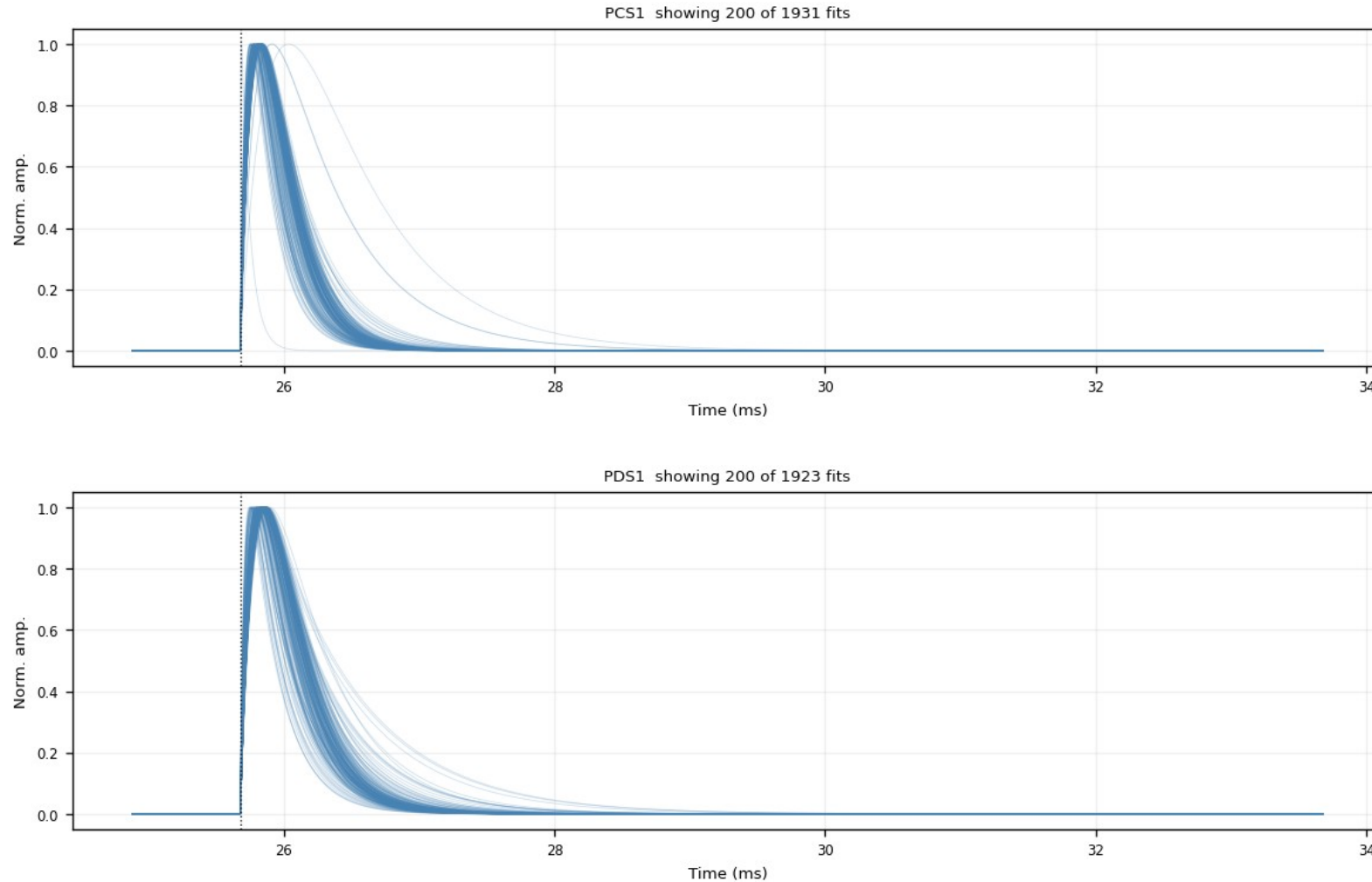
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



Backup · Z7 · fitted_curves_overlay: after $\text{NRMSE} \leq 0.4$ (2/6)

figure: zip7_fitted_curves_overlay_nrmse0.4.png, channel panels 3–4 of 11, top to bottom

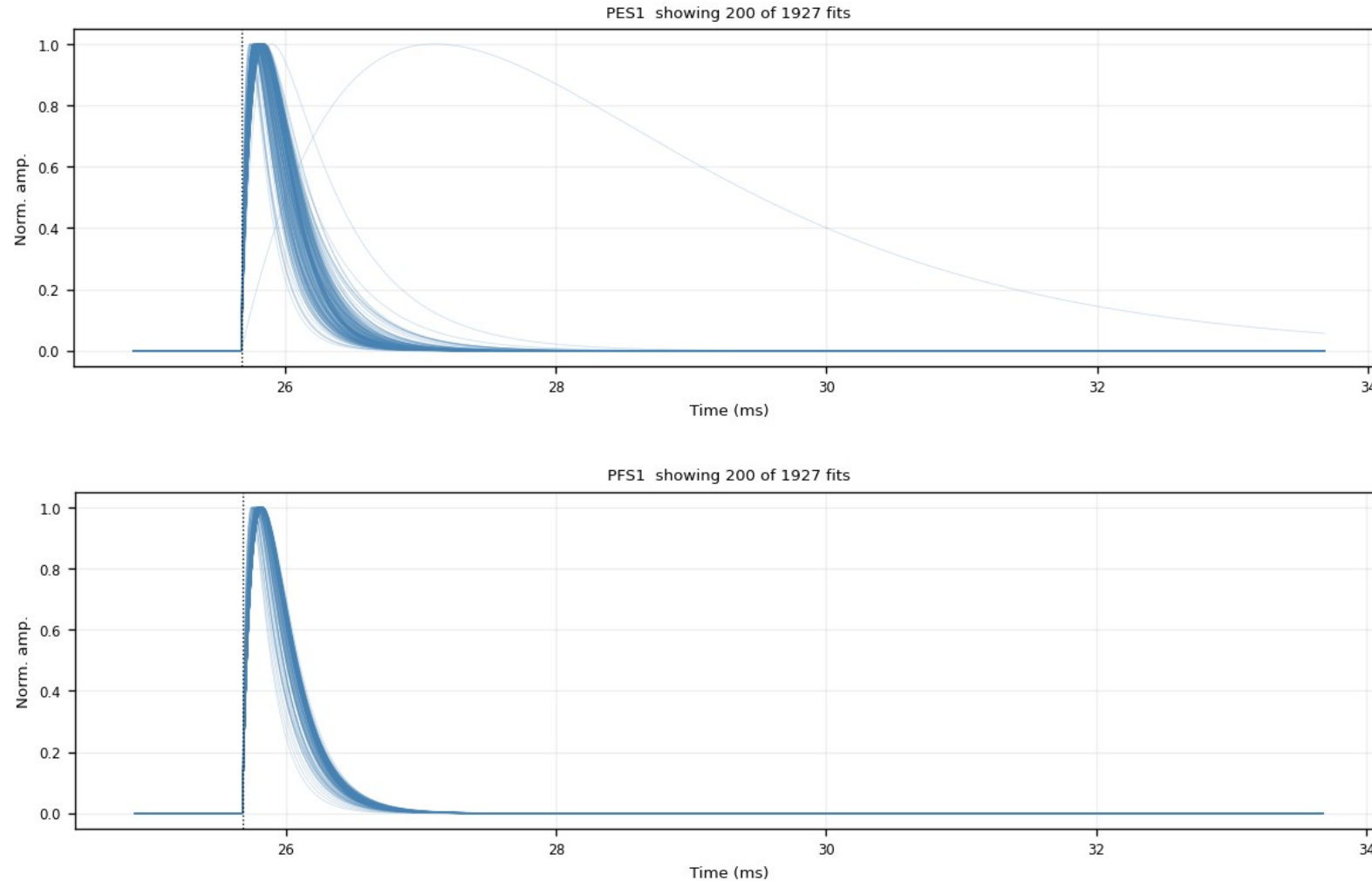
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



Backup · Z7 · fitted_curves_overlay: after $\text{NRMSE} \leq 0.4$ (3/6)

figure: zip7_fitted_curves_overlay_nrmse0.4.png, channel panels 5–6 of 11, top to bottom

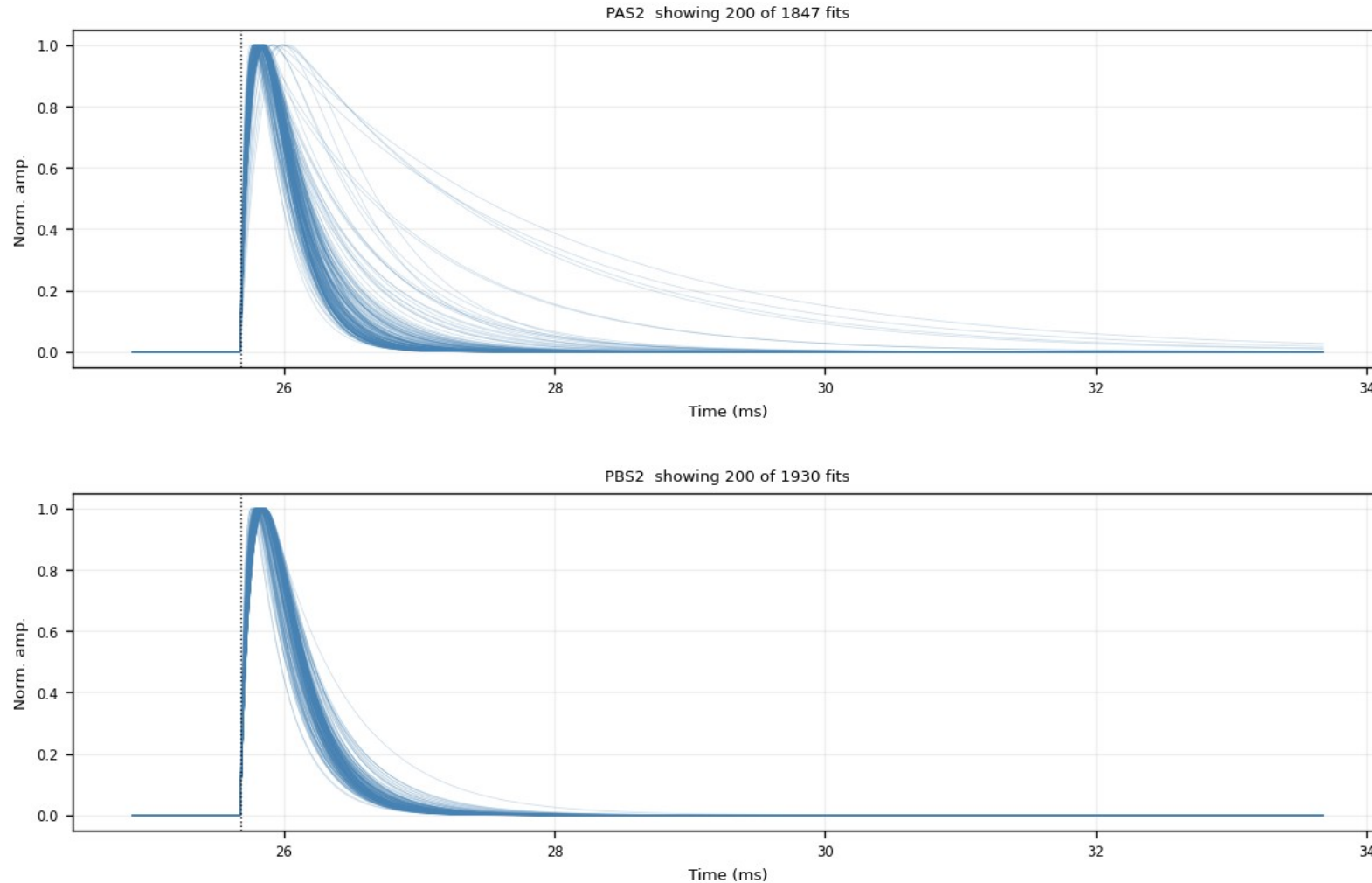
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



Backup · Z7 · fitted_curves_overlay: after $\text{NRMSE} \leq 0.4$ (4/6)

figure: zip7_fitted_curves_overlay_nrmse0.4.png, channel panels 7–8 of 11, top to bottom

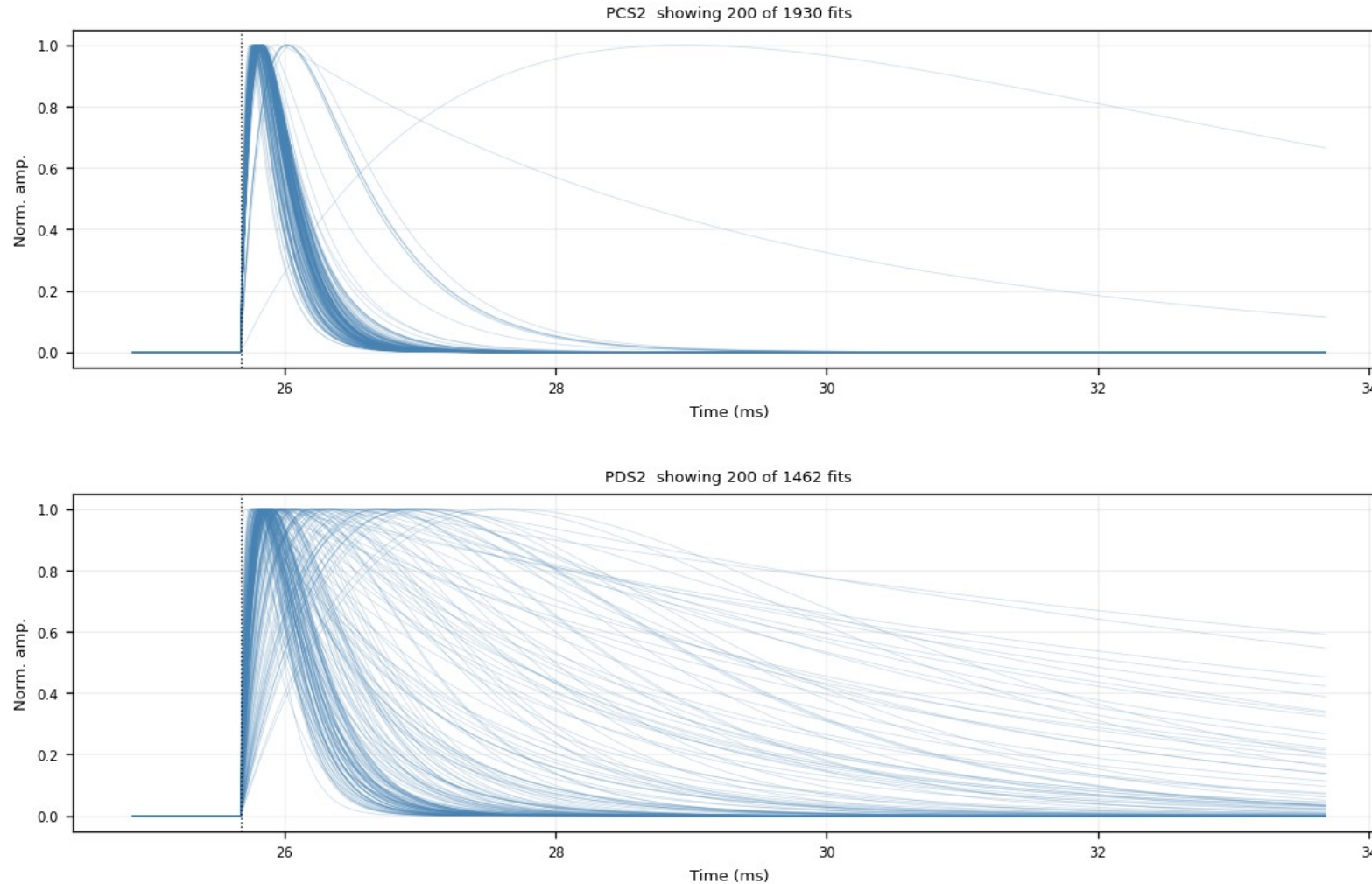
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



Backup · Z7 · fitted_curves_overlay: after $\text{NRMSE} \leq 0.4$ (5/6)

figure: zip7_fitted_curves_overlay_nrmse0.4.png, channel panels 9–10 of 11, top to bottom

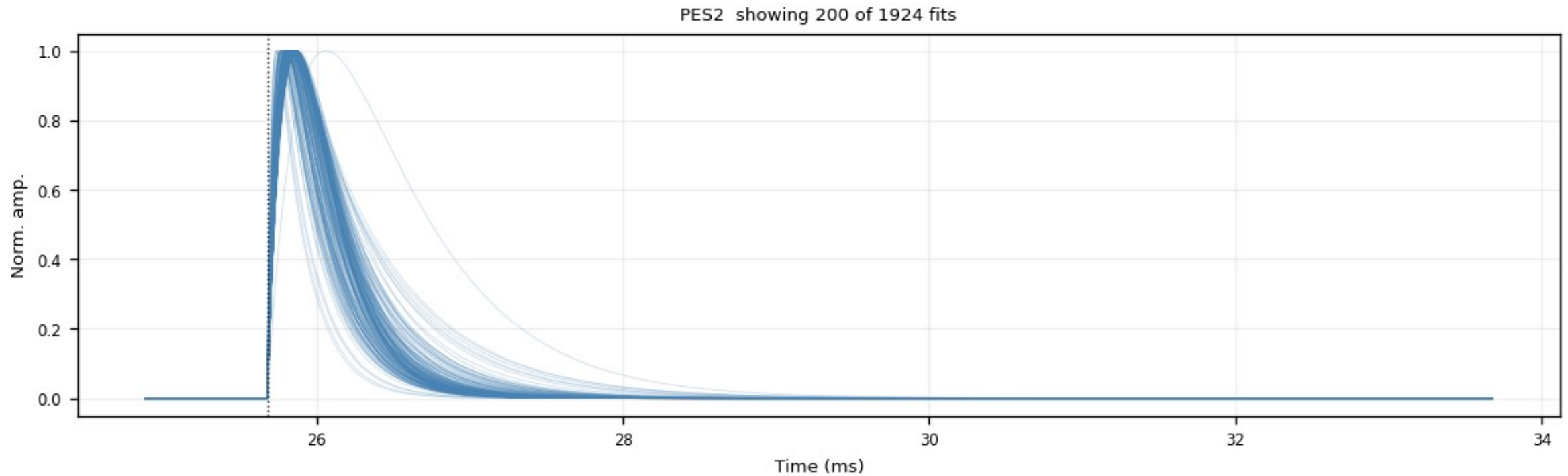
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



Backup · Z7 · fitted_curves_overlay: after $\text{NRMSE} \leq 0.4$ (6/6)

figure: zip7_fitted_curves_overlay_nrmse0.4.png, channel panels 11–11 of 11, top to bottom

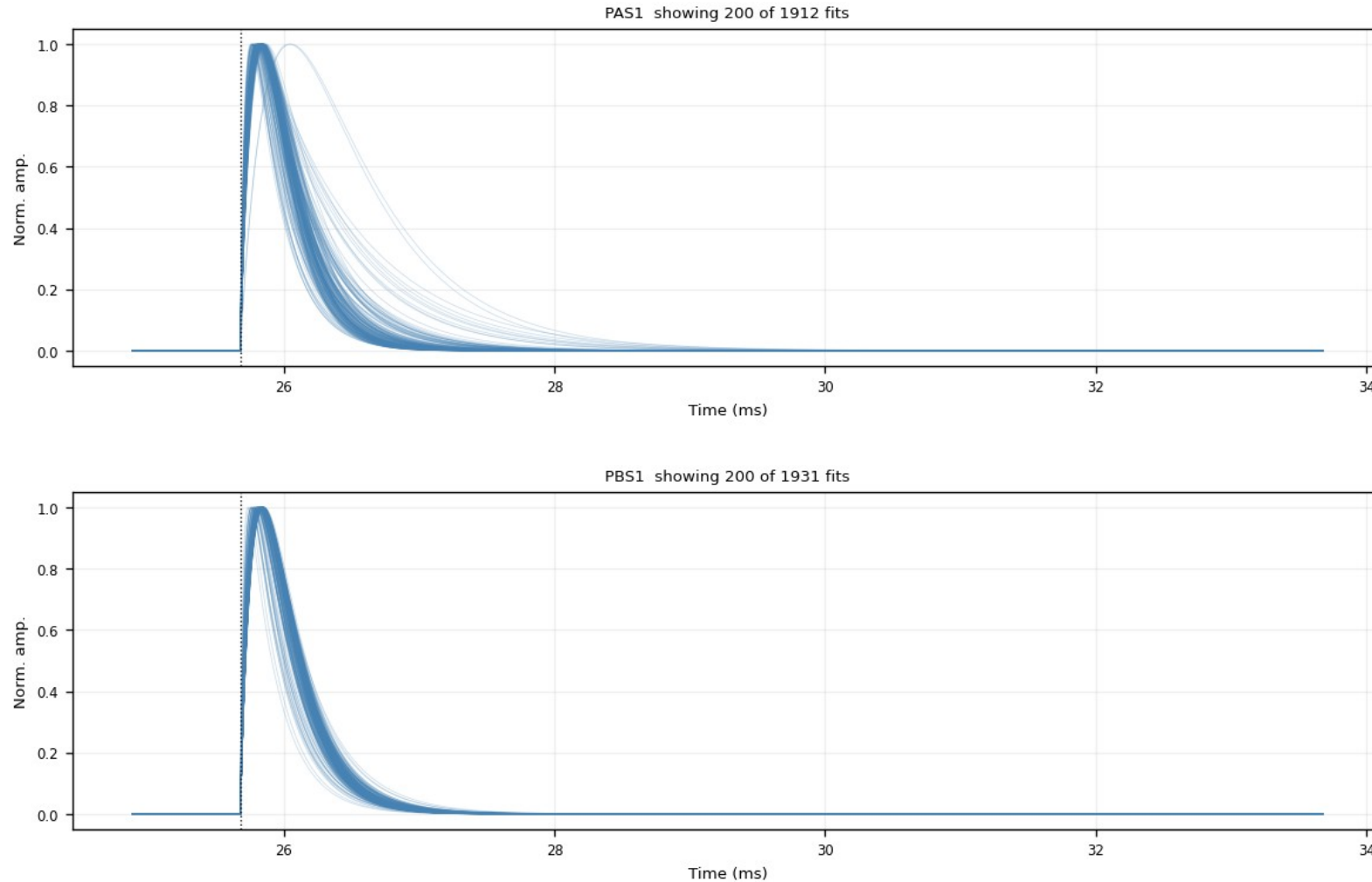
Same fan after the NRMSE cut: the noise fan collapses, the fast-pulse bundle remains.



Backup · Z7 · fitted_curves_overlay: after both cuts (1/6)

figure: zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 1–2 of 11, top to bottom

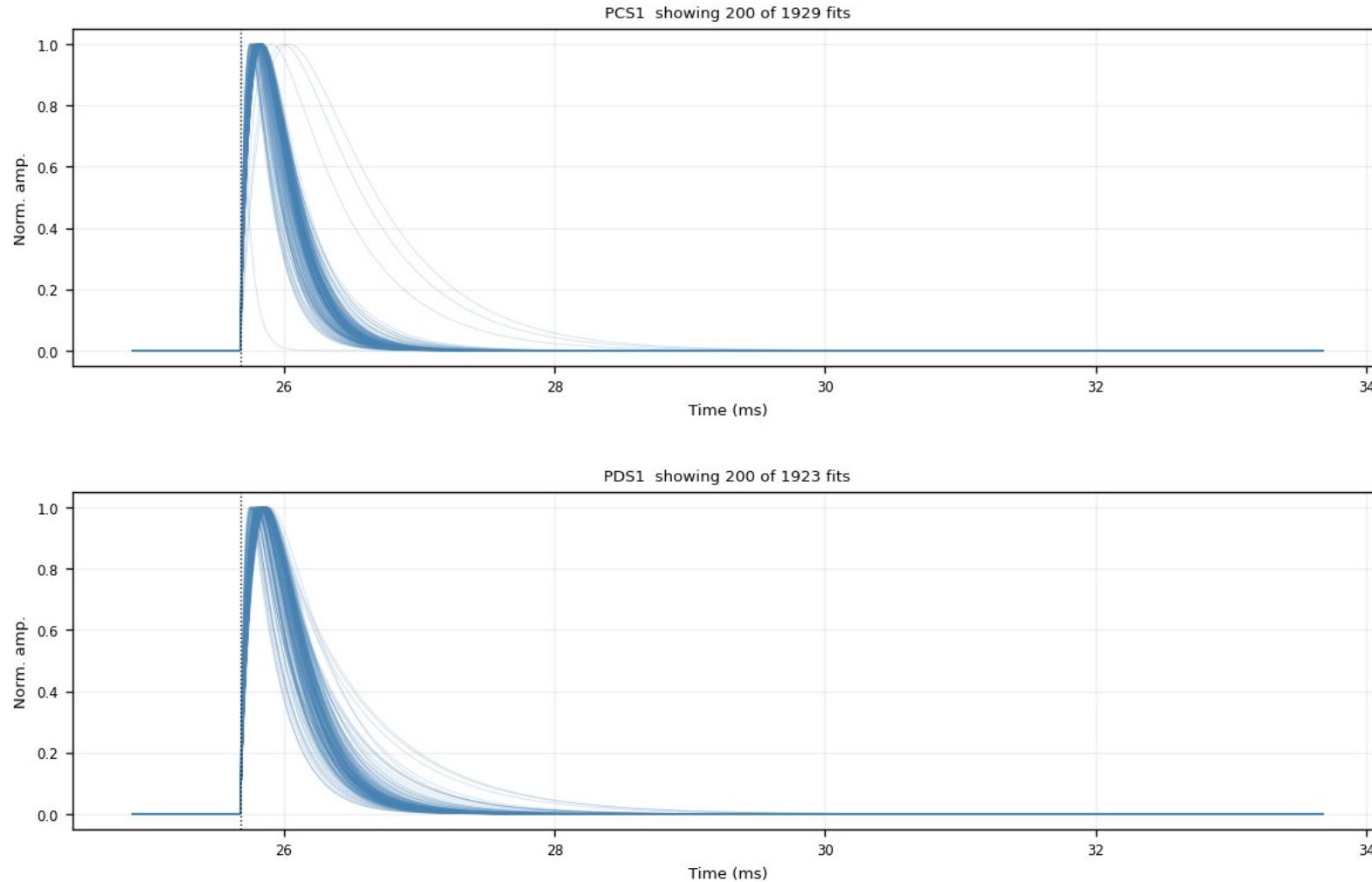
After $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{rise}} \leq 0.3$ ms, the exact PCA input population: the clean fast-pulse bundle only.



Backup · Z7 · fitted_curves_overlay: after both cuts (2/6)

figure: zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 3–4 of 11, top to bottom

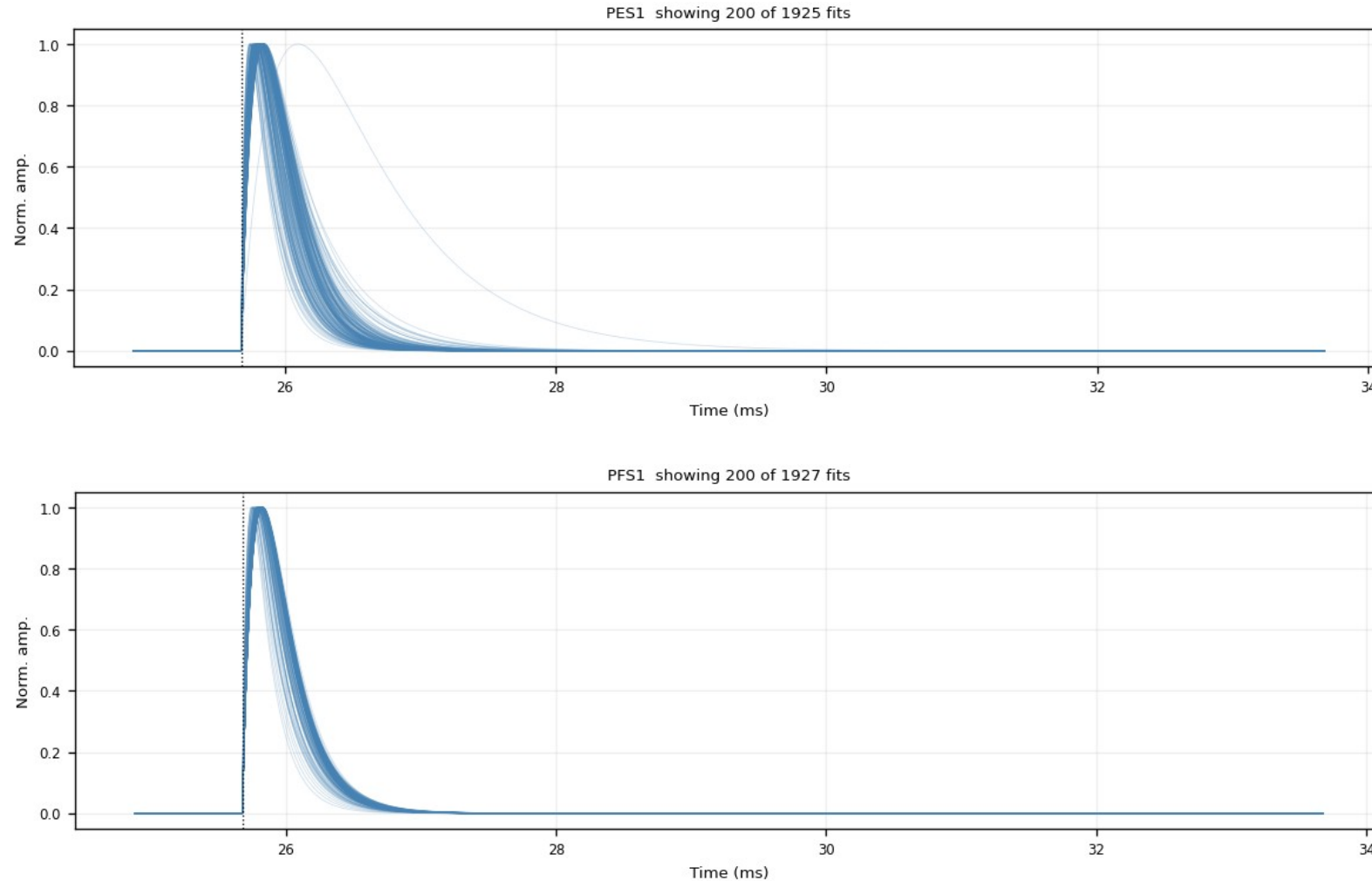
After $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{rise}} \leq 0.3$ ms, the exact PCA input population: the clean fast-pulse bundle only.



Backup · Z7 · fitted_curves_overlay: after both cuts (3/6)

figure: zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 5–6 of 11, top to bottom

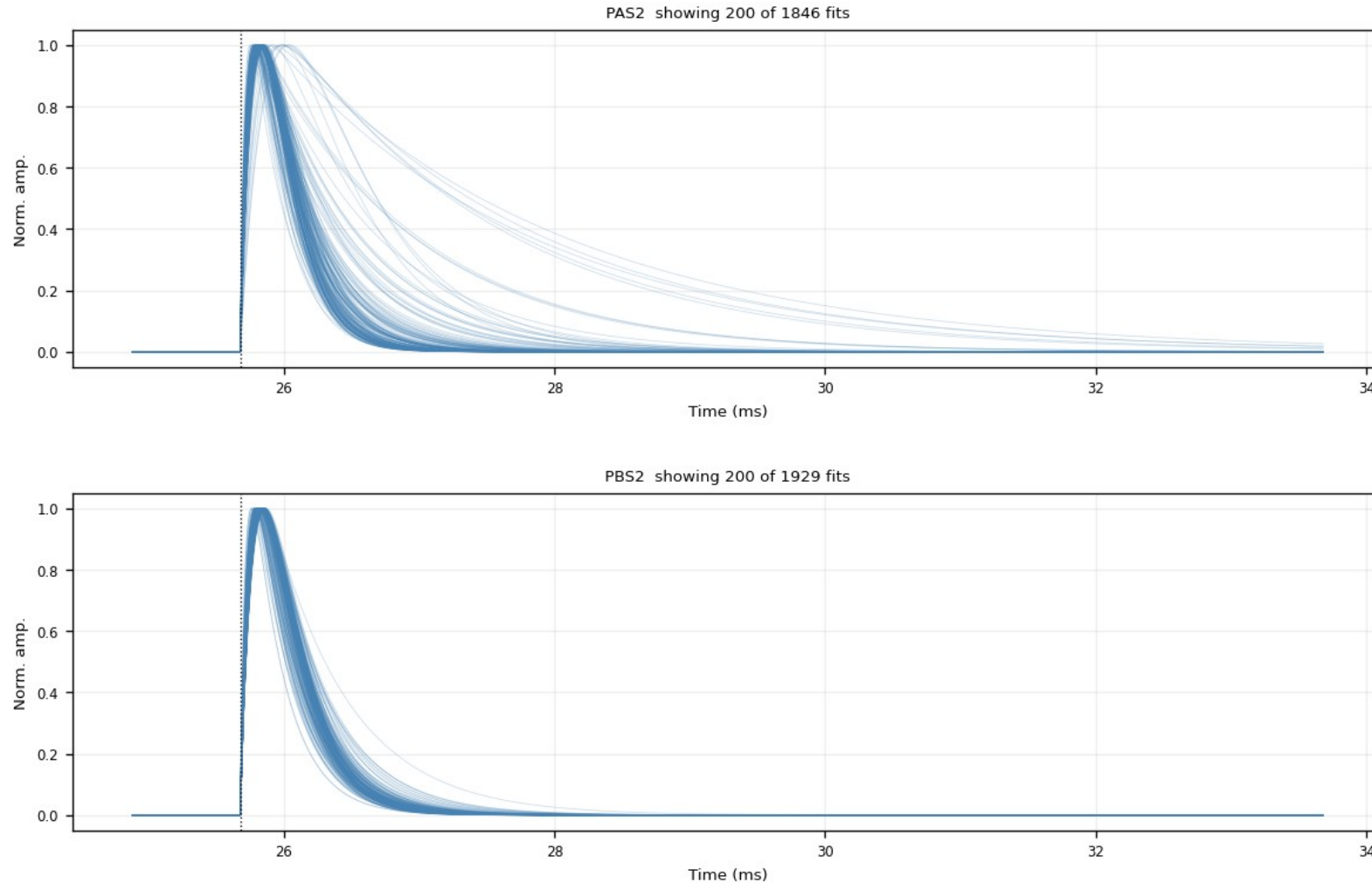
After $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{rise}} \leq 0.3$ ms, the exact PCA input population: the clean fast-pulse bundle only.



Backup · Z7 · fitted_curves_overlay: after both cuts (4/6)

figure: zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 7–8 of 11, top to bottom

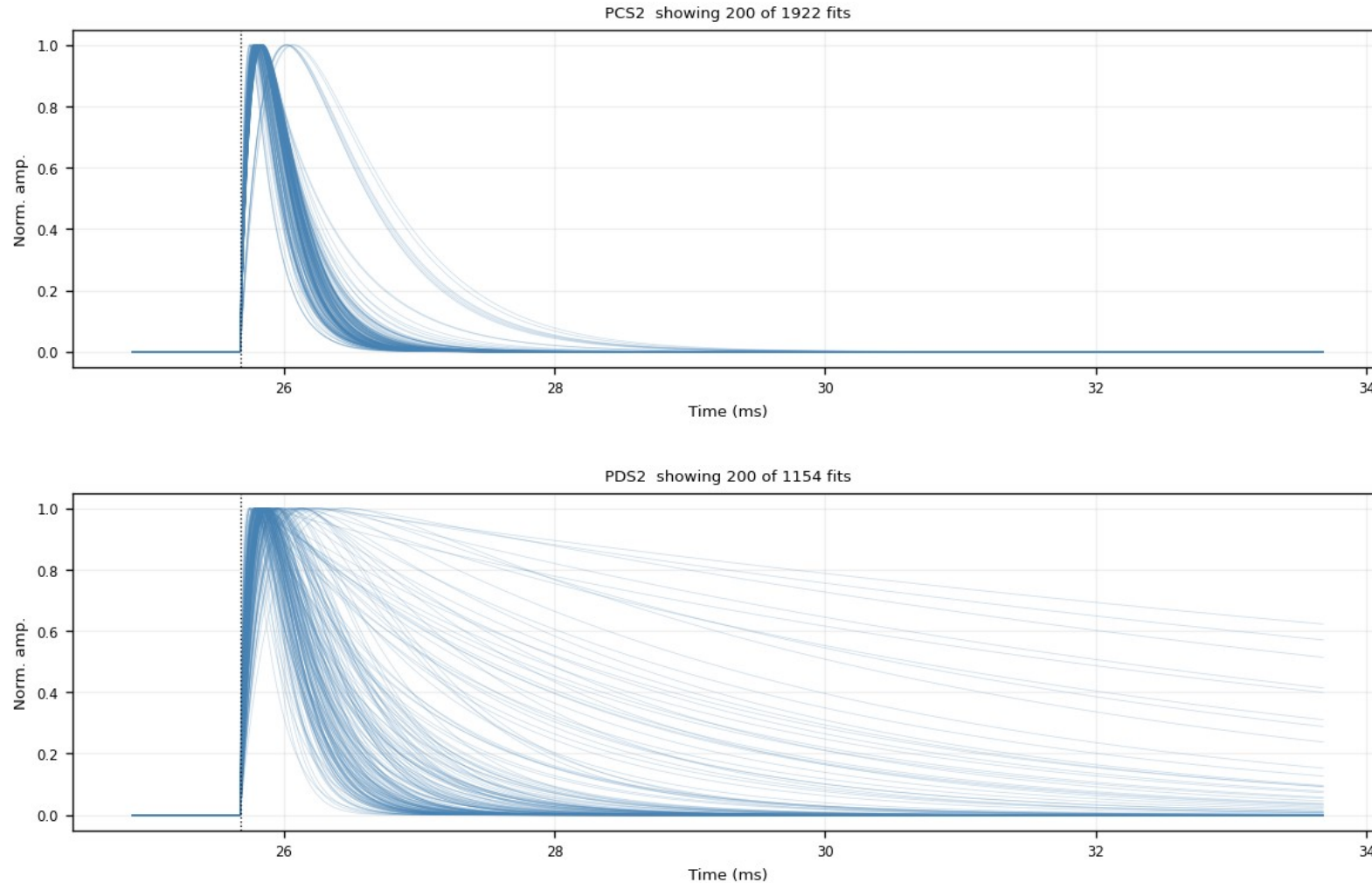
After $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{rise}} \leq 0.3$ ms, the exact PCA input population: the clean fast-pulse bundle only.



Backup · Z7 · fitted_curves_overlay: after both cuts (5/6)

figure: zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 9–10 of 11, top to bottom

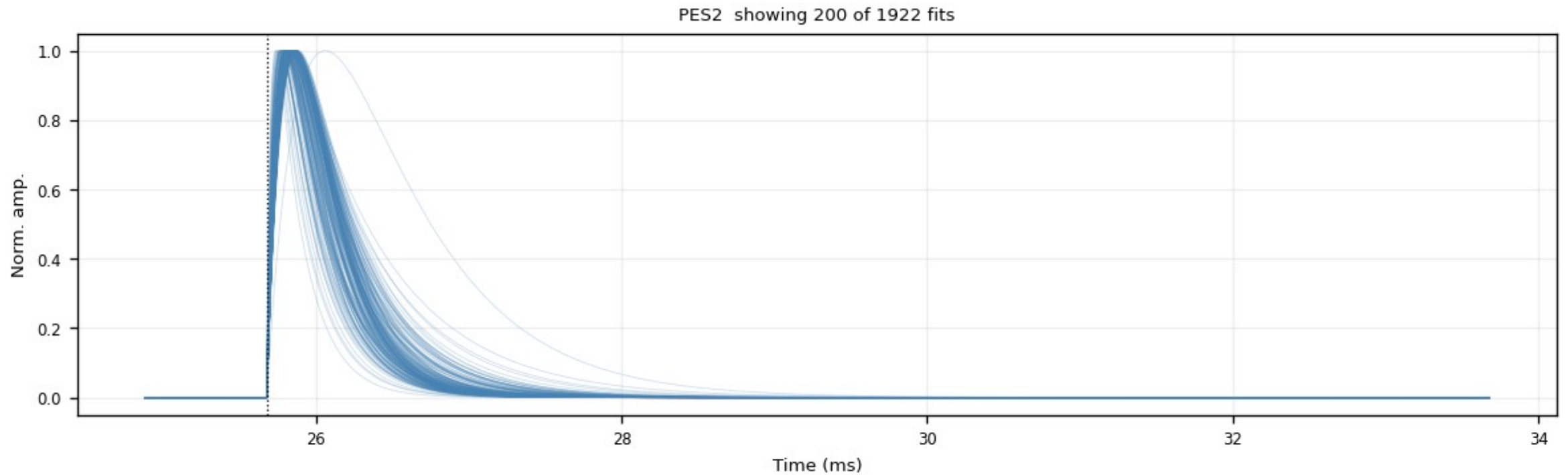
After $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{rise}} \leq 0.3$ ms, the exact PCA input population: the clean fast-pulse bundle only.



Backup · Z7 · fitted_curves_overlay: after both cuts (6/6)

figure: zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png, channel panels 11–11 of 11, top to bottom

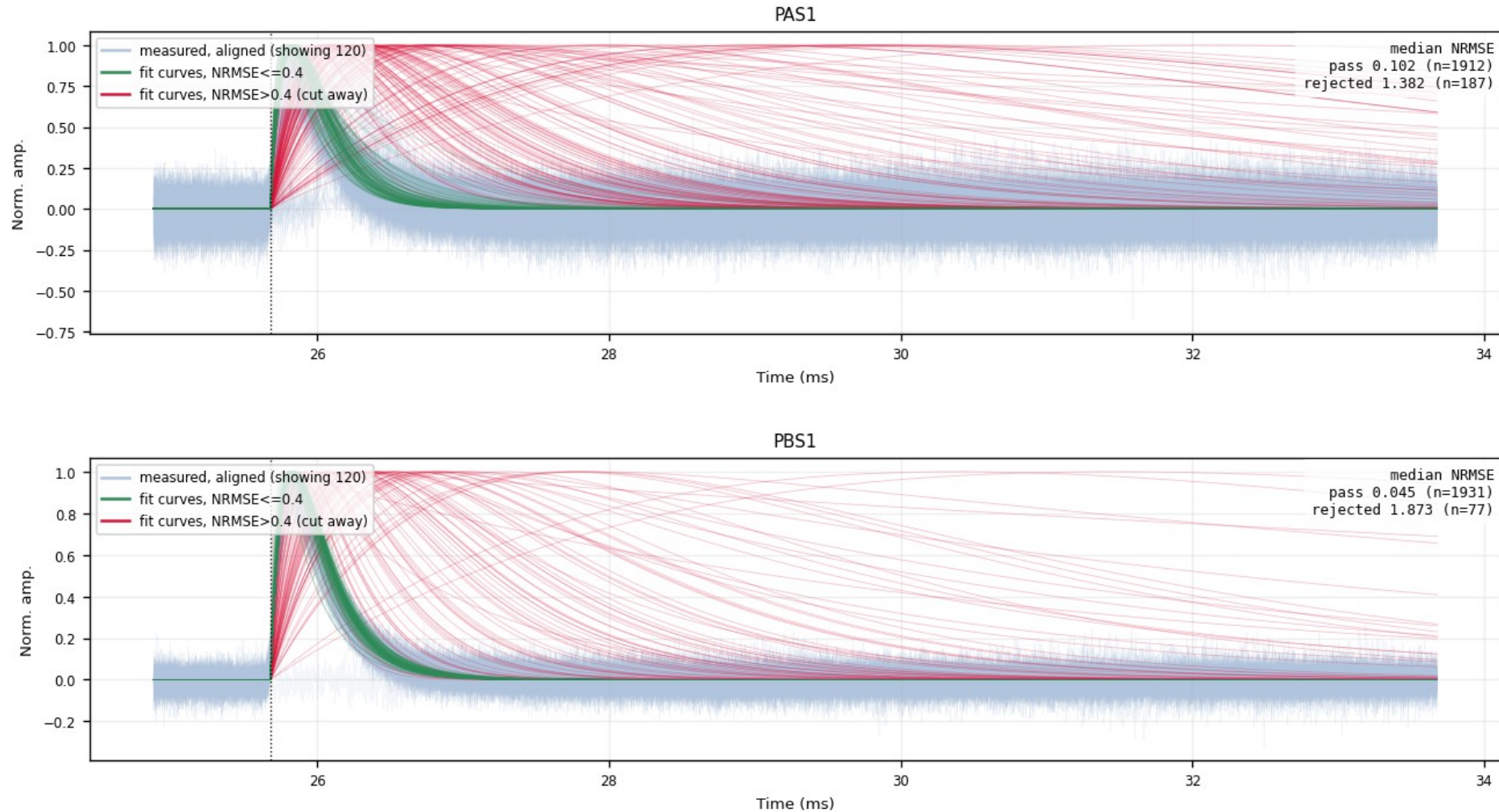
After $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{rise}} \leq 0.3$ ms, the exact PCA input population: the clean fast-pulse bundle only.



Backup · Z7 · overlay_fan_cut: kept vs rejected fits (1/6)

figure: zip7_overlay_fan_cut_nrmse0.4.png, channel panels 1–2 of 11, top to bottom

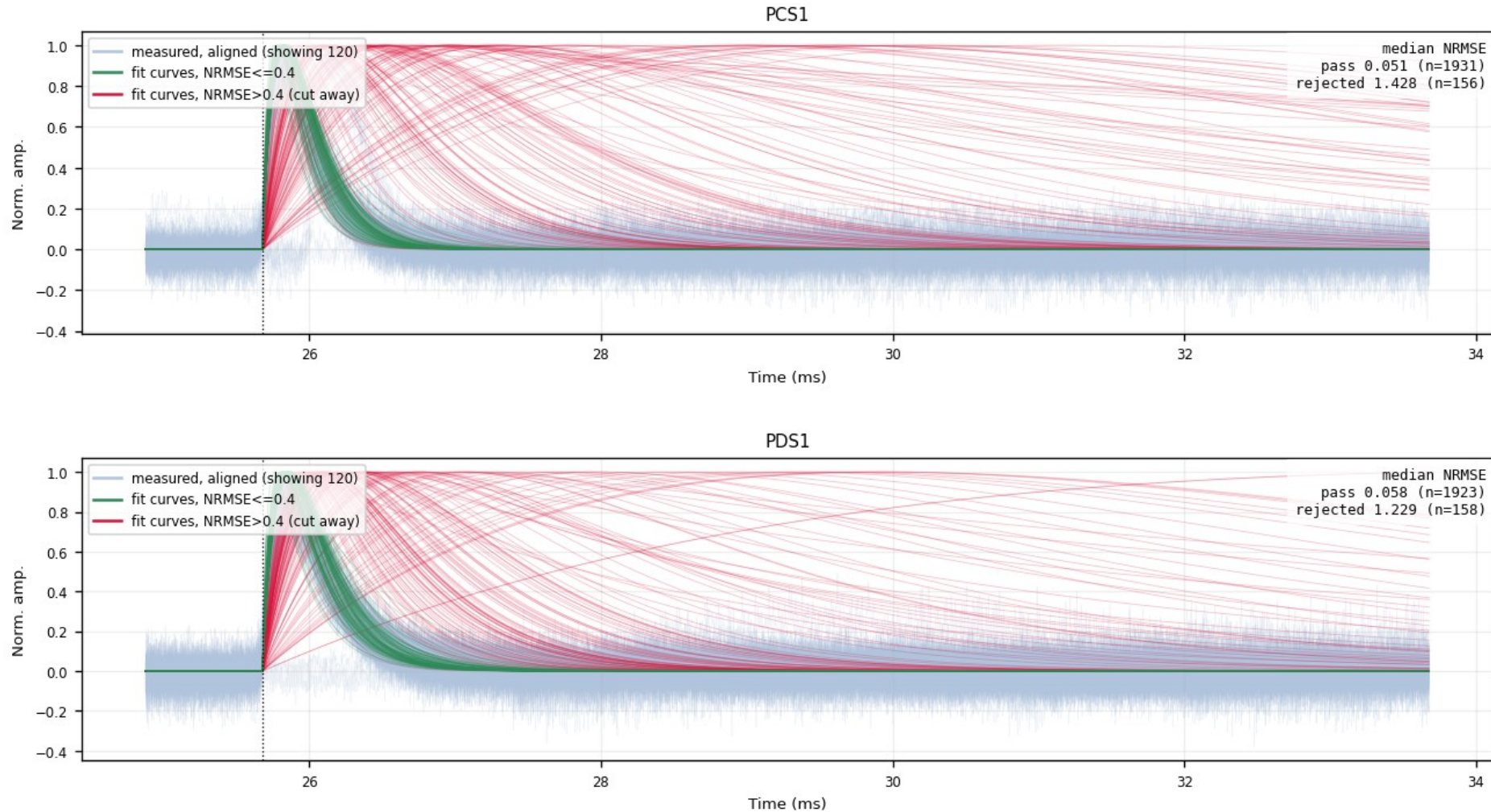
Gray-blue = aligned measured traces; green = fits with $\text{NRMSE} \leq 0.4$ (kept); red = $\text{NRMSE} > 0.4$ (cut away). Median NRMSE and counts of both populations stamped per channel.



Backup · Z7 · overlay_fan_cut: kept vs rejected fits (2/6)

figure: zip7_overlay_fan_cut_nrmse0.4.png, channel panels 3–4 of 11, top to bottom

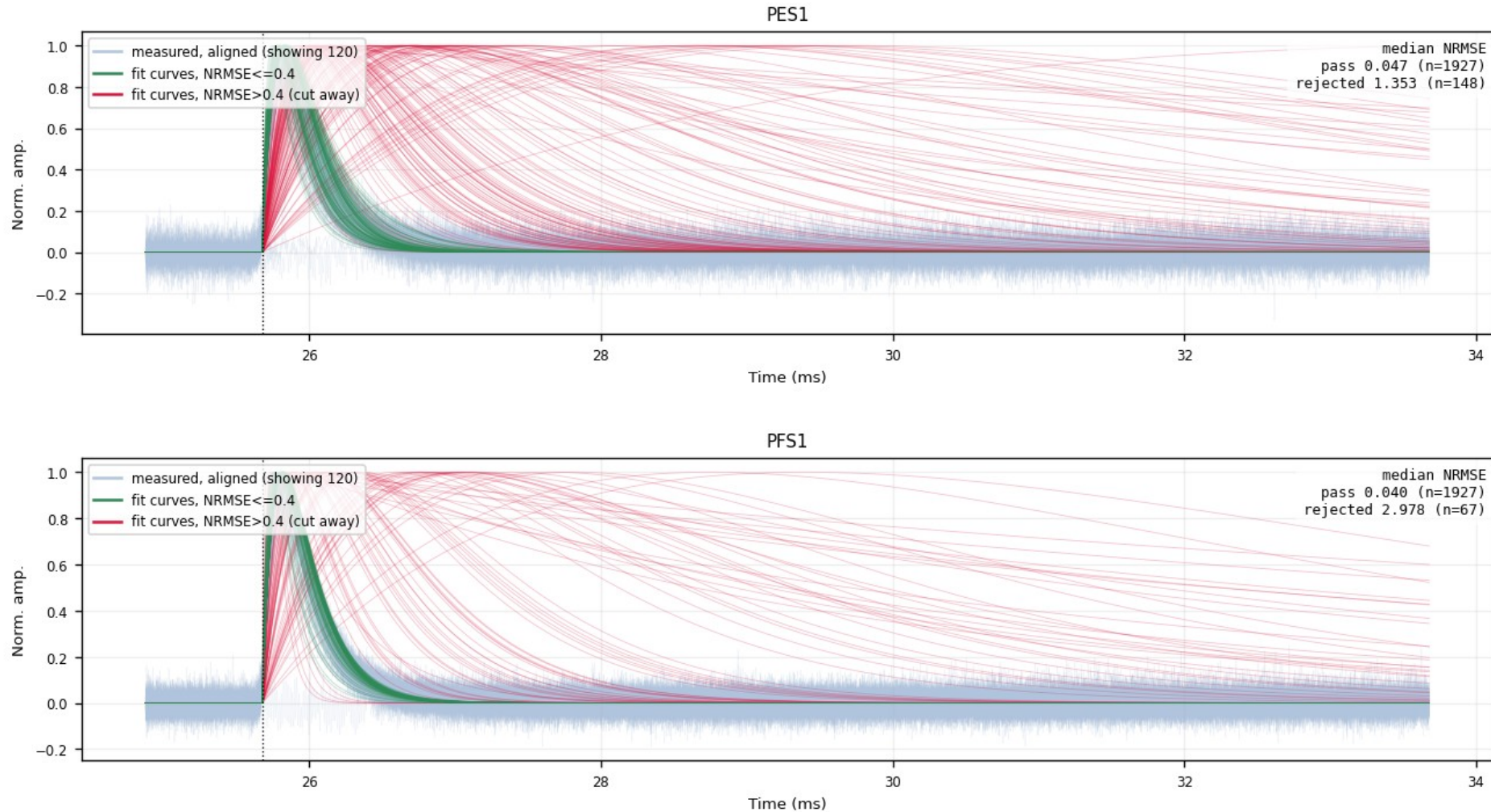
Gray-blue = aligned measured traces; green = fits with $\text{NRMSE} \leq 0.4$ (kept); red = $\text{NRMSE} > 0.4$ (cut away). Median NRMSE and counts of both populations stamped per channel.



Backup · Z7 · overlay_fan_cut: kept vs rejected fits (3/6)

figure: zip7_overlay_fan_cut_nrmse0.4.png, channel panels 5–6 of 11, top to bottom

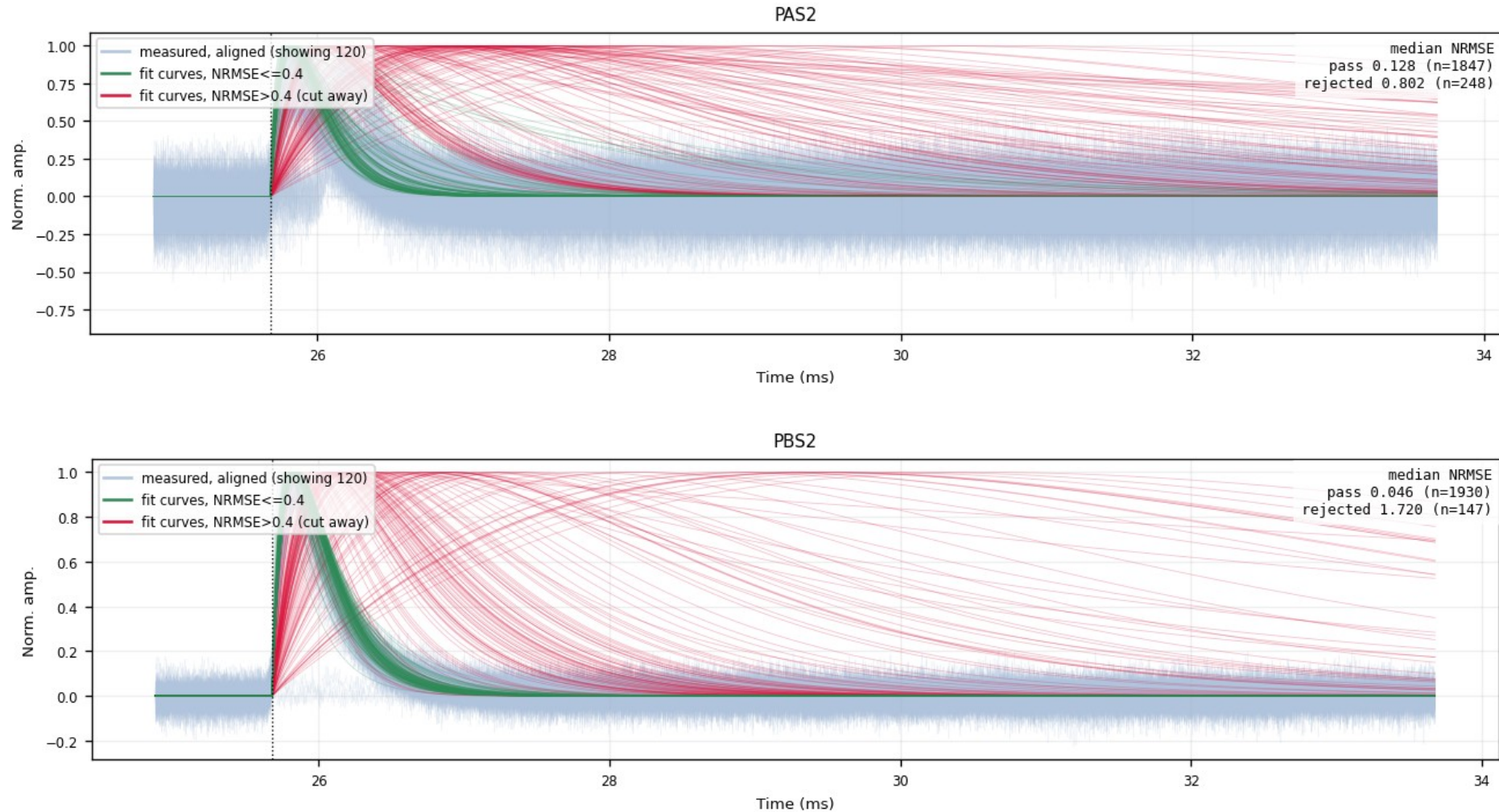
Gray-blue = aligned measured traces; green = fits with $\text{NRMSE} \leq 0.4$ (kept); red = $\text{NRMSE} > 0.4$ (cut away). Median NRMSE and counts of both populations stamped per channel.



Backup · Z7 · overlay_fan_cut: kept vs rejected fits (4/6)

figure: zip7_overlay_fan_cut_nrmse0.4.png, channel panels 7–8 of 11, top to bottom

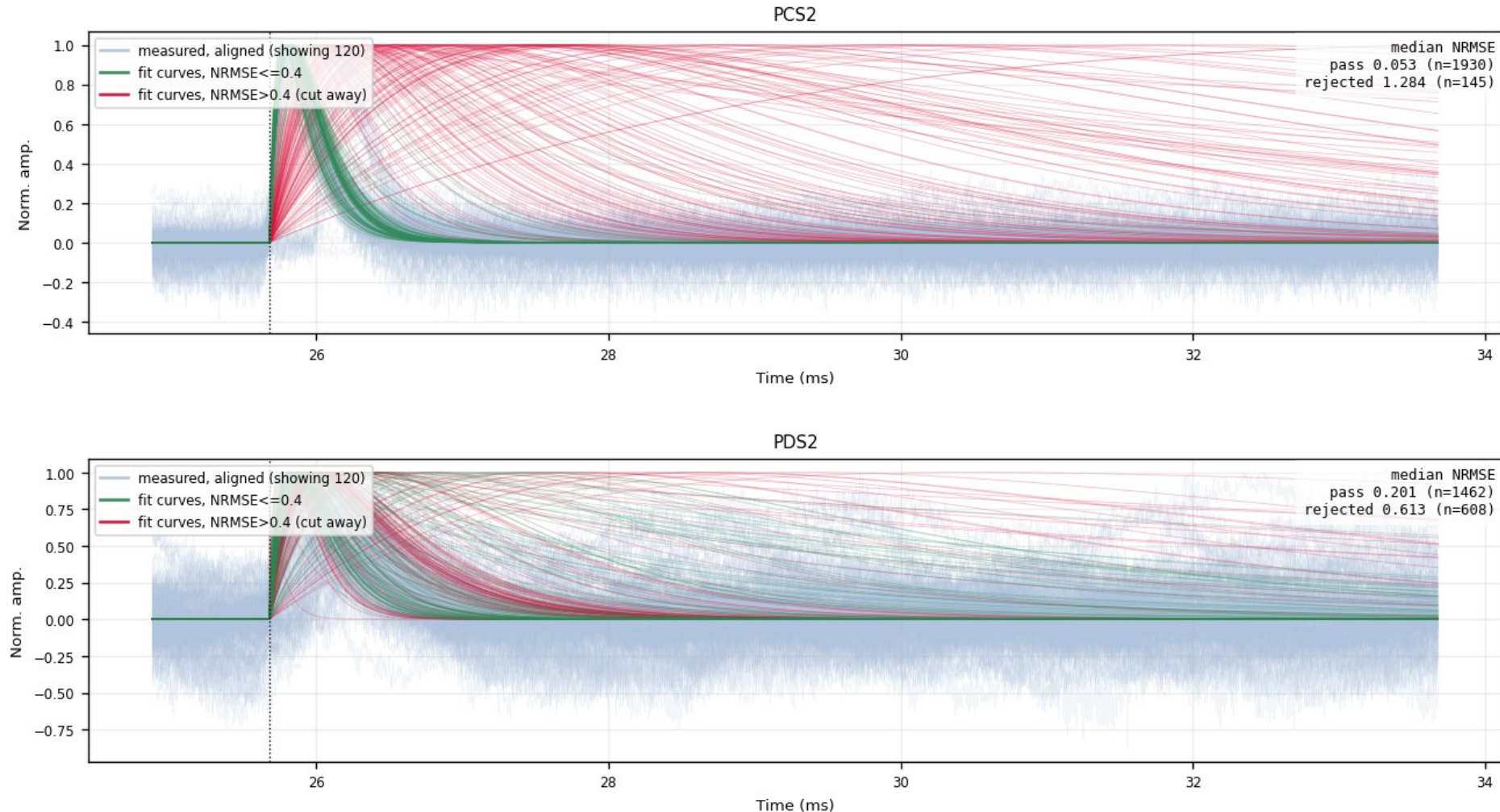
Gray-blue = aligned measured traces; green = fits with $\text{NRMSE} \leq 0.4$ (kept); red = $\text{NRMSE} > 0.4$ (cut away). Median NRMSE and counts of both populations stamped per channel.



Backup · Z7 · overlay_fan_cut: kept vs rejected fits (5/6)

figure: zip7_overlay_fan_cut_nrmse0.4.png, channel panels 9–10 of 11, top to bottom

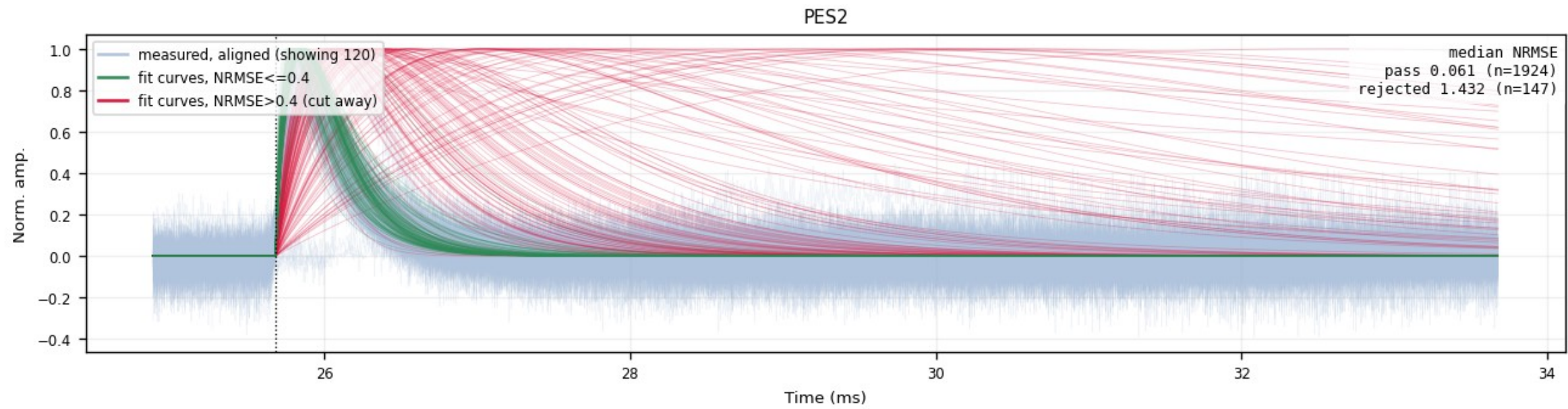
Gray-blue = aligned measured traces; green = fits with $\text{NRMSE} \leq 0.4$ (kept); red = $\text{NRMSE} > 0.4$ (cut away). Median NRMSE and counts of both populations stamped per channel.



Backup · Z7 · overlay_fan_cut: kept vs rejected fits (6/6)

figure: zip7_overlay_fan_cut_nrmse0.4.png, channel panels 11–11 of 11, top to bottom

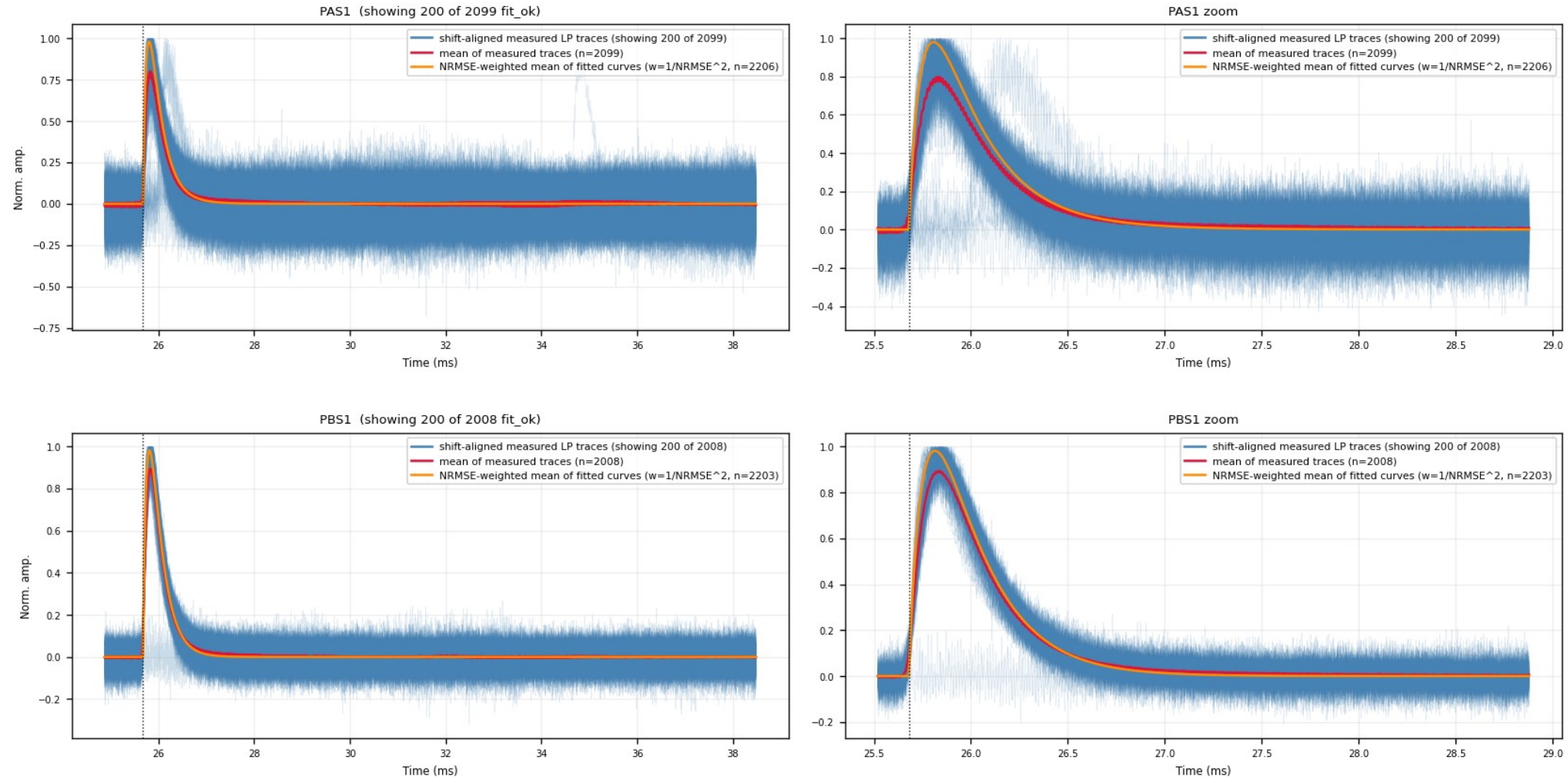
Gray-blue = aligned measured traces; green = fits with $\text{NRMSE} \leq 0.4$ (kept); red = $\text{NRMSE} > 0.4$ (cut away). Median NRMSE and counts of both populations stamped per channel.



Backup · Z7 · aligned_overlay: aligned measured traces (1/6)

figure: zip7_lp_aligned_overlay.png, channel panels 1–2 of 11, top to bottom

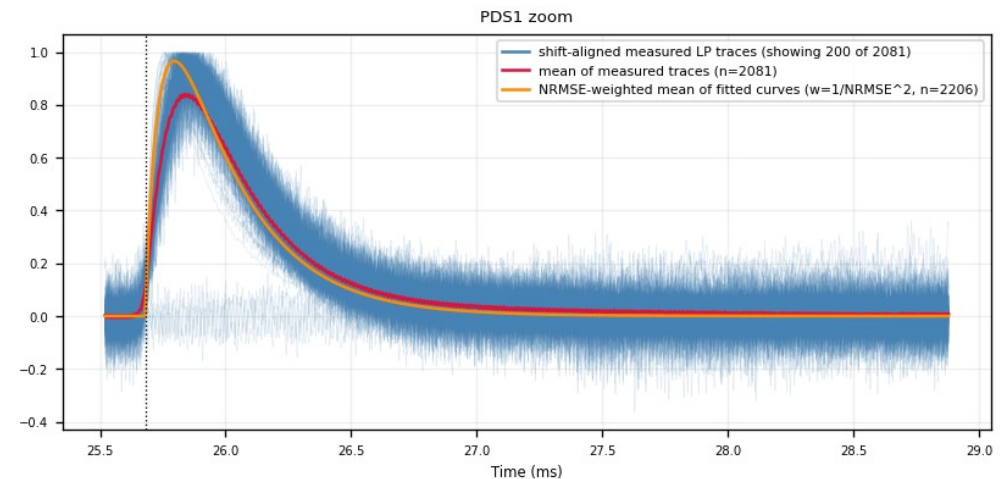
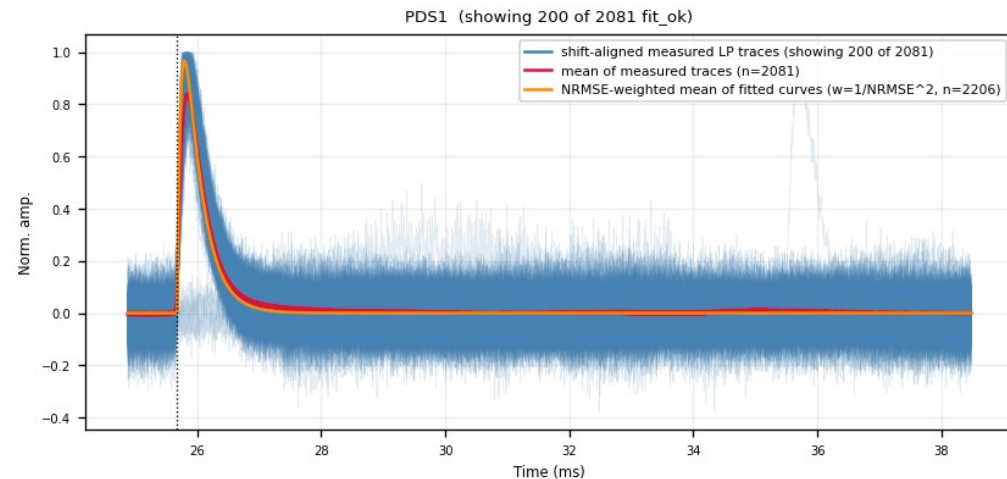
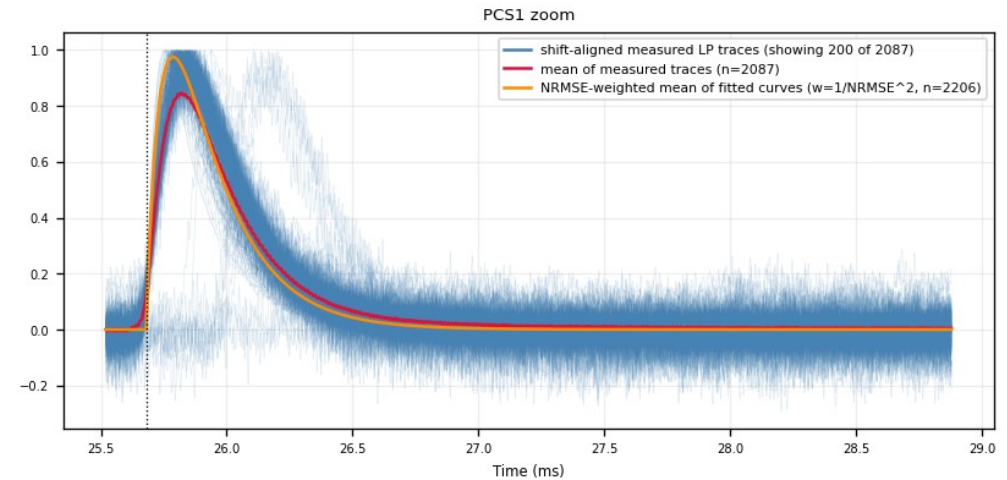
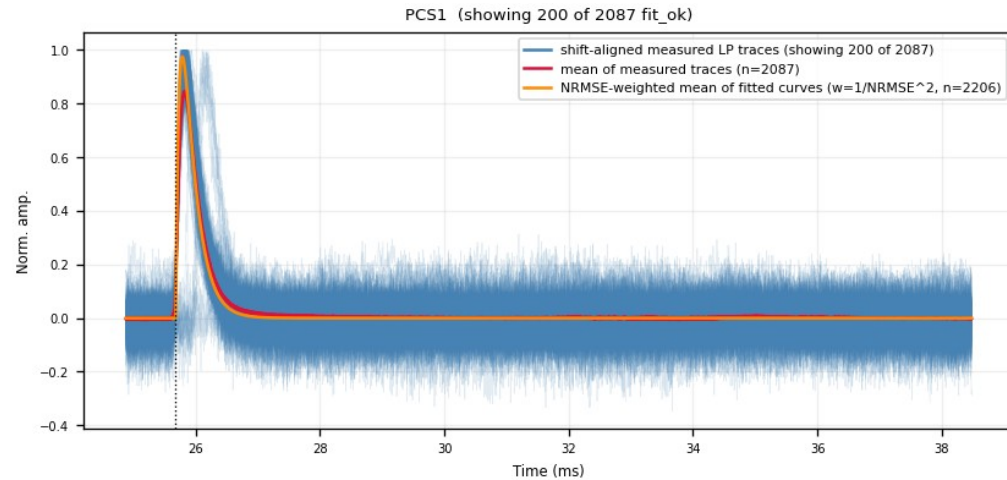
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Backup · Z7 · aligned_overlay: aligned measured traces (2/6)

figure: zip7_lp_aligned_overlay.png, channel panels 3–4 of 11, top to bottom

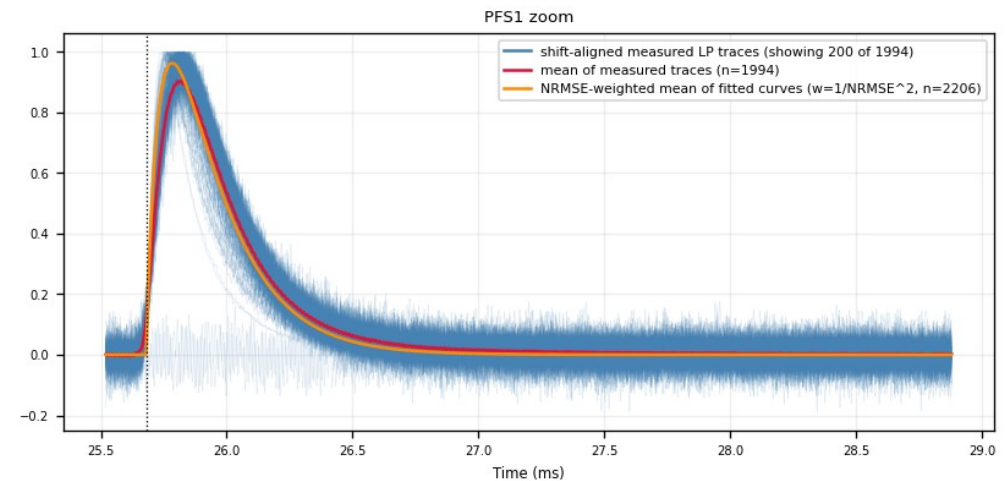
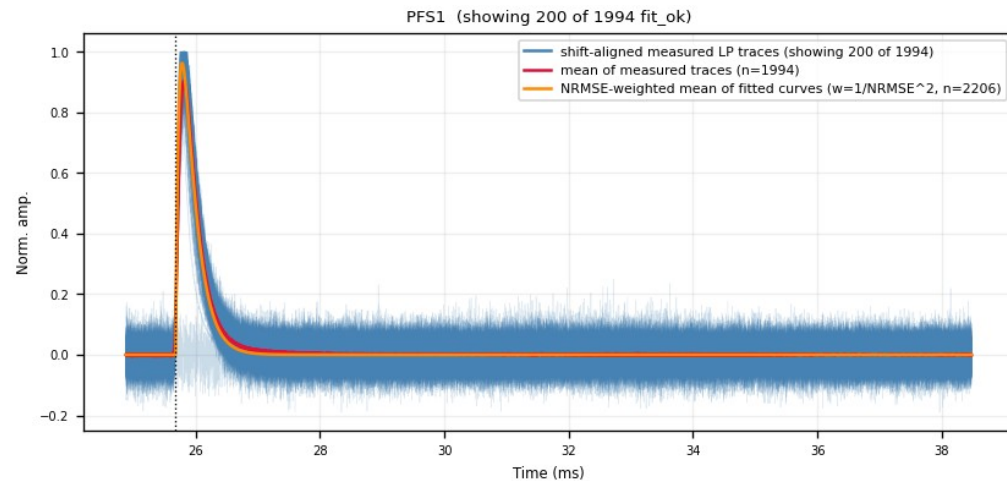
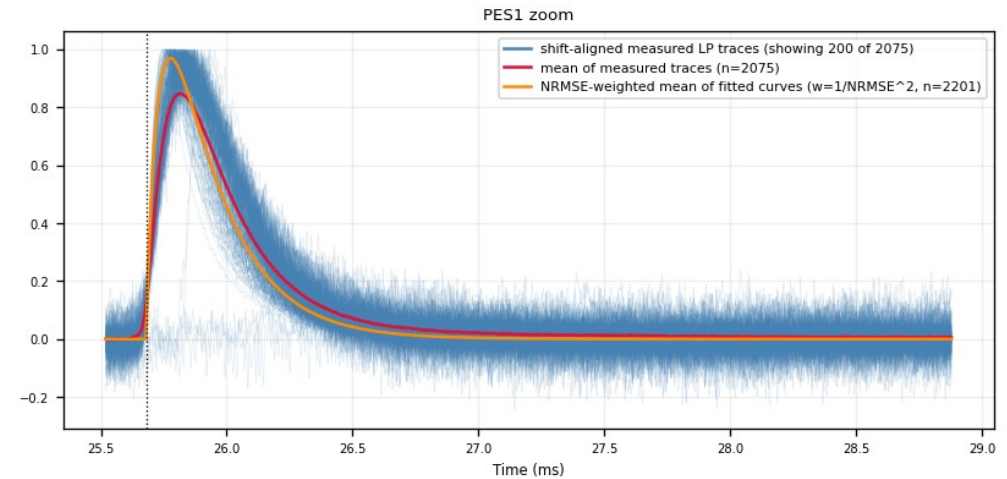
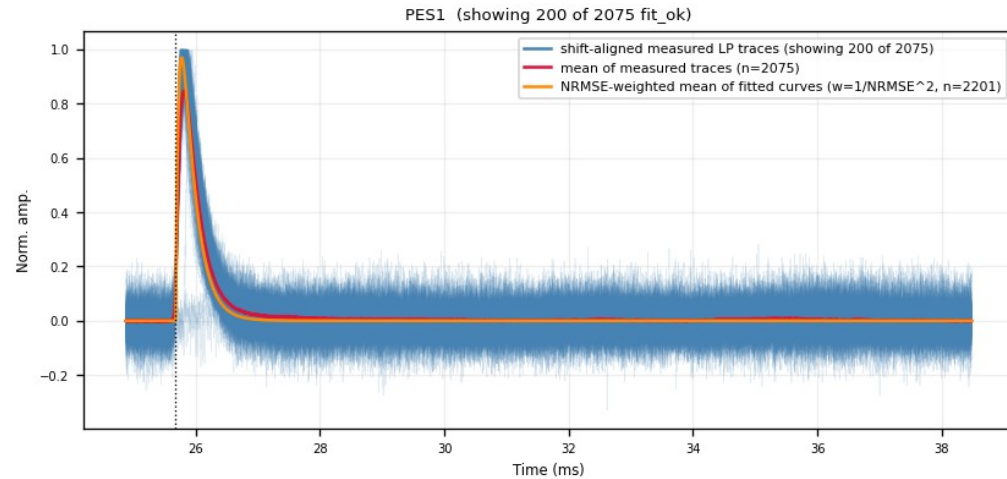
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Backup · Z7 · aligned_overlay: aligned measured traces (3/6)

figure: zip7_lp_aligned_overlay.png, channel panels 5–6 of 11, top to bottom

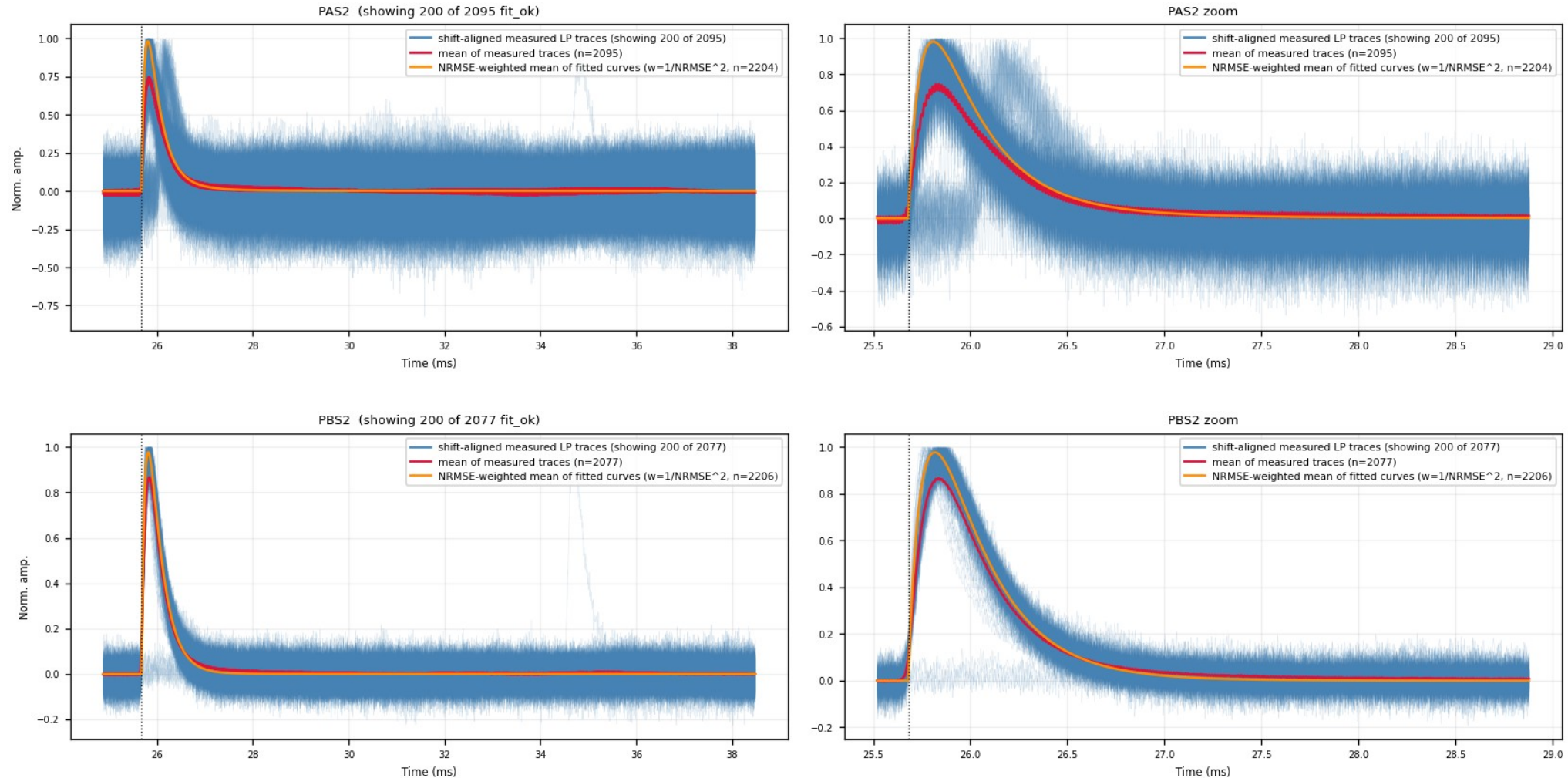
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Backup · Z7 · aligned_overlay: aligned measured traces (4/6)

figure: zip7_lp_aligned_overlay.png, channel panels 7–8 of 11, top to bottom

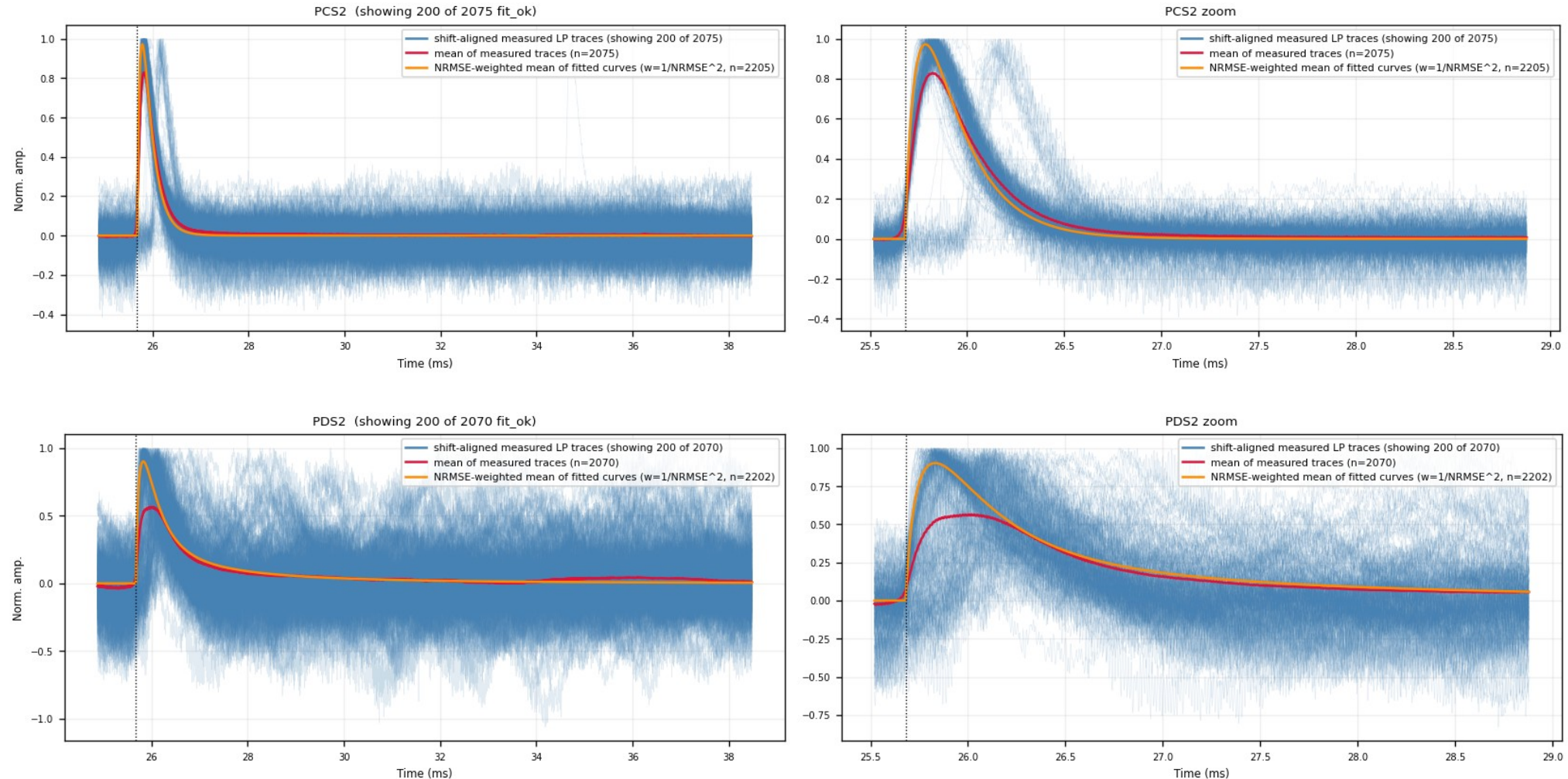
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Backup · Z7 · aligned_overlay: aligned measured traces (5/6)

figure: zip7_lp_aligned_overlay.png, channel panels 9–10 of 11, top to bottom

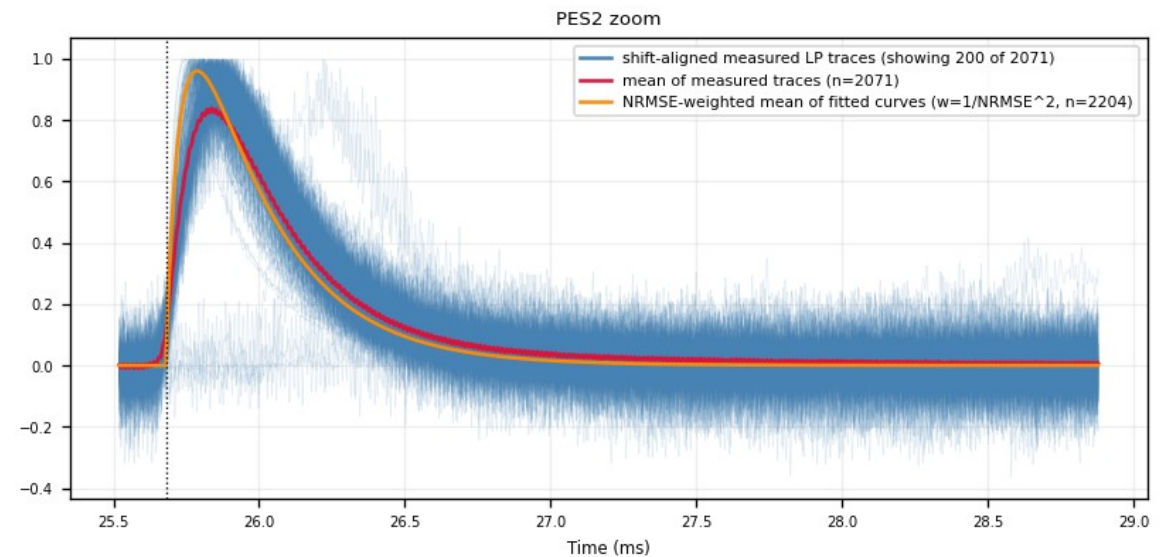
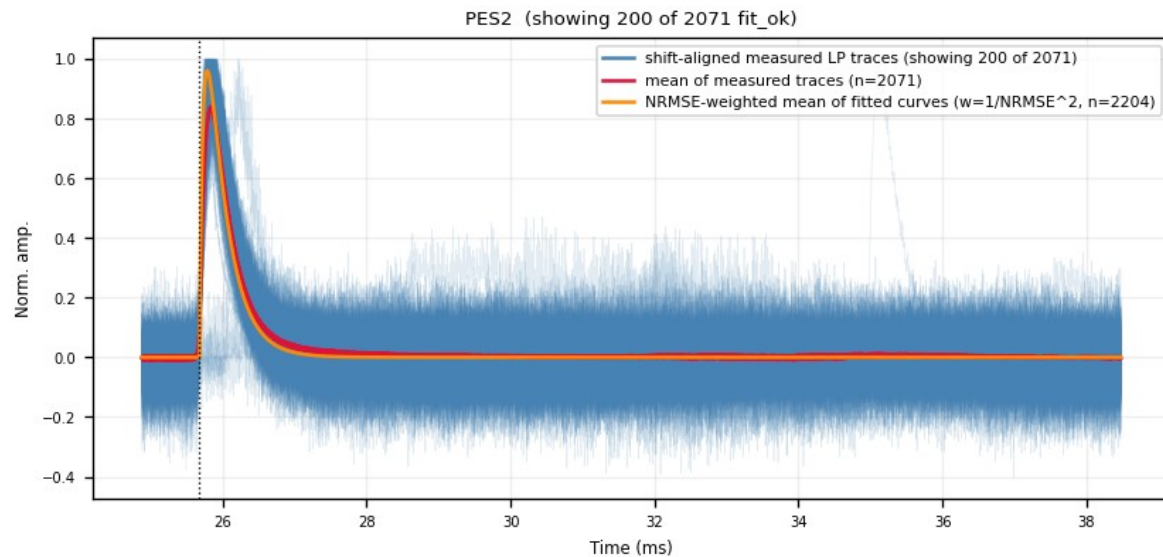
Blue = measured LP traces shifted by (fitted pretrigger – 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Backup · Z7 · aligned_overlay: aligned measured traces (6/6)

figure: zip7_lp_aligned_overlay.png, channel panels 11–11 of 11, top to bottom

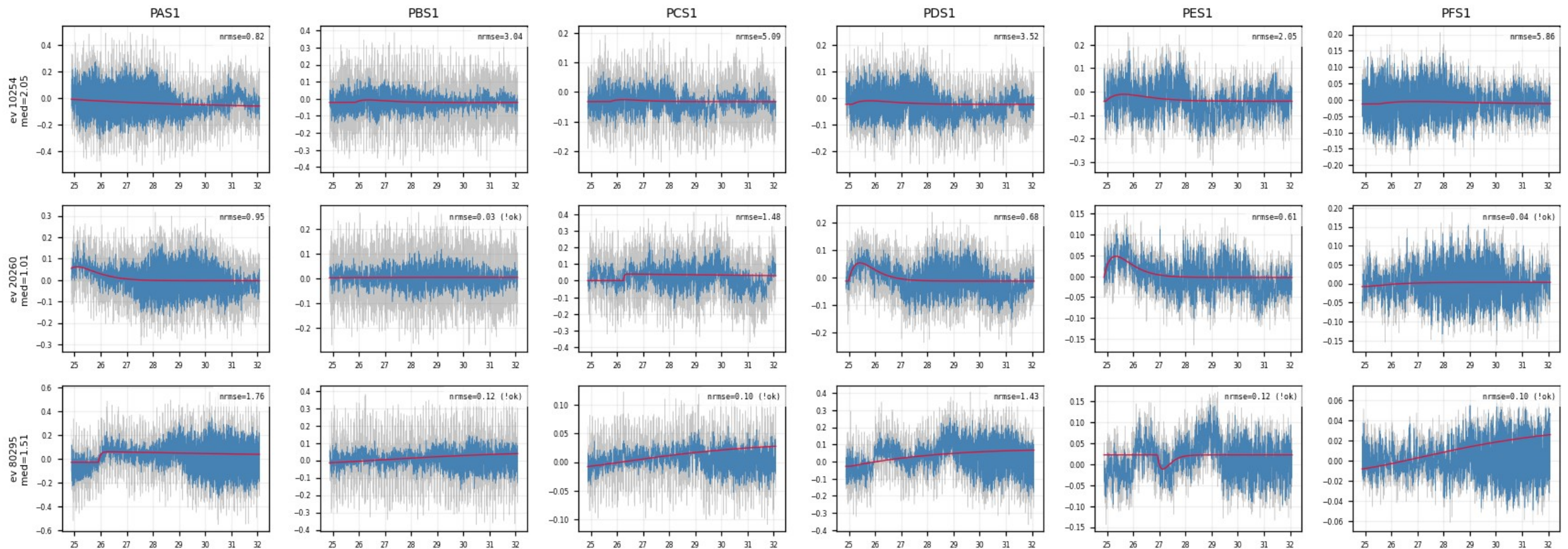
Blue = measured LP traces shifted by (fitted pretrigger - 16050); red = mean of all fit_ok events; orange = NRMSE-weighted mean of the fitted curves. Rise edges line up at 16050.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (1/10)

figure: zip7_slow_rise_events.png, event rows 1–3 of 15, left half of the channels

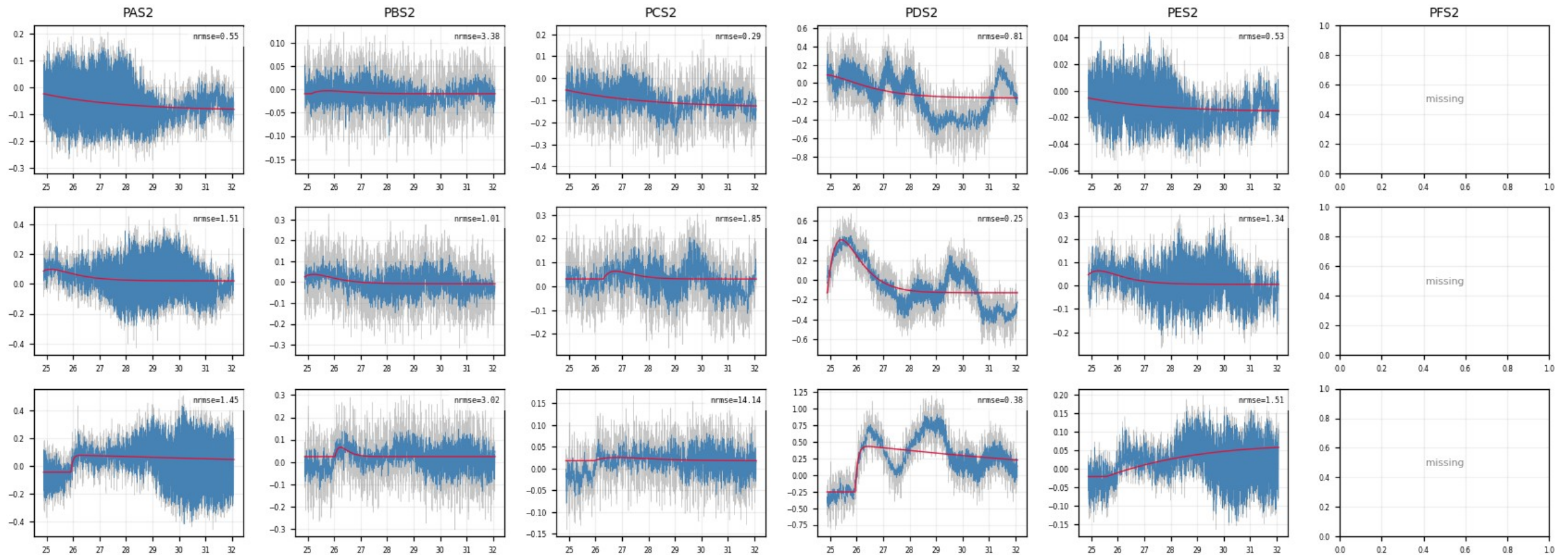
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (2/10)

figure: zip7_slow_rise_events.png, event rows 1–3 of 15, right half of the channels

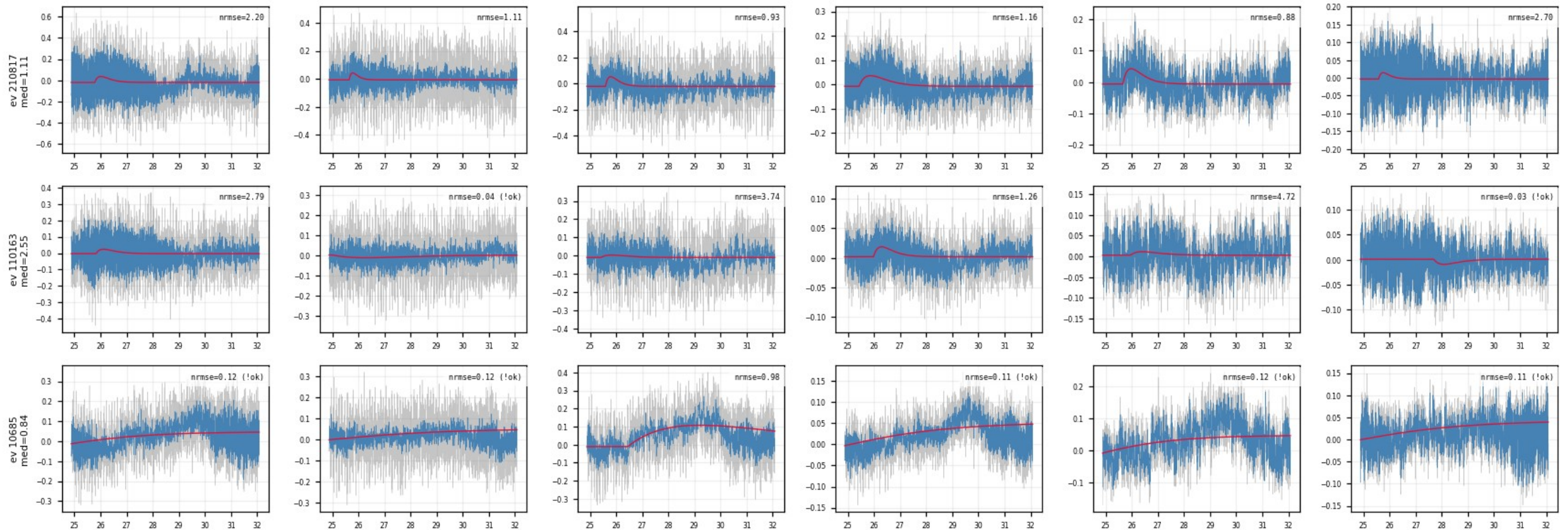
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (3/10)

figure: zip7_slow_rise_events.png, event rows 4–6 of 15, left half of the channels

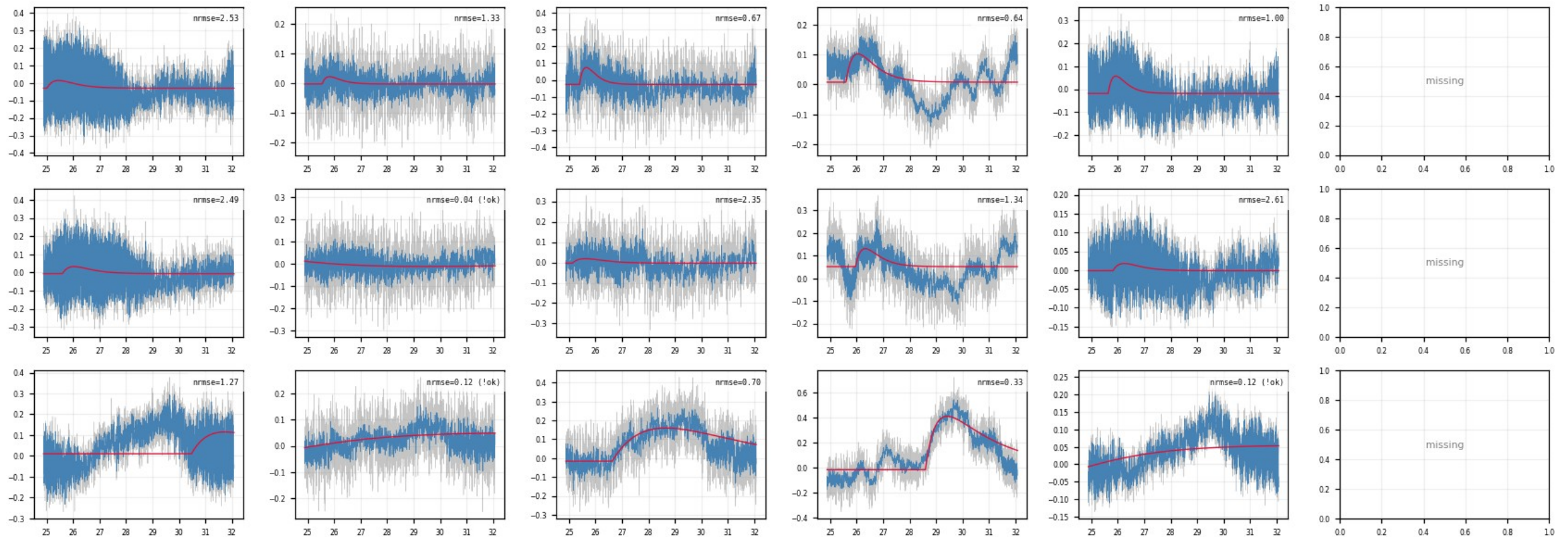
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (4/10)

figure: zip7_slow_rise_events.png, event rows 4–6 of 15, right half of the channels

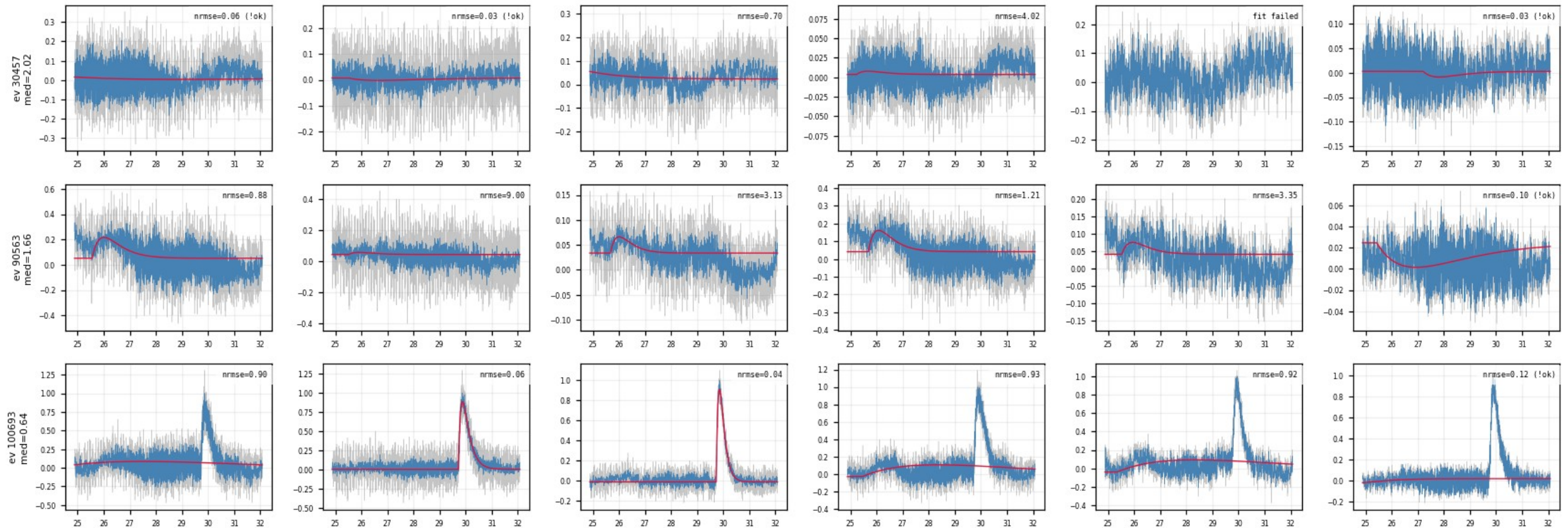
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (5/10)

figure: zip7_slow_rise_events.png, event rows 7–9 of 15, left half of the channels

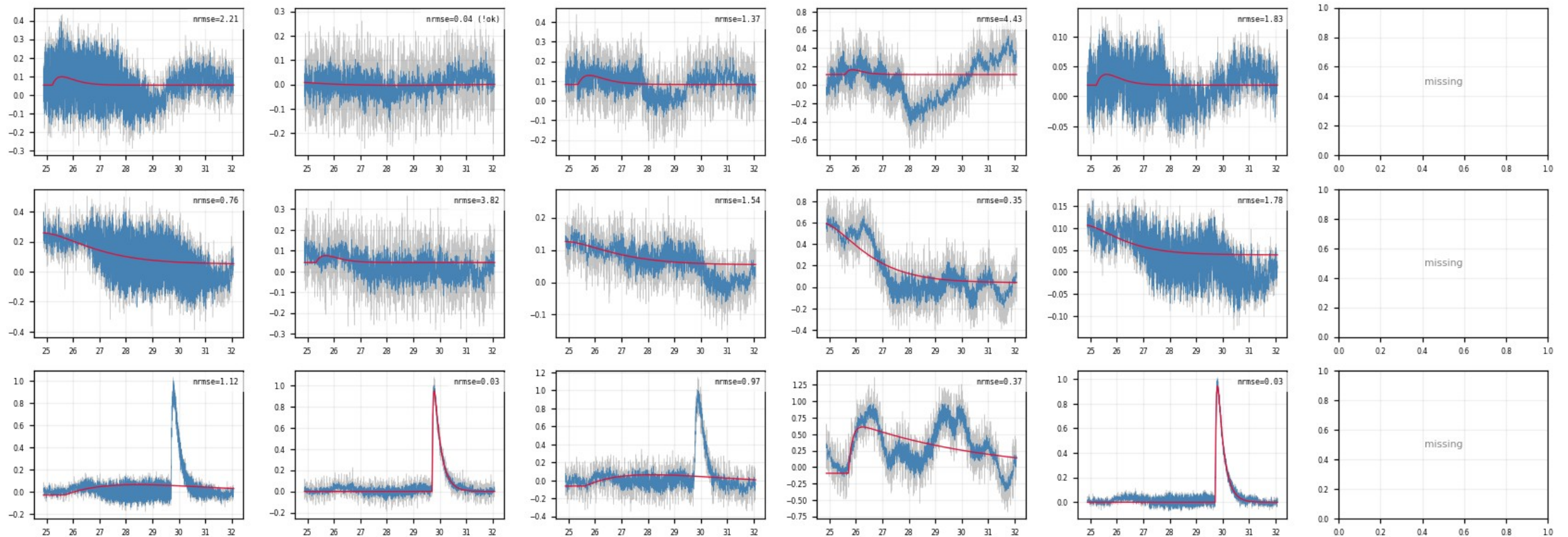
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (6/10)

figure: zip7_slow_rise_events.png, event rows 7–9 of 15, right half of the channels

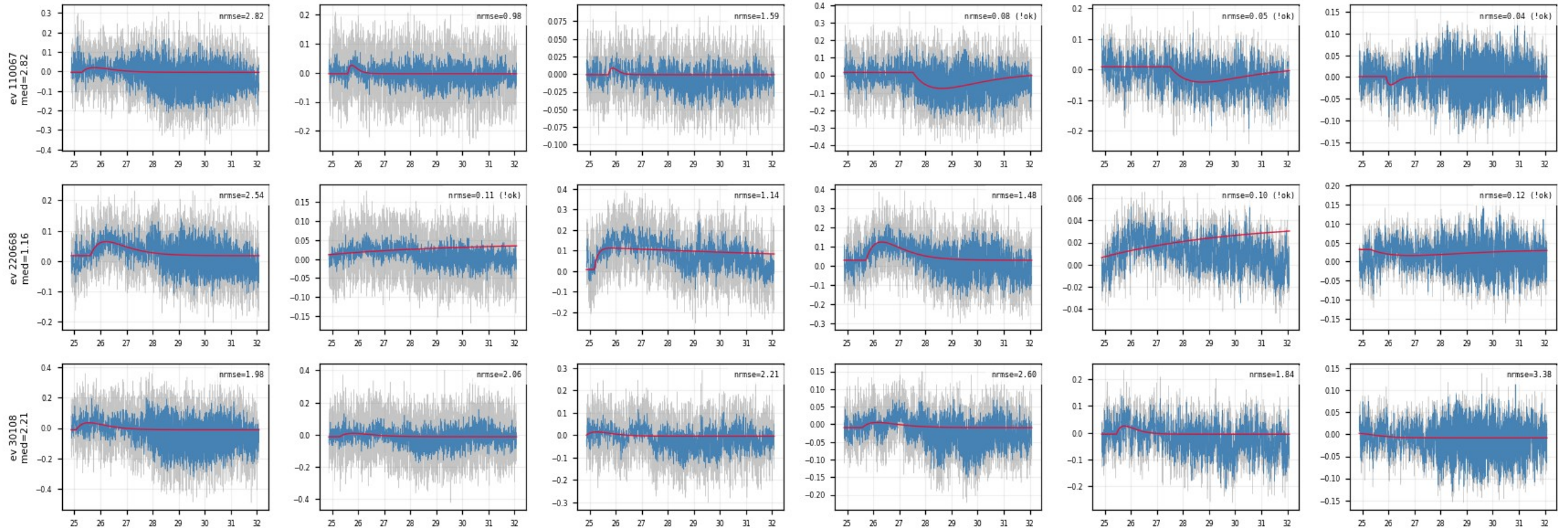
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (7/10)

figure: zip7_slow_rise_events.png, event rows 10–12 of 15, left half of the channels

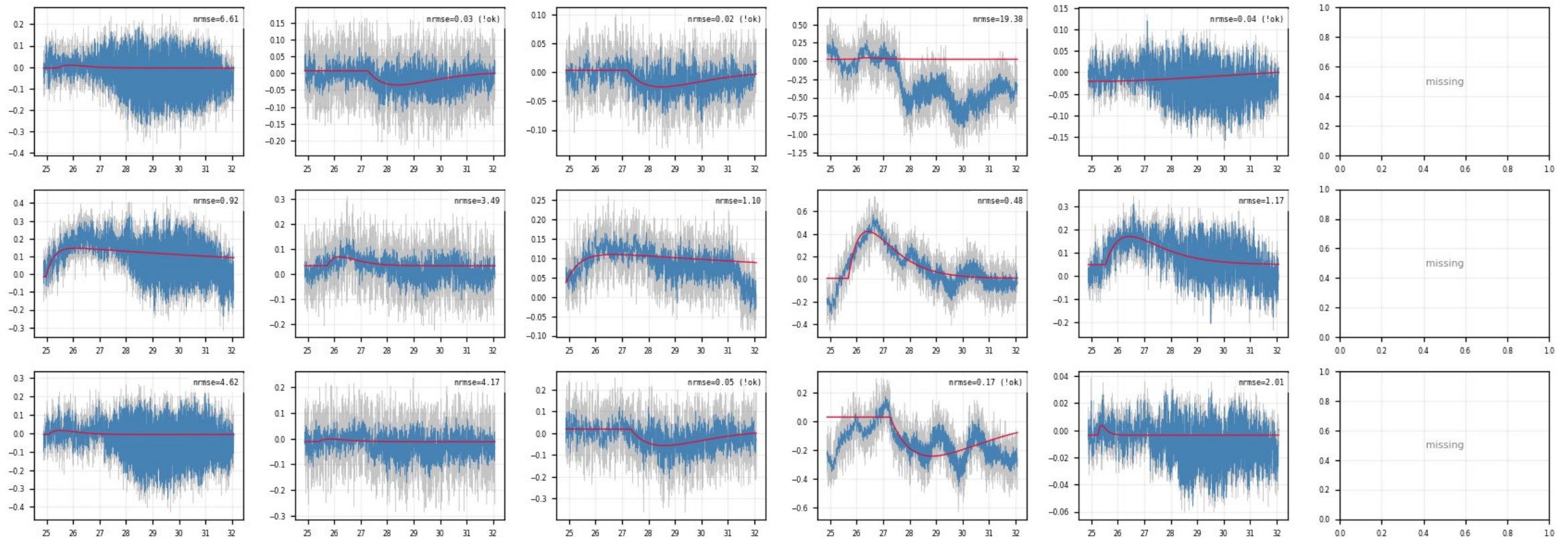
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (8/10)

figure: zip7_slow_rise_events.png, event rows 10–12 of 15, right half of the channels

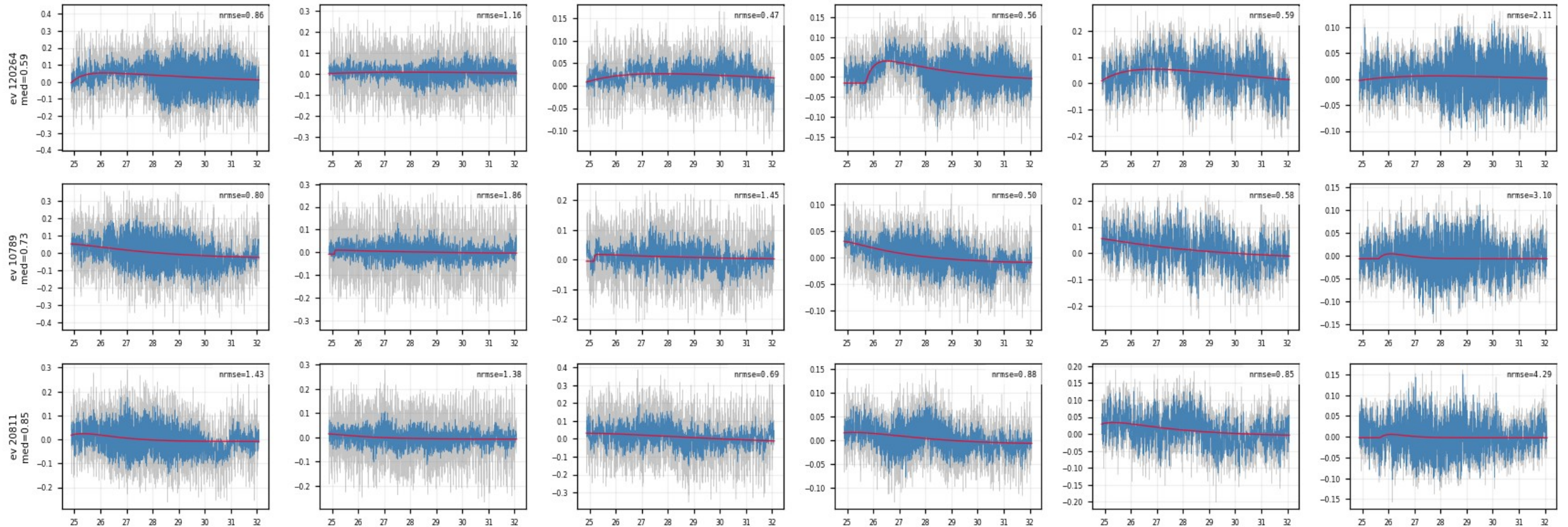
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (9/10)

figure: zip7_slow_rise_events.png, event rows 13–15 of 15, left half of the channels

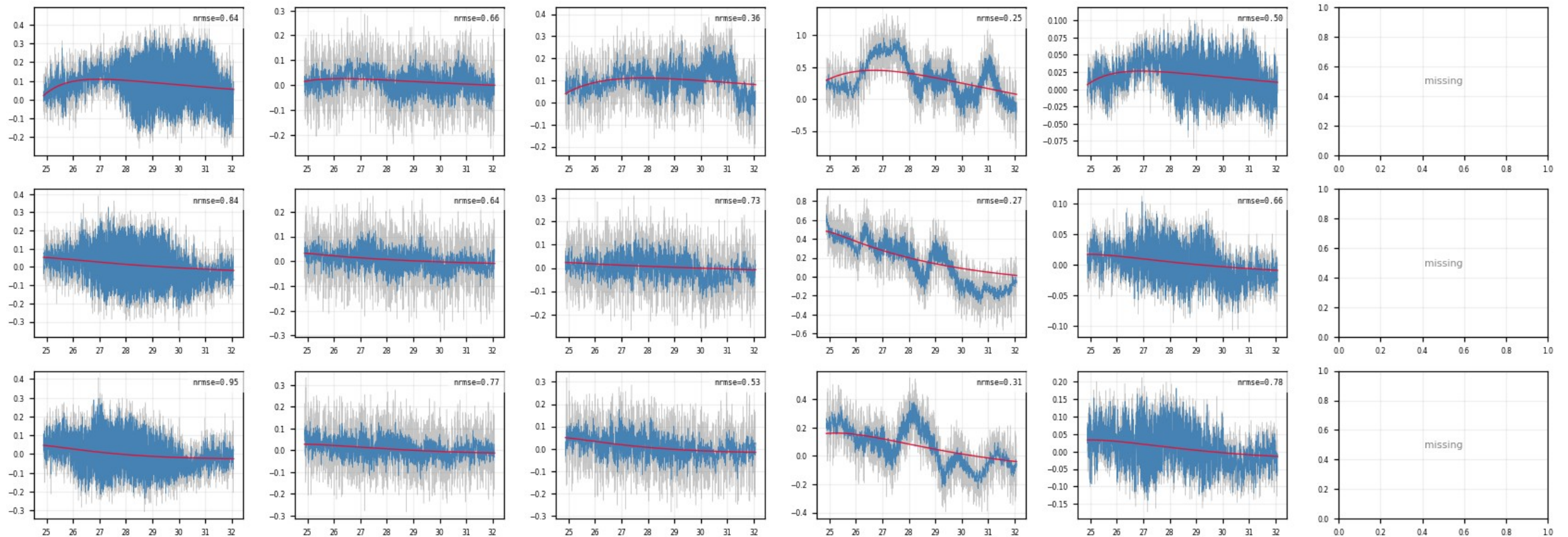
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · slow_rise_events: NRMSE-rejected events (10/10)

figure: zip7_slow_rise_events.png, event rows 13–15 of 15, right half of the channels

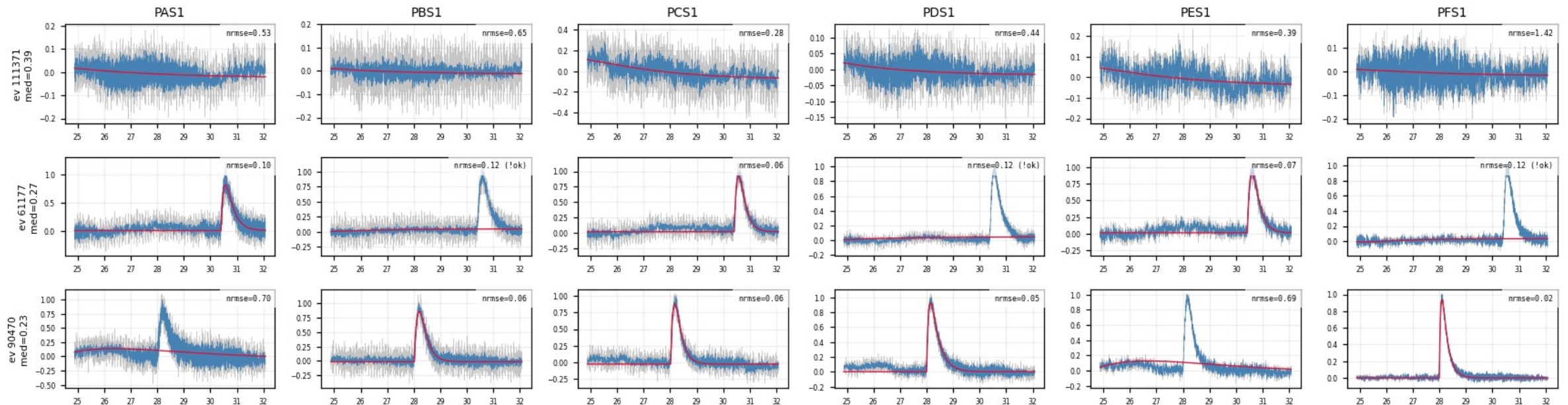
Events with median NRMSE > 0.4 across channels: the raw traces show no pulse in any channel: noise triggers, not physics.



Backup · Z7 · shadow_events: well-fit slow-rise events (1/2)

figure: zip7_shadow_events.png, event rows 1–3 of 3, left half of the channels

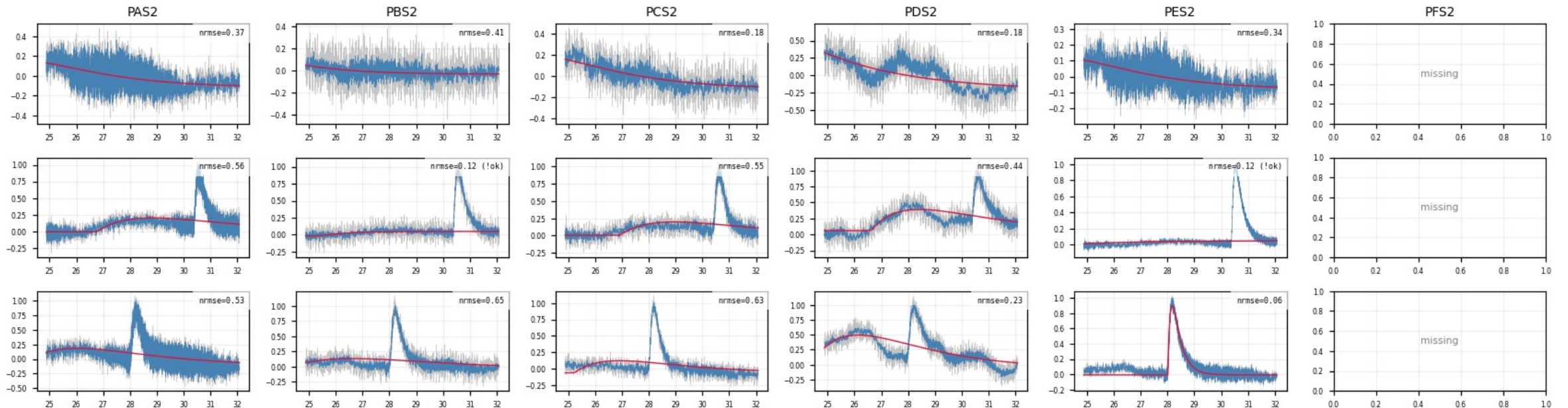
Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} > 0.2$ ms: genuinely slow, well-fit pulses; kept, and exactly what the NxM multi-templates are for.



Backup · Z7 · shadow_events: well-fit slow-rise events (2/2)

figure: zip7_shadow_events.png, event rows 1–3 of 3, right half of the channels

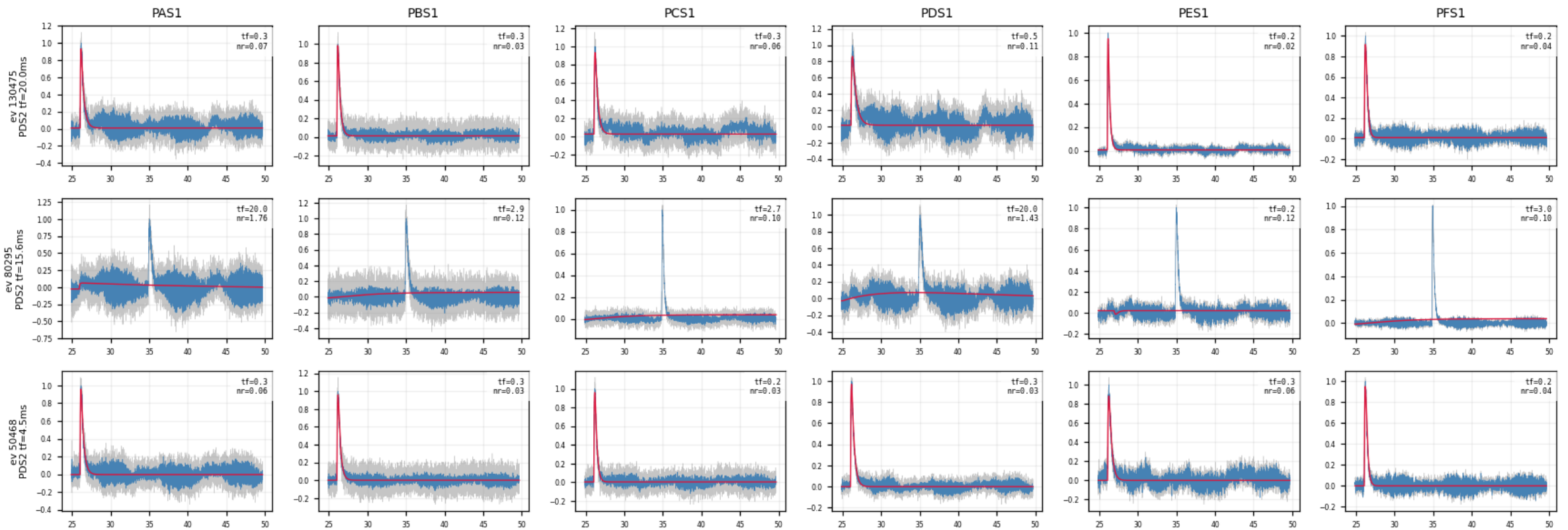
Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} > 0.2$ ms: genuinely slow, well-fit pulses; kept, and exactly what the NxM multi-templates are for.



Backup · Z7 · slow_fall_events: long- τ _fall study (PDS2) (1/8)

figure: zip7_PDS2_slow_fall.png, event rows 1–3 of 10, left half of the channels

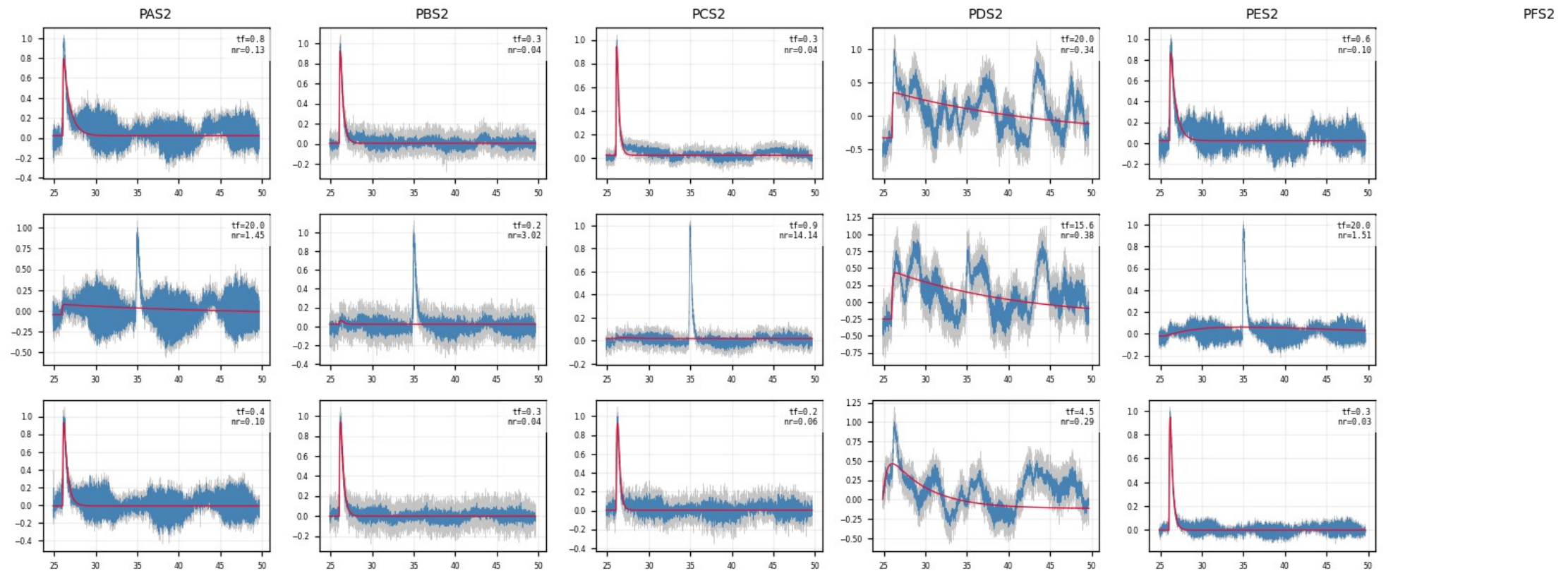
10 random events with $\tau_{\text{fall}} > 1.5$ ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · slow_fall_events: long- τ _fall study (PDS2) (2/8)

figure: zip7_PDS2_slow_fall.png, event rows 1–3 of 10, right half of the channels

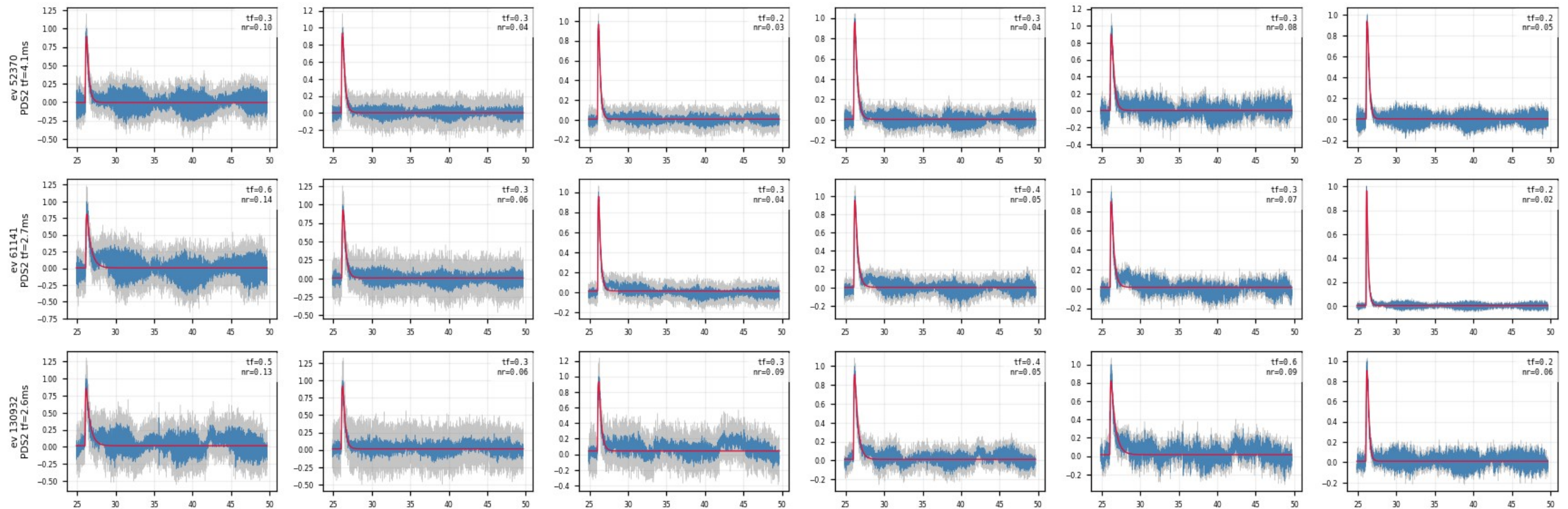
10 random events with $\tau_{\text{fall}} > 1.5$ ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · slow_fall_events: long- τ _fall study (PDS2) (3/8)

figure: zip7_PDS2_slow_fall.png, event rows 4–6 of 10, left half of the channels

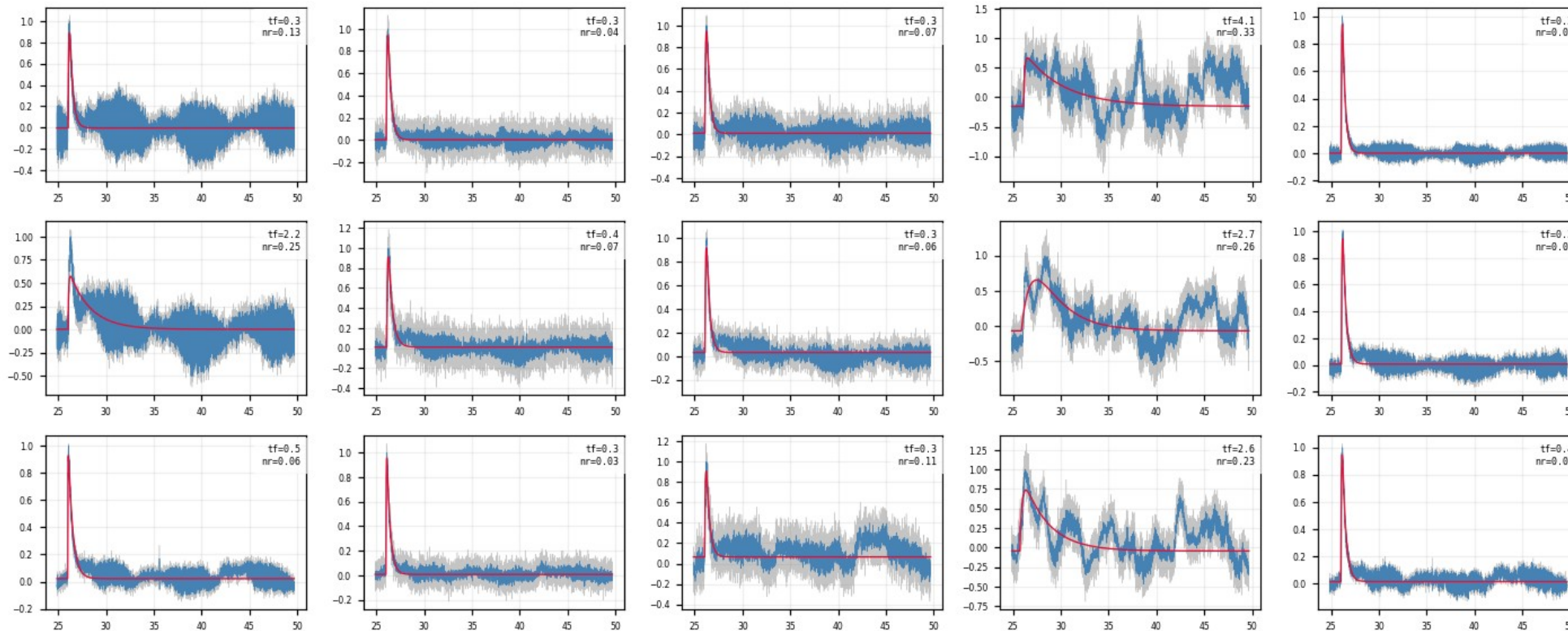
10 random events with $\tau_{\text{fall}} > 1.5$ ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · slow_fall_events: long- τ _fall study (PDS2) (4/8)

figure: zip7_PDS2_slow_fall.png, event rows 4–6 of 10, right half of the channels

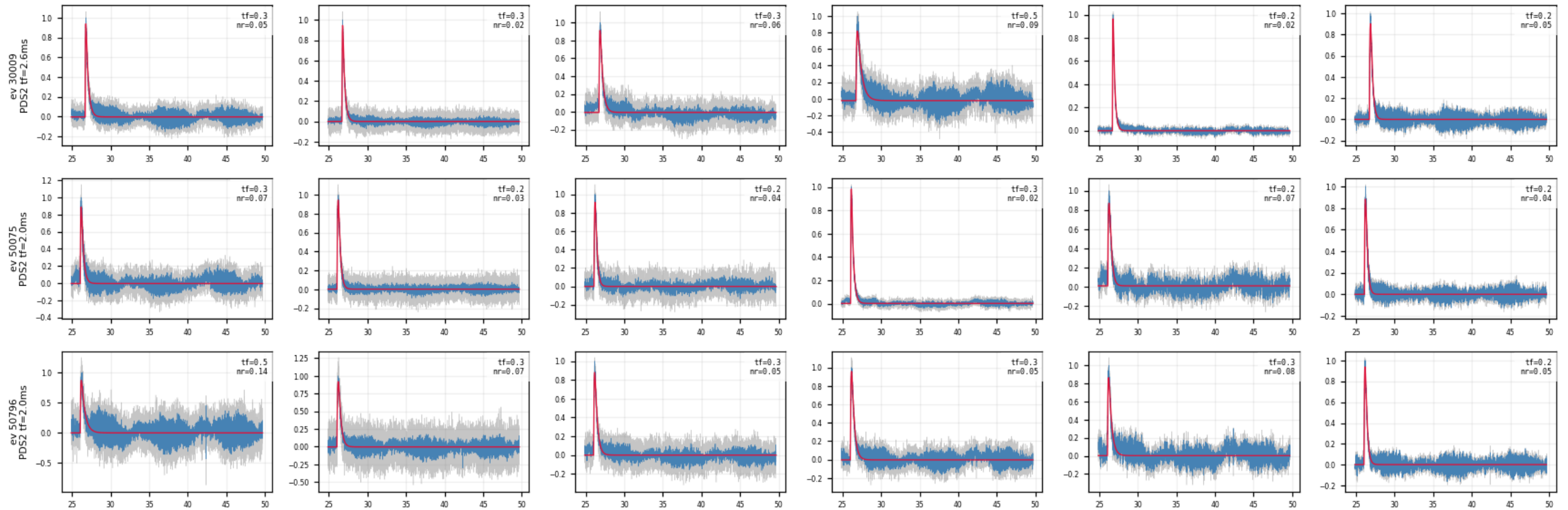
10 random events with $\tau_{\text{fall}} > 1.5$ ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · slow_fall_events: long- τ _fall study (PDS2) (5/8)

figure: zip7_PDS2_slow_fall.png, event rows 7–9 of 10, left half of the channels

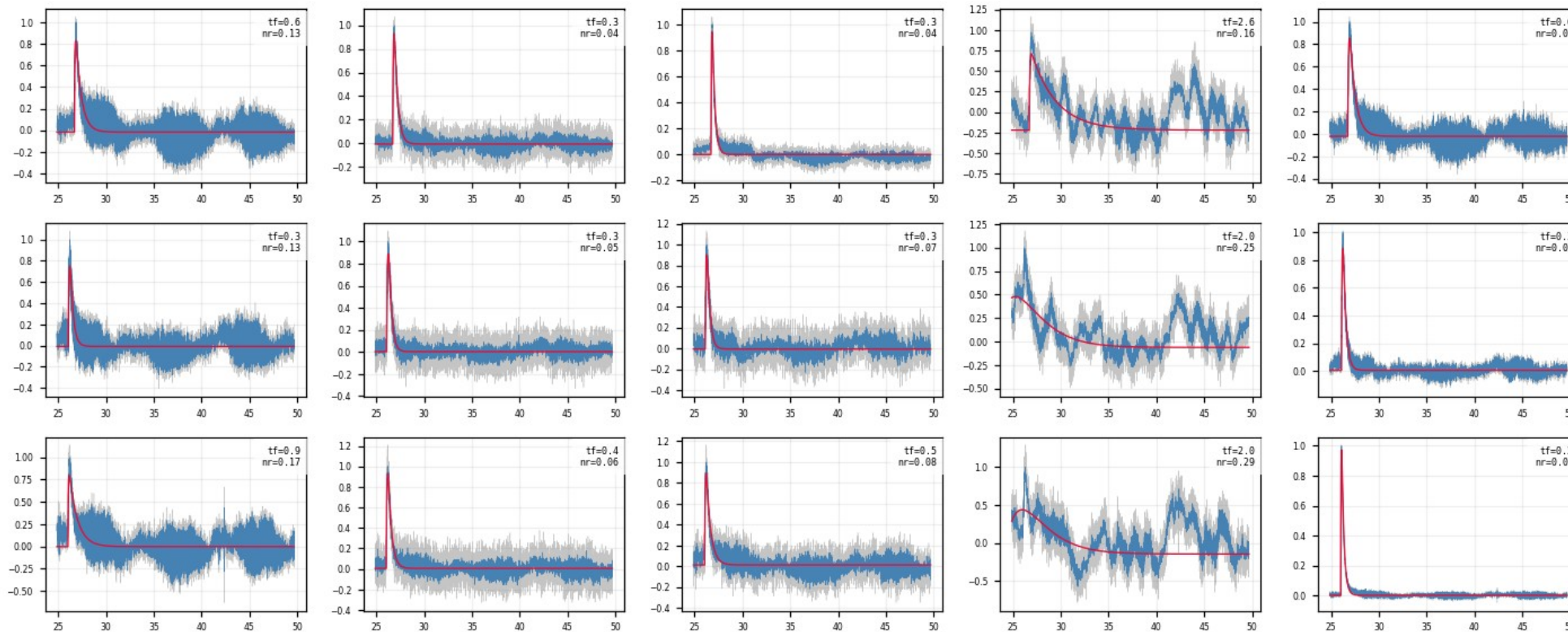
10 random events with $\tau_{\text{fall}} > 1.5$ ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · slow_fall_events: long- τ _fall study (PDS2) (6/8)

figure: zip7_PDS2_slow_fall.png, event rows 7–9 of 10, right half of the channels

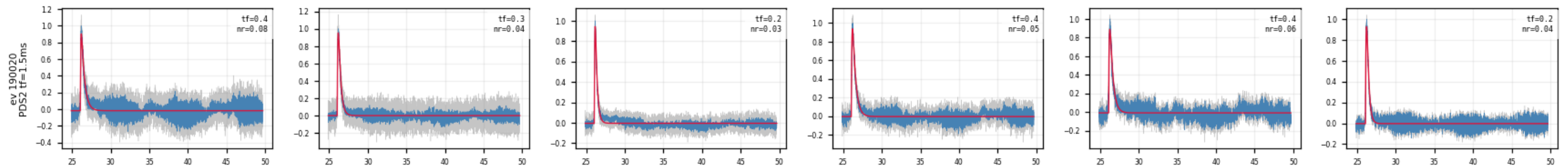
10 random events with $\tau_{\text{fall}} > 1.5$ ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · slow_fall_events: long- τ _fall study (PDS2) (7/8)

figure: zip7_PDS2_slow_fall.png, event rows 10–10 of 10, left half of the channels

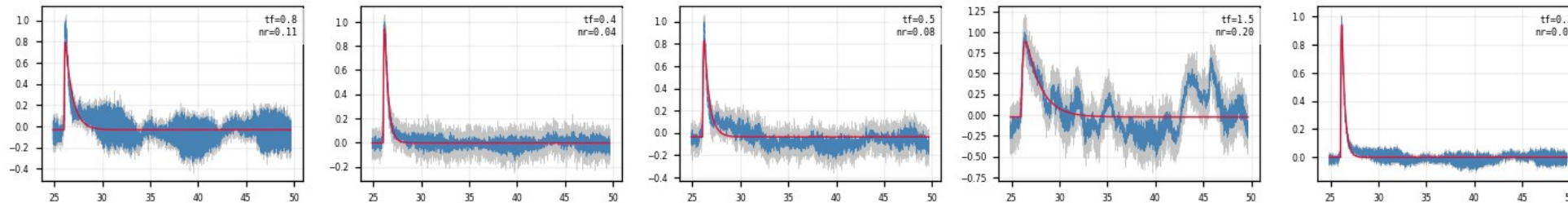
10 random events with $\tau_{\text{fall}} > 1.5$ ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · slow_fall_events: long- τ _fall study (PDS2) (8/8)

figure: zip7_PDS2_slow_fall.png, event rows 10–10 of 10, right half of the channels

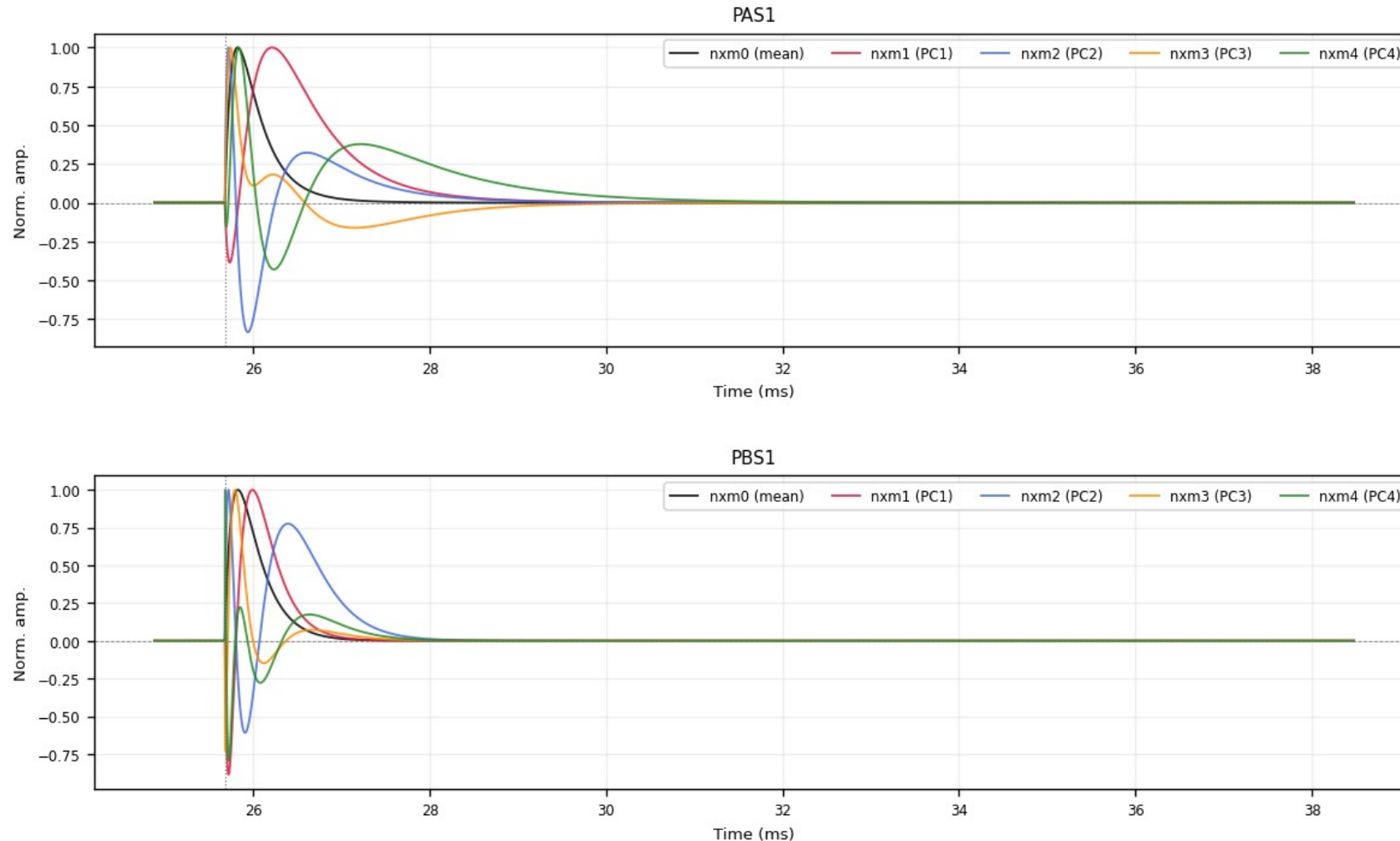
10 random events with $\tau_{\text{fall}} > 1.5$ ms drawn in every channel: 11 channels show normal fast pulses; the long fall is the PDS2 low-frequency hardware noise, not slow physics.



Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (1/6)

figure: zip7_pca_templates.png, channel panels 1–2 of 11, top to bottom

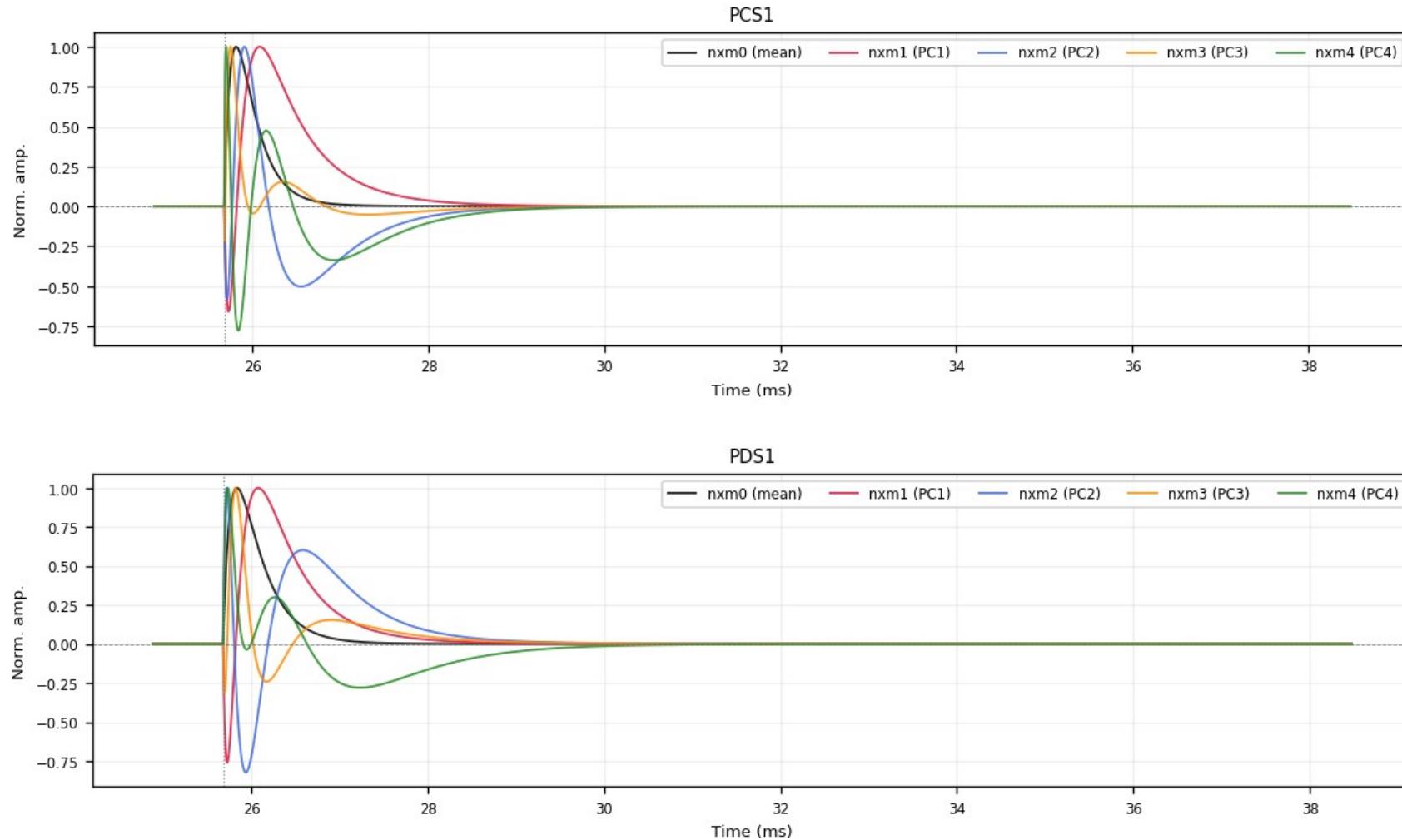
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.



Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (2/6)

figure: zip7_pca_templates.png, channel panels 3–4 of 11, top to bottom

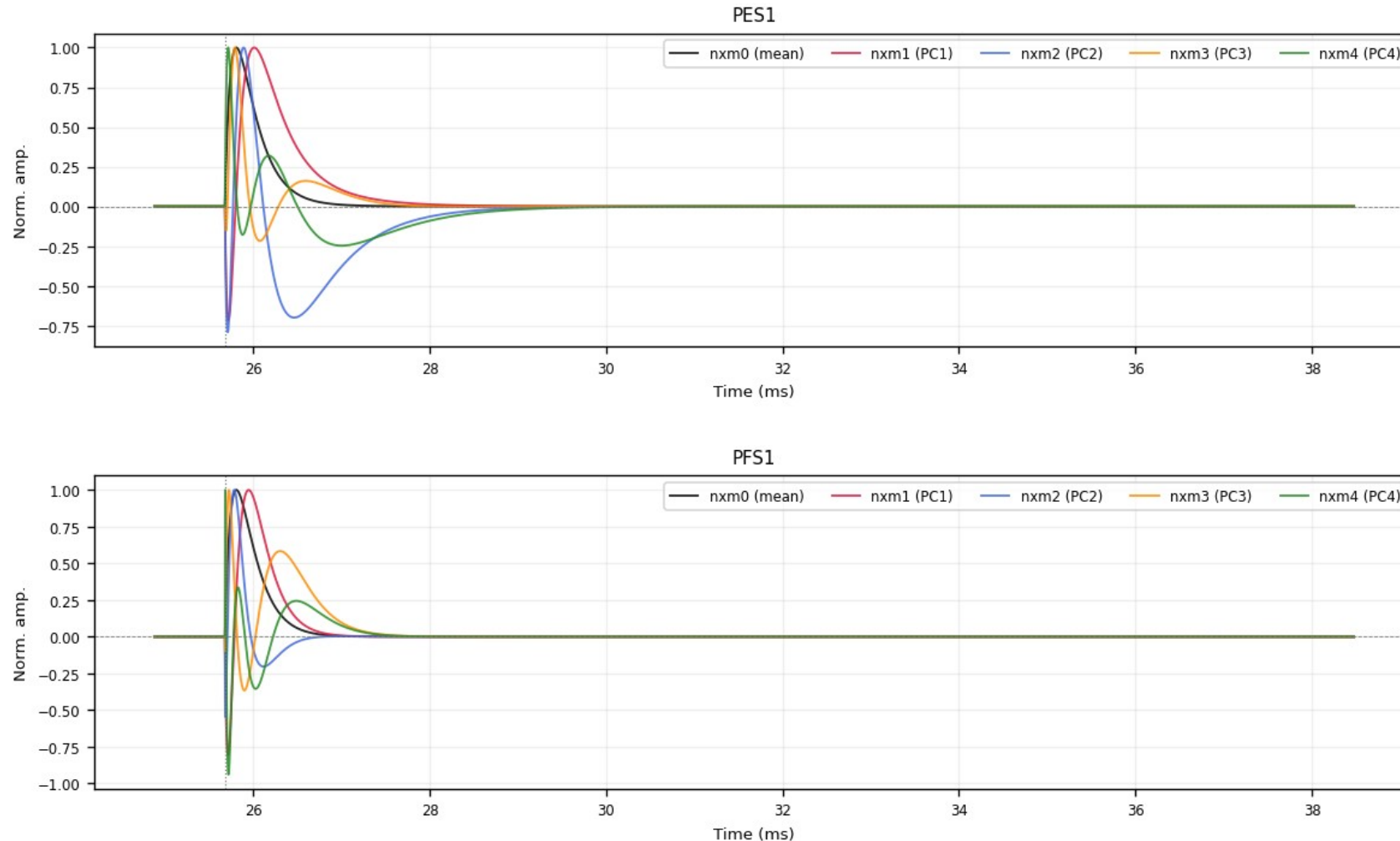
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.



Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (3/6)

figure: zip7_pca_templates.png, channel panels 5–6 of 11, top to bottom

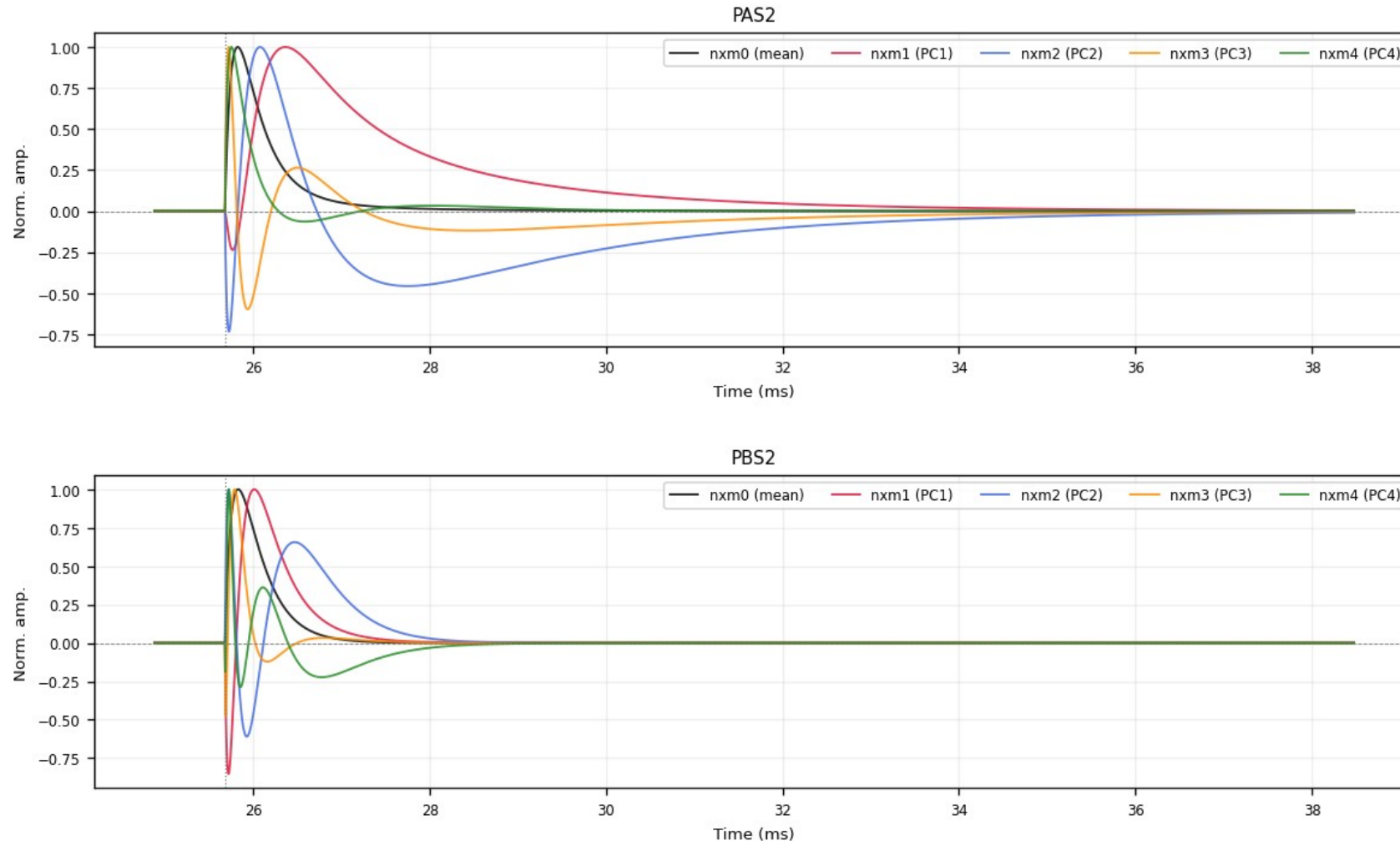
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.



Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (4/6)

figure: zip7_pca_templates.png, channel panels 7–8 of 11, top to bottom

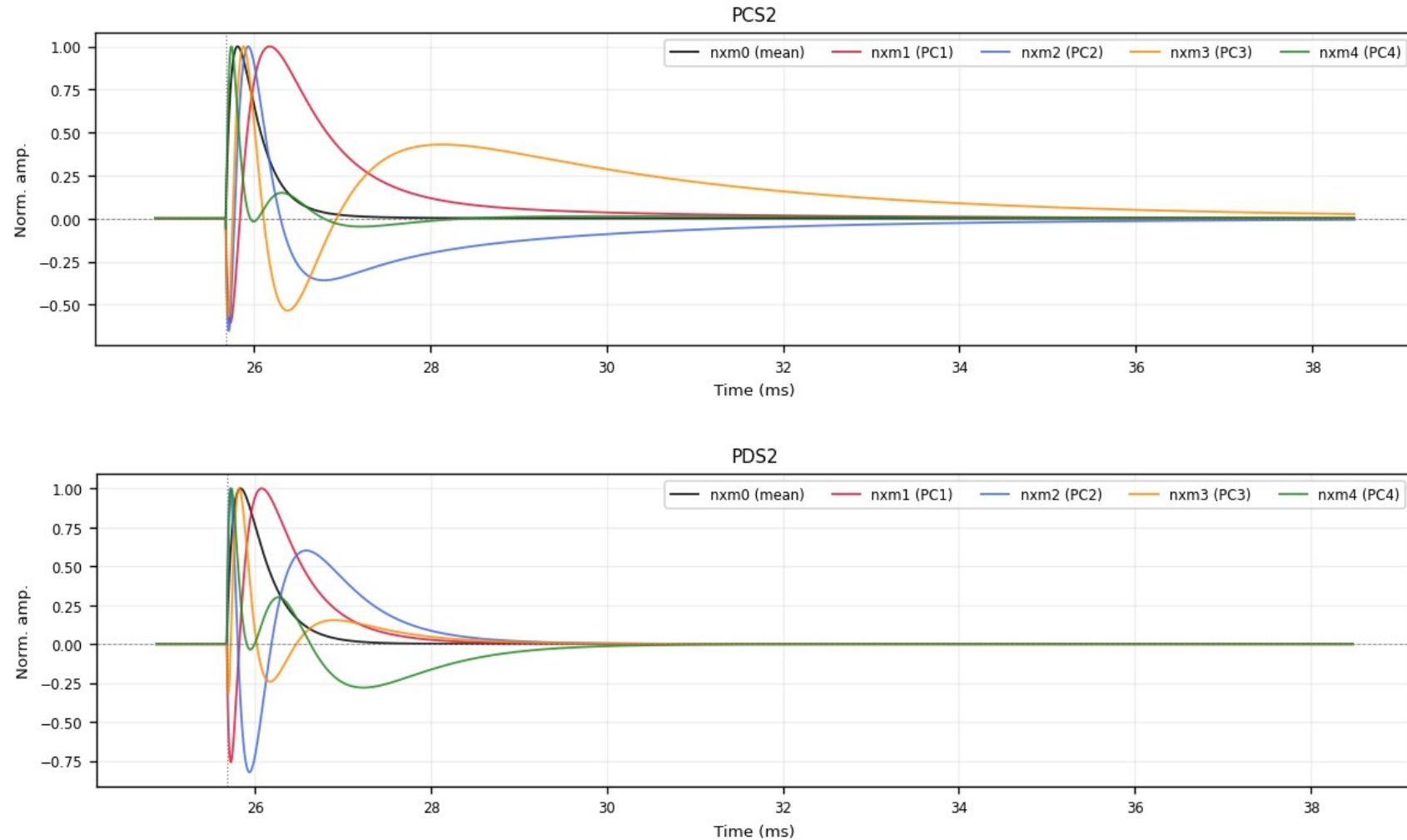
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.



Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (5/6)

figure: zip7_pca_templates.png, channel panels 9–10 of 11, top to bottom

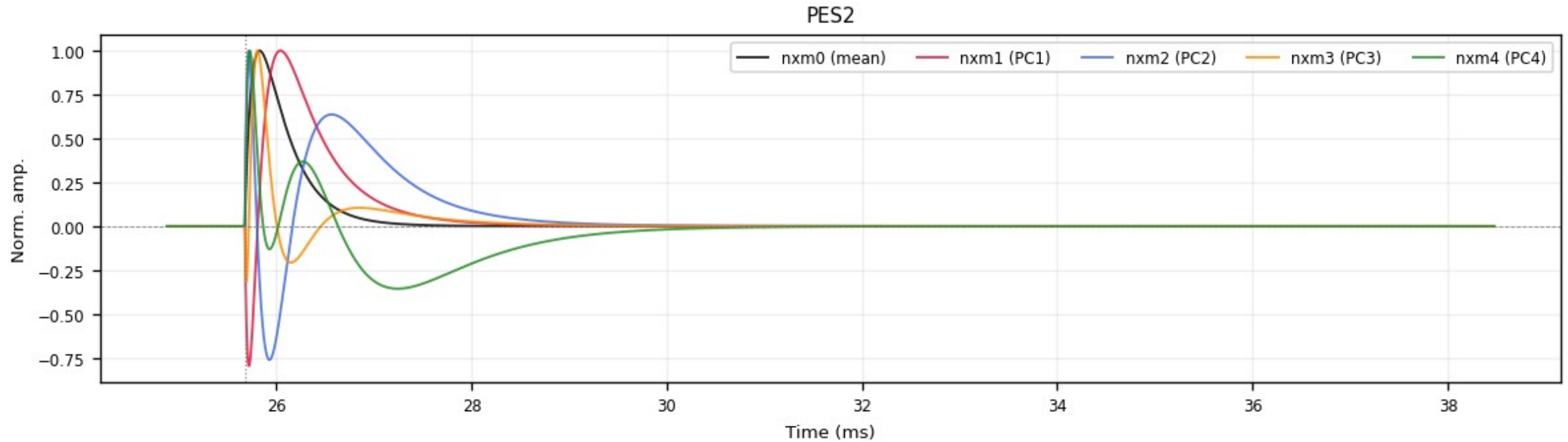
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.



Backup · Z7 · deliverables/nxm: PCA templates nxm0–4 (6/6)

figure: zip7_pca_templates.png, channel panels 11–11 of 11, top to bottom

nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized; PC1+PC2 capture 96–98% of the shape variance. PDS2 replaced by the PDS1 template in the delivered files.



Backup · Z7 · 1x1 summed templates PT / PS1 / PS2

figures: deliverables/1x1/plots/{PT,PS1,PS2}/zip7_*.png

Single-template (1x1) summed curves: peak-normalized average of the per-channel nxm0 templates: PT = all channels, PS1 / PS2 = side-1 / side-2 only.

